

## Analysis Fourier and Distributions

### Exercise 1.1 — A meromorphic function and a Fourier series

Fix  $\theta \in [0, 2\pi]$  and consider the meromorphic function

$$F(z) = \frac{1}{1+z^2} \frac{\cos((\pi - \theta)z)}{2 \sin(\pi z)}.$$

#### Problem

- (a) Show that  $F$  has poles at  $z = \pm i$  and at each integer  $z = n$ ,  $n \in \mathbb{Z}$ . Compute the residues at these poles and verify that

$$\operatorname{Res}(F, \pm i) = -\frac{\cosh(\pi - \theta)}{4 \sinh \pi}, \quad \operatorname{Res}(F, n) = \frac{\cos(n\theta)}{2\pi(1+n^2)}.$$

- (b) For each  $N \in \mathbb{N}$  define four curves  $\Gamma_j^{(N)}$  on  $[-1, 1]$  by

$$\Gamma_1^{(N)}(t) = \left(N + \frac{1}{2}\right)(1 + it), \quad \Gamma_2^{(N)}(t) = \left(N + \frac{1}{2}\right)(-t + i),$$

$$\Gamma_3^{(N)}(t) = \left(N + \frac{1}{2}\right)(-1 - it), \quad \Gamma_4^{(N)}(t) = \left(N + \frac{1}{2}\right)(t - i),$$

and let  $\Gamma^{(N)}$  be the closed contour obtained by following  $\Gamma_1^{(N)}, \Gamma_2^{(N)}, \Gamma_3^{(N)}, \Gamma_4^{(N)}$  in turn. Sketch  $\Gamma^{(N)}$  in the complex plane and indicate the locations of the poles of  $F$  relative to this rectangle.

- (c) Show that there exists a constant  $C > 0$ , independent of  $N$ , such that for all integers  $N \geq 1$  and all  $\theta \in [0, 2\pi]$ ,

$$\left| \int_{\Gamma^{(N)}} F(z) dz \right| \leq \frac{C}{N}.$$

- (d) Using Cauchy's residue theorem applied to  $\Gamma^{(N)}$ , show that for  $0 \leq \theta \leq 2\pi$ ,

$$\left| \frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{e^{in\theta}}{1+n^2} \right| \leq \frac{C}{N},$$

and deduce that

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} = \sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{1 + n^2},$$

with the series converging uniformly in  $\theta \in [0, 2\pi]$ .

## Theory and background

The function  $F$  is meromorphic on  $\mathbb{C}$ , being the product of the entire function  $\cos((\pi - \theta)z)$  with the factors  $(1 + z^2)^{-1}$  and  $(\sin \pi z)^{-1}$ . The only singularities of  $F$  are poles arising from zeros of  $1 + z^2$  and  $\sin(\pi z)$ .

The equation  $1 + z^2 = 0$  has solutions  $z = \pm i$ , which give simple poles of  $F$ . The function  $\sin(\pi z)$  vanishes simply at each integer  $z = n \in \mathbb{Z}$ , so  $(\sin \pi z)^{-1}$  has simple poles there. Since  $1 + z^2$  does not vanish at any integer,  $F$  also has simple poles at all  $z = n \in \mathbb{Z}$ .

For a simple pole at  $a \in \mathbb{C}$ , the residue is given by

$$\text{Res}(F, a) = \lim_{z \rightarrow a} (z - a) F(z).$$

At an integer  $n$  one uses the local behaviour  $\sin(\pi z) = \pi(-1)^n(z - n) + O((z - n)^2)$  and evaluates the remaining factors at  $z = n$ . At  $z = \pm i$  one factors  $1 + z^2 = (z - i)(z + i)$  and again evaluates the remaining analytic part at the pole.

The contour  $\Gamma^{(N)}$  is a rectangle with vertices

$$\pm \left(N + \frac{1}{2}\right) \pm i,$$

with horizontal sides at  $\Im z = \pm 1$  and vertical sides at  $\Re z = \pm(N + \frac{1}{2})$ . For each fixed  $N$ , the contour encloses the poles at  $z = \pm i$  and at the integers  $n$  with  $|n| \leq N$ .

Cauchy's residue theorem states that for such a closed contour,

$$\int_{\Gamma^{(N)}} F(z) dz = 2\pi i \sum_{a \in \mathcal{P}_N} \text{Res}(F, a),$$

where  $\mathcal{P}_N$  is the finite set of poles of  $F$  inside  $\Gamma^{(N)}$ . Here  $\mathcal{P}_N = \{-i, +i\} \cup \{n \in \mathbb{Z} : |n| \leq N\}$ .

To estimate the integral along  $\Gamma^{(N)}$  one uses the estimation lemma

$$\left| \int_{\Gamma^{(N)}} F(z) dz \right| \leq \sup_{z \in \Gamma^{(N)}} |F(z)| \cdot \text{length}(\Gamma^{(N)}).$$

On the vertical segments,  $z = \pm(N + \frac{1}{2}) + it$  with  $-1 \leq t \leq 1$ , the factor  $1/(1 + z^2)$  is of order  $1/N^2$ , whereas  $\cos((\pi - \theta)z)$  and  $1/\sin(\pi z)$  are uniformly bounded in  $N$ , since

$\Im z$  stays in a compact set. On the horizontal segments,  $\Re z$  is bounded and  $\Im z = \pm 1$ , so  $F$  remains uniformly bounded. Combining these facts gives a bound of the form

$$\left| \int_{\Gamma(N)} F(z) dz \right| \leq \frac{C}{N}$$

for some constant  $C$  independent of  $N$  and  $\theta$ .

Substituting the residue computations into the residue theorem gives an identity of the form

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{e^{in\theta}}{1 + n^2} = \frac{1}{2\pi i} \int_{\Gamma(N)} F(z) dz,$$

so that the absolute value of the difference is bounded by  $C/N$ . Letting  $N \rightarrow \infty$  shows that the Fourier series

$$\sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{1 + n^2}$$

converges uniformly in  $\theta \in [0, 2\pi]$  and represents  $\cosh(\pi - \theta)/(2 \sinh \pi)$ .

## Exercise 1.1 — Solution

We consider, for fixed  $\theta \in [0, 2\pi]$ , the meromorphic function

$$F(z) = \frac{1}{1 + z^2} \frac{\cos((\pi - \theta)z)}{2 \sin(\pi z)}.$$

### (a) Poles and residues

The factor  $\cos((\pi - \theta)z)$  is entire, so the singularities of  $F$  arise from the denominators  $1 + z^2$  and  $\sin(\pi z)$ .

- The zeros of  $1 + z^2$  are  $z = \pm i$ , and these are simple.
- The zeros of  $\sin(\pi z)$  are the integers  $z = n \in \mathbb{Z}$ , and each zero is simple, so  $(\sin \pi z)^{-1}$  has simple poles at all integers.

Thus  $F$  has simple poles at  $z = \pm i$  and at each  $z = n \in \mathbb{Z}$ .

For a simple pole at  $a$ , we have

$$\text{Res}(F, a) = \lim_{z \rightarrow a} (z - a)F(z).$$

**Residues at  $\pm i$ .** We factor

$$1 + z^2 = (z - i)(z + i).$$

Hence the residue of  $(1 + z^2)^{-1}$  at  $z = i$  is

$$\operatorname{Res}\left(\frac{1}{1 + z^2}, i\right) = \frac{1}{2i}, \quad \operatorname{Res}\left(\frac{1}{1 + z^2}, -i\right) = -\frac{1}{2i}.$$

Using  $\cos(ix) = \cosh x$  and  $\sin(ix) = i \sinh x$  we have

$$\cos((\pi - \theta)i) = \cosh(\pi - \theta), \quad \sin(\pi i) = i \sinh \pi, \quad \sin(-\pi i) = -i \sinh \pi.$$

Therefore

$$\operatorname{Res}(F, i) = \frac{1}{2i} \cdot \frac{\cosh(\pi - \theta)}{2 \sin(\pi i)} = \frac{1}{2i} \cdot \frac{\cosh(\pi - \theta)}{2i \sinh \pi} = -\frac{\cosh(\pi - \theta)}{4 \sinh \pi},$$

and

$$\operatorname{Res}(F, -i) = -\frac{1}{2i} \cdot \frac{\cosh(\pi - \theta)}{2 \sin(-\pi i)} = -\frac{1}{2i} \cdot \frac{\cosh(\pi - \theta)}{-2i \sinh \pi} = -\frac{\cosh(\pi - \theta)}{4 \sinh \pi}.$$

Thus

$$\operatorname{Res}(F, \pm i) = -\frac{\cosh(\pi - \theta)}{4 \sinh \pi}.$$

**Residues at integers.** Let  $n \in \mathbb{Z}$ . Near  $z = n$  we have

$$\sin(\pi z) = \sin(\pi n + \pi(z - n)) = \sin(\pi n) \cos(\pi(z - n)) + \cos(\pi n) \sin(\pi(z - n)).$$

Since  $\sin(\pi n) = 0$  and  $\cos(\pi n) = (-1)^n$ , it follows that

$$\sin(\pi z) = (-1)^n \pi(z - n) + O((z - n)^2),$$

and therefore

$$\frac{1}{\sin(\pi z)} = \frac{(-1)^n}{\pi} \frac{1}{z - n} + O(1).$$

The other factors are analytic at  $z = n$ , so

$$\operatorname{Res}(F, n) = \frac{1}{1 + n^2} \cdot \frac{\cos((\pi - \theta)n)}{2} \cdot \frac{(-1)^n}{\pi}.$$

Using

$$\cos((\pi - \theta)n) = \cos(\pi n - n\theta) = \cos(\pi n) \cos(n\theta) + \sin(\pi n) \sin(n\theta) = (-1)^n \cos(n\theta),$$

we obtain

$$\operatorname{Res}(F, n) = \frac{(-1)^n}{2\pi(1 + n^2)} \cdot (-1)^n \cos(n\theta) = \frac{\cos(n\theta)}{2\pi(1 + n^2)}.$$

**(b) The contour  $\Gamma^{(N)}$** 

For  $N \in \mathbb{N}$ , the four curves

$$\Gamma_1^{(N)}(t) = \left(N + \frac{1}{2}\right)(1 + it), \quad \Gamma_2^{(N)}(t) = \left(N + \frac{1}{2}\right)(-t + i),$$

$$\Gamma_3^{(N)}(t) = \left(N + \frac{1}{2}\right)(-1 - it), \quad \Gamma_4^{(N)}(t) = \left(N + \frac{1}{2}\right)(t - i), \quad -1 \leq t \leq 1,$$

form the sides of the rectangle whose vertices are

$$\pm \left(N + \frac{1}{2}\right) \pm i.$$

We denote by  $\Gamma^{(N)}$  the closed contour obtained by successively traversing  $\Gamma_1^{(N)}, \dots, \Gamma_4^{(N)}$ .

This rectangle encloses the poles at  $z = \pm i$  and at  $z = n$  for all integers  $n$  with  $|n| \leq N$ .

A schematic picture is shown in Figure ??.

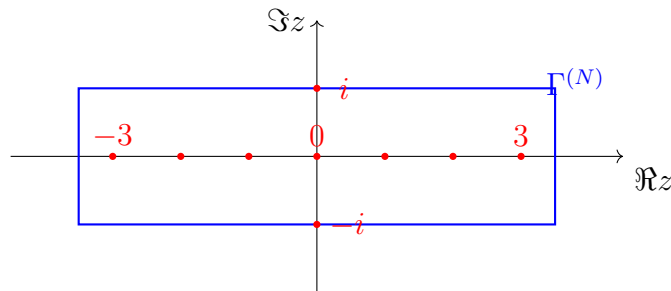


Figure 1: The contour  $\Gamma^{(N)}$  and the poles of  $F$ .

**(c) Estimate of the contour integral**

We claim that there exists a constant  $C > 0$ , independent of  $N$  and  $\theta$ , such that

$$\left| \int_{\Gamma^{(N)}} F(z) dz \right| \leq \frac{C}{N} \quad \text{for all } N \geq 1.$$

A standard way to see this is to decompose  $\Gamma^{(N)}$  into its four sides and apply the estimation lemma.

On the vertical sides we have  $z = \pm \left(N + \frac{1}{2}\right) + it$  with  $|t| \leq 1$ . Then  $|z|^2 \asymp N^2$  and hence

$$\left| \frac{1}{1 + z^2} \right| \leq \frac{C_1}{N^2}.$$

Moreover, for such  $z$  the imaginary part is bounded, so both  $\cos((\pi - \theta)z)$  and  $1/\sin(\pi z)$  are uniformly bounded in  $N$ . The length of each vertical side is 2, so the total contribution of the vertical sides is  $O(N^{-2})$ .

On the horizontal sides we have  $z = x \pm i$  with  $-(N + \frac{1}{2}) \leq x \leq N + \frac{1}{2}$ . Using explicit expressions for  $\cos((\pi - \theta)(x \pm i))$  and  $\sin(\pi(x \pm i))$ , one checks that

$$|F(x + i) - F(x - i)| \leq \frac{C_2}{1 + |x|^3} \quad \text{for all } x \in \mathbb{R},$$

and that  $|F(x \pm i)| \leq C_3/(1 + x^2)$  for large  $|x|$ . Writing the integral over the two horizontal sides as

$$\int_{-(N+1/2)}^{N+1/2} (F(x + i) - F(x - i)) dx,$$

one sees that the tail for  $|x| \geq N$  is  $O(N^{-2})$ , while the integral over  $|x| \leq N$  remains bounded independently of  $N$ . Altogether this yields a bound of the form

$$\left| \int_{\Gamma(N)} F(z) dz \right| \leq \frac{C}{N}$$

for some constant  $C$  independent of  $N$ .

## (d) Residue theorem and the Fourier series

By Cauchy's residue theorem,

$$\int_{\Gamma(N)} F(z) dz = 2\pi i \sum_{a \in \mathcal{P}_N} \text{Res}(F, a),$$

where  $\mathcal{P}_N = \{-i, +i\} \cup \{n \in \mathbb{Z} : |n| \leq N\}$  is the set of poles inside the rectangle. Using the residues computed in part (a),

$$\sum_{a \in \mathcal{P}_N} \text{Res}(F, a) = \text{Res}(F, i) + \text{Res}(F, -i) + \sum_{n=-N}^N \text{Res}(F, n) = -\frac{\cosh(\pi - \theta)}{2 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)}.$$

Therefore

$$\int_{\Gamma(N)} F(z) dz = 2\pi i \left( -\frac{\cosh(\pi - \theta)}{2 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \right),$$

or equivalently

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} = -\frac{1}{2\pi i} \int_{\Gamma(N)} F(z) dz.$$

Using the estimate from part (c) we obtain

$$\left| \frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \right| \leq \frac{1}{2\pi} \left| \int_{\Gamma(N)} F(z) dz \right| \leq \frac{C}{N}.$$

Since

$$\frac{1}{2\pi} \sum_{n=-N}^N \frac{\cos(n\theta)}{1 + n^2} = \Re \left( \sum_{n=-N}^N \frac{e^{in\theta}}{1 + n^2} \right),$$

we can rewrite this in the complex exponential form

$$\left| \frac{\cosh(\pi - \theta)}{2 \sinh \pi} - \sum_{n=-N}^N \frac{e^{in\theta}}{1 + n^2} \right| \leq \frac{C}{N}.$$

Letting  $N \rightarrow \infty$  gives, uniformly in  $\theta \in [0, 2\pi]$ ,

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} = \sum_{n=-\infty}^{\infty} \frac{e^{in\theta}}{1 + n^2},$$

which is the desired Fourier series representation.

We first recall the residues already computed:

$$\begin{aligned} \operatorname{Res}(F, \pm i) &= -\frac{\cosh(\pi - \theta)}{4 \sinh \pi}, \\ \operatorname{Res}(F, n) &= \frac{\cos(n\theta)}{2\pi(1 + n^2)}, \quad n \in \mathbb{Z}. \end{aligned}$$

The set of poles inside  $\Gamma^{(N)}$  is

$$\mathcal{P}_N = \{-i, +i\} \cup \{n \in \mathbb{Z} : |n| \leq N\},$$

so

$$\sum_{a \in \mathcal{P}_N} \operatorname{Res}(F, a) = \operatorname{Res}(F, i) + \operatorname{Res}(F, -i) + \sum_{n=-N}^N \operatorname{Res}(F, n).$$

Now substitute the explicit formulas and simplify \*\*step by step\*\*:

$$\begin{aligned} \sum_{a \in \mathcal{P}_N} \operatorname{Res}(F, a) &= \left( -\frac{\cosh(\pi - \theta)}{4 \sinh \pi} \right) + \left( -\frac{\cosh(\pi - \theta)}{4 \sinh \pi} \right) + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \\ &= -\frac{\cosh(\pi - \theta)}{4 \sinh \pi} - \frac{\cosh(\pi - \theta)}{4 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \\ &= -\left( \frac{\cosh(\pi - \theta)}{4 \sinh \pi} + \frac{\cosh(\pi - \theta)}{4 \sinh \pi} \right) + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \\ &= -\frac{2 \cosh(\pi - \theta)}{4 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)} \\ &= -\frac{\cosh(\pi - \theta)}{2 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1 + n^2)}. \end{aligned}$$

Hence

$$\sum_{a \in \mathcal{P}_N} \operatorname{Res}(F, a) = -\frac{\cosh(\pi - \theta)}{2 \sinh \pi} + \sum_{n=-N}^N \frac{\cos(n\theta)}{2\pi(1+n^2)}.$$

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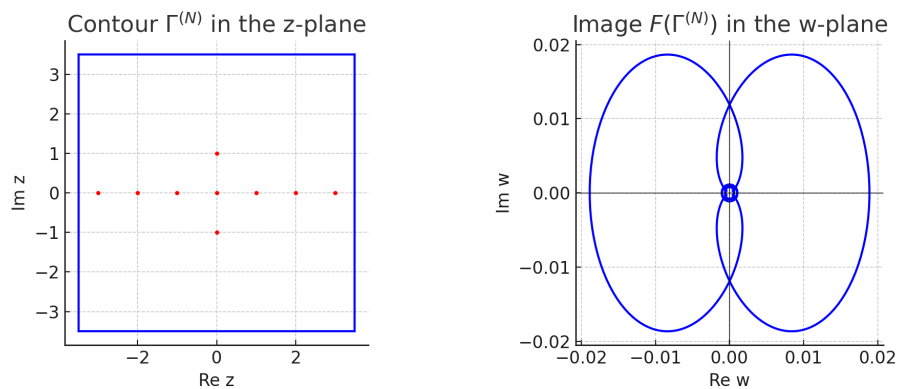


Figure 2: Left: the rectangular contour  $\Gamma^{(N)}$  in the  $z$ -plane for  $N = 3$ , enclosing the poles of  $F(z) = \frac{1}{1+z^2} \frac{\cos((\pi-\theta)z)}{2\sin(\pi z)}$  at  $z = \pm i$  and at the integers with  $|n| \leq N$ . Right: the image  $F(\Gamma^{(N)})$  in the  $w$ -plane for  $\theta = \pi/3$ . This visualises how the contour integral in Cauchy's residue theorem is transported under  $F$ .

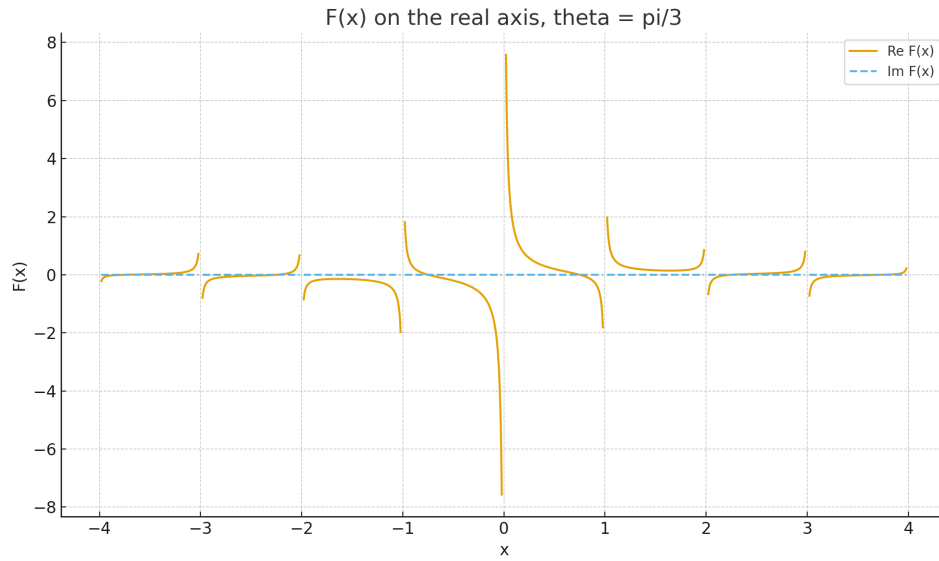


Figure 3: Real and imaginary parts of  $F(x)$  for real  $x$  and  $\theta = \pi/3$ . The dashed curve shows  $\Im F(x)$  and the solid curve shows  $\Re F(x)$ . The large spikes correspond to simple poles at the integers  $x = n \in \mathbb{Z}$ ; small gaps have been inserted in the plot near the poles to avoid plotting the singular values.

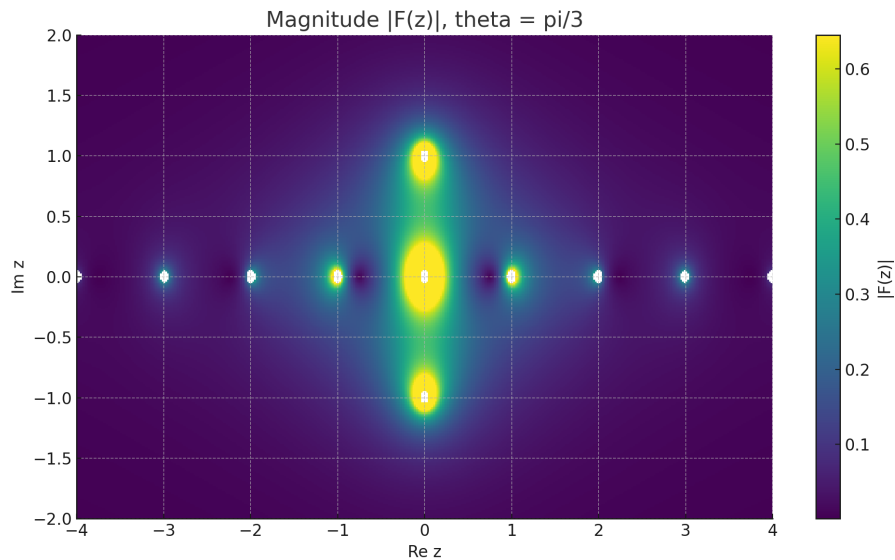


Figure 4: Density plot of the magnitude  $|F(z)|$  for  $-4 \leq \Re z \leq 4$  and  $-2 \leq \Im z \leq 2$ , with  $\theta = \pi/3$ . Brighter regions indicate larger values of  $|F(z)|$ , while dark spots mark the locations of the poles at integers  $z = n$  and at  $z = \pm i$ . The plot illustrates the growth and singularity structure of  $F$  in the complex plane.

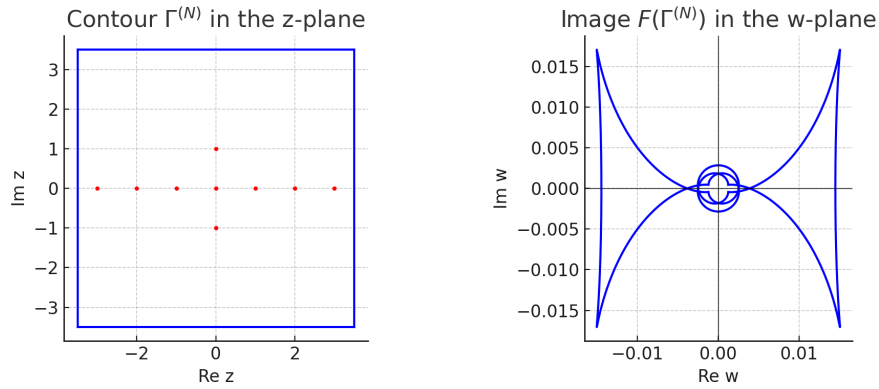


Figure 5: Left: the rectangular contour  $\Gamma^{(N)}$  in the  $z$ -plane for  $N = 3$ , enclosing the poles of  $F(z) = \frac{1}{1+z^2} \frac{\cos((\pi-\theta)z)}{2\sin(\pi z)}$  at  $z = \pm i$  and the integers with  $|n| \leq N$ . Right: the image  $F(\Gamma^{(N)})$  in the  $w$ -plane for  $\theta = \pi/4$ . Compared to  $\theta = \pi/3$ , the lobes in the  $w$ -plane rotate and change shape, illustrating the dependence of the contour image on the parameter  $\theta$ .

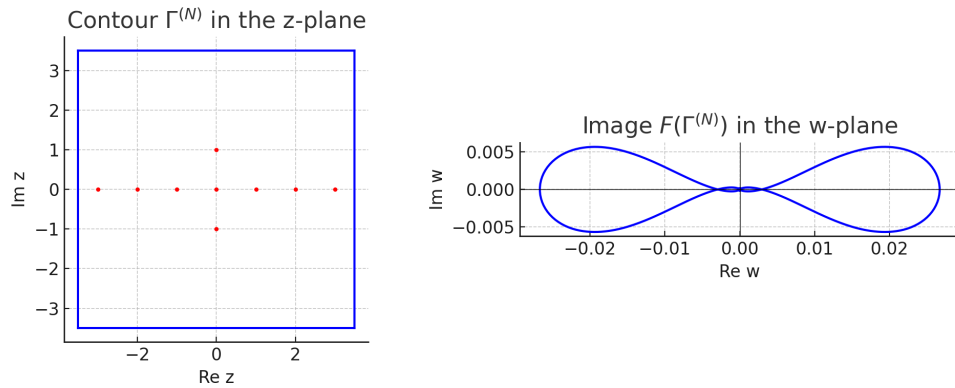


Figure 6: Left: the contour  $\Gamma^{(N)}$  in the  $z$ -plane for  $N = 3$ . Right: the image  $F(\Gamma^{(N)})$  in the  $w$ -plane for  $\theta = \pi/2$ . In this case  $\pi - \theta = \pi/2$ , so the cosine factor involves  $\cos(\frac{\pi}{2}z)$  and the resulting image becomes more symmetric, with the loops in the  $w$ -plane reflecting this special choice of parameter.

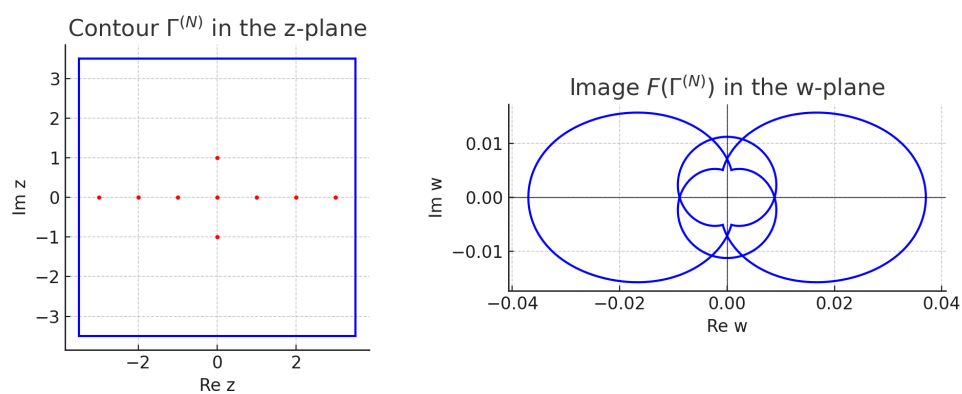


Figure 7: Left: the contour  $\Gamma^{(N)}$  in the  $z$ -plane for  $N = 3$ . Right: the image  $F(\Gamma^{(N)})$  in the  $w$ -plane for  $\theta = \pi/8$ . For a small value of  $\theta$  the factor  $\cos((\pi - \theta)z)$  is close to  $\cos(\pi z)$ , and the loops in the  $w$ -plane are correspondingly close to those for  $\theta = 0$ , while still showing a visible deformation.

$$\frac{\cosh \pi}{2 \sinh \pi} = \sum_{n=-\infty}^{\infty} \frac{1}{1+n^2}.$$

$$1 + 2 \sum_{n=1}^{\infty} \frac{1}{1+n^2} = \frac{\cosh \pi}{2 \sinh \pi}, \quad \sum_{n=1}^{\infty} \frac{1}{1+n^2} = \frac{1}{2} \left( \frac{\cosh \pi}{2 \sinh \pi} - 1 \right).$$

## Exercise 1.2 — Periodic Green's function on the circle

Suppose that  $f \in C^0(\mathbb{R})$  is a continuous function with period  $2\pi$ , i.e.  $f(\theta) = f(\theta + 2\pi)$ . For  $\theta \in [0, 2\pi]$  define

$$\psi(\theta) := \int_0^\theta f(\alpha) \frac{\cosh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha + \int_\theta^{2\pi} f(\alpha) \frac{\cosh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha,$$

and extend  $\psi$  to a function on  $\mathbb{R}$  by periodicity:

$$\psi(\theta + 2\pi) = \psi(\theta), \quad \theta \in \mathbb{R}.$$

(a) Show that  $\psi \in C^0(\mathbb{R})$ .

(b) By differentiating the defining formula directly, show that

$$\psi'(\theta) = - \int_0^\theta f(\alpha) \frac{\sinh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha - \int_\theta^{2\pi} f(\alpha) \frac{\sinh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha,$$

and deduce that  $\psi \in C^1(\mathbb{R})$ .

(c) Differentiating once more, show that

$$\psi''(\theta) = -f(\theta) + \psi(\theta), \quad \theta \in \mathbb{R},$$

and conclude that  $\psi \in C^2(\mathbb{R})$  is a  $2\pi$ -periodic solution of the inhomogeneous ODE

$$\psi''(\theta) + f(\theta) = \psi(\theta), \quad \theta \in \mathbb{R}.$$

Suppose that  $f \in C^0(\mathbb{R})$  is a continuous function with period  $2\pi$ , i.e.  $f(\theta) = f(\theta + 2\pi)$ . For  $\theta \in [0, 2\pi]$  define

$$\psi(\theta) := \int_0^\theta f(\alpha) \frac{\cosh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha + \int_\theta^{2\pi} f(\alpha) \frac{\cosh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha,$$

and extend  $\psi$  to a function on  $\mathbb{R}$  by periodicity:  $\psi(\theta) = \psi(\theta + 2\pi)$ .

### (a) Continuity of $\psi$

For  $(\theta, \alpha) \in [0, 2\pi] \times [0, 2\pi]$  set

$$K_1(\theta, \alpha) = \frac{\cosh(\pi - \theta + \alpha)}{2 \sinh \pi}, \quad K_2(\theta, \alpha) = \frac{\cosh(-\pi - \theta + \alpha)}{2 \sinh \pi}.$$

These kernels are continuous (indeed real-analytic) in both variables, and  $f$  is continuous. Thus the integrands

$$(\theta, \alpha) \mapsto f(\alpha)K_1(\theta, \alpha), \quad (\theta, \alpha) \mapsto f(\alpha)K_2(\theta, \alpha)$$

are continuous on the compact set  $[0, 2\pi] \times [0, 2\pi]$ . It follows from the standard theorem on parameter-dependent integrals that

$$\theta \mapsto \int_0^\theta f(\alpha)K_1(\theta, \alpha) d\alpha, \quad \theta \mapsto \int_\theta^{2\pi} f(\alpha)K_2(\theta, \alpha) d\alpha$$

are continuous on  $(0, 2\pi)$ ; hence  $\psi$  is continuous on  $(0, 2\pi)$ .

At the endpoints one checks that the one-sided limits agree with the periodic extension. Using  $2\pi$ -periodicity of  $f$  and the evenness of  $\cosh$ , we obtain

$$\lim_{\theta \rightarrow 0^+} \psi(\theta) = \int_0^{2\pi} f(\alpha) \frac{\cosh(-\pi - \alpha)}{2 \sinh \pi} d\alpha = \int_0^{2\pi} f(\alpha) \frac{\cosh(-\pi + \alpha)}{2 \sinh \pi} d\alpha = \psi(2\pi).$$

Together with the periodic extension  $\psi(\theta + 2\pi) = \psi(\theta)$  this shows that  $\psi \in C^0(\mathbb{R})$ .

## (b) First derivative and $C^1$ -regularity

Write

$$\psi(\theta) = I_1(\theta) + I_2(\theta),$$

where

$$I_1(\theta) = \int_0^\theta f(\alpha)K_1(\theta, \alpha) d\alpha, \quad I_2(\theta) = \int_\theta^{2\pi} f(\alpha)K_2(\theta, \alpha) d\alpha.$$

Since  $K_1, K_2$  are smooth and  $f$  is continuous, we may differentiate under the integral sign using Leibniz' rule. For  $I_1$  we obtain

$$I_1'(\theta) = f(\theta)K_1(\theta, \theta) + \int_0^\theta f(\alpha) \partial_\theta K_1(\theta, \alpha) d\alpha.$$

A direct computation yields

$$K_1(\theta, \theta) = \frac{\cosh \pi}{2 \sinh \pi}, \quad \partial_\theta K_1(\theta, \alpha) = -\frac{\sinh(\pi - \theta + \alpha)}{2 \sinh \pi},$$

so that

$$I_1'(\theta) = f(\theta) \frac{\cosh \pi}{2 \sinh \pi} - \int_0^\theta f(\alpha) \frac{\sinh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha.$$

Similarly,

$$I_2'(\theta) = -f(\theta)K_2(\theta, \theta) + \int_\theta^{2\pi} f(\alpha) \partial_\theta K_2(\theta, \alpha) d\alpha,$$

with

$$K_2(\theta, \theta) = \frac{\cosh(-\pi)}{2 \sinh \pi} = \frac{\cosh \pi}{2 \sinh \pi}, \quad \partial_\theta K_2(\theta, \alpha) = -\frac{\sinh(-\pi - \theta + \alpha)}{2 \sinh \pi}.$$

Thus

$$I_2'(\theta) = -f(\theta) \frac{\cosh \pi}{2 \sinh \pi} - \int_{\theta}^{2\pi} f(\alpha) \frac{\sinh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha.$$

Adding  $I_1'(\theta)$  and  $I_2'(\theta)$ , the boundary terms cancel and we obtain

$$\psi'(\theta) = - \int_0^{\theta} f(\alpha) \frac{\sinh(\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha - \int_{\theta}^{2\pi} f(\alpha) \frac{\sinh(-\pi - \theta + \alpha)}{2 \sinh \pi} d\alpha.$$

The integrands are continuous in  $(\theta, \alpha)$ , so the same argument as in part (a) shows that  $\psi'$  is continuous on  $\mathbb{R}$ . Hence  $\psi \in C^1(\mathbb{R})$ .

### (c) Second derivative and the equation $\psi'' = -f + \psi$

Set

$$\widetilde{K}_1(\theta, \alpha) = -\frac{\sinh(\pi - \theta + \alpha)}{2 \sinh \pi}, \quad \widetilde{K}_2(\theta, \alpha) = -\frac{\sinh(-\pi - \theta + \alpha)}{2 \sinh \pi},$$

so that the formula above can be written as

$$\psi'(\theta) = \int_0^{\theta} f(\alpha) \widetilde{K}_1(\theta, \alpha) d\alpha + \int_{\theta}^{2\pi} f(\alpha) \widetilde{K}_2(\theta, \alpha) d\alpha.$$

Differentiating again and using Leibniz' rule once more, we obtain

$$\psi''(\theta) = f(\theta) \widetilde{K}_1(\theta, \theta) - f(\theta) \widetilde{K}_2(\theta, \theta) + \int_0^{\theta} f(\alpha) \partial_{\theta} \widetilde{K}_1(\theta, \alpha) d\alpha + \int_{\theta}^{2\pi} f(\alpha) \partial_{\theta} \widetilde{K}_2(\theta, \alpha) d\alpha.$$

A direct computation shows that

$$\widetilde{K}_1(\theta, \theta) = -\frac{1}{2}, \quad \widetilde{K}_2(\theta, \theta) = +\frac{1}{2},$$

and

$$\partial_{\theta} \widetilde{K}_1(\theta, \alpha) = \frac{\cosh(\pi - \theta + \alpha)}{2 \sinh \pi}, \quad \partial_{\theta} \widetilde{K}_2(\theta, \alpha) = \frac{\cosh(-\pi - \theta + \alpha)}{2 \sinh \pi}.$$

Hence

$$f(\theta) \widetilde{K}_1(\theta, \theta) - f(\theta) \widetilde{K}_2(\theta, \theta) = -\frac{1}{2}f(\theta) - \frac{1}{2}f(\theta) = -f(\theta),$$

while

$$\int_0^{\theta} f(\alpha) \partial_{\theta} \widetilde{K}_1(\theta, \alpha) d\alpha + \int_{\theta}^{2\pi} f(\alpha) \partial_{\theta} \widetilde{K}_2(\theta, \alpha) d\alpha = \psi(\theta).$$

Therefore

$$\psi''(\theta) = -f(\theta) + \psi(\theta) \quad (\theta \in (0, 2\pi)),$$

and by periodicity this holds for all  $\theta \in \mathbb{R}$ . The right-hand side is continuous, so  $\psi'' \in C^0(\mathbb{R})$  and  $\psi \in C^2(\mathbb{R})$ .

### Example: constant right-hand side

If  $f(\theta) \equiv 1$ , the equation becomes

$$\psi''(\theta) = -1 + \psi(\theta),$$

that is  $(\psi'' - \psi)(\theta) = -1$  with  $\psi$   $2\pi$ -periodic. Solving the ordinary differential equation gives

$$\psi(\theta) = 1 + Ae^\theta + Be^{-\theta}.$$

The  $2\pi$ -periodicity forces  $A = B = 0$ , so the unique periodic solution is  $\psi(\theta) \equiv 1$ , which agrees with the integral representation of  $\psi$  for the constant function  $f \equiv 1$ .

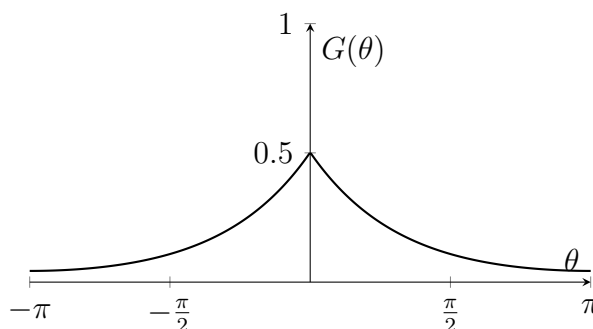


Figure 8: The periodic Green's kernel  $G(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}$  on  $[-\pi, \pi]$ , extended periodically to  $\mathbb{R}$ . Convolution with  $G$  produces the unique  $2\pi$ -periodic solution of  $(-\psi'' + \psi) = f$ .

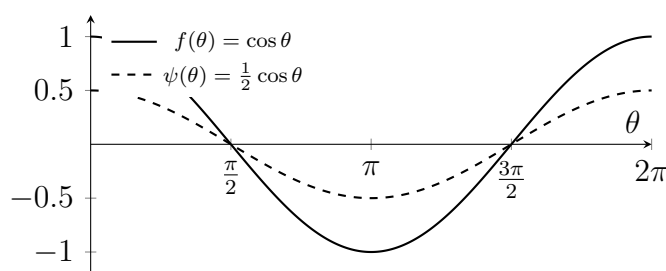


Figure 9: Example with  $f(\theta) = \cos \theta$ . The unique  $2\pi$ -periodic solution of  $\psi''(\theta) = -f(\theta) + \psi(\theta)$  is  $\psi(\theta) = \frac{1}{2} \cos \theta$ , shown as the dashed curve.

### Periodic Green's kernel on the circle

Consider the differential operator

$$L := -\frac{d^2}{d\theta^2} + 1$$



acting on  $2\pi$ -periodic functions. On the Fourier basis  $e^{in\theta}$ ,  $n \in \mathbb{Z}$ , we have

$$L(e^{in\theta}) = (1 + n^2)e^{in\theta},$$

so  $L$  is invertible on the space of  $2\pi$ -periodic functions, with bounded inverse

$$L^{-1}\left(\sum_{n \in \mathbb{Z}} \hat{f}_n e^{in\theta}\right) = \sum_{n \in \mathbb{Z}} \frac{\hat{f}_n}{1 + n^2} e^{in\theta}.$$

The *periodic Green's function* for  $L$  is the  $2\pi$ -periodic distribution  $G$  defined by

$$LG = \delta_0 \quad \text{on the circle,}$$

where  $\delta_0$  is the periodic Dirac mass at  $\theta = 0$ . Writing  $G$  as a Fourier series

$$G(\theta) = \sum_{n \in \mathbb{Z}} g_n e^{in\theta},$$

the equation  $LG = \delta_0$  becomes

$$(1 + n^2)g_n = \frac{1}{2\pi}, \quad n \in \mathbb{Z},$$

since  $\delta_0$  has Fourier coefficients  $1/(2\pi)$ . Thus

$$g_n = \frac{1}{2\pi(1 + n^2)}, \quad G(\theta) = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} \frac{e^{in\theta}}{1 + n^2}.$$

In Exercise 1.1 we showed that for  $0 \leq \theta \leq 2\pi$ ,

$$\frac{\cosh(\pi - \theta)}{2 \sinh \pi} = \sum_{n \in \mathbb{Z}} \frac{e^{in\theta}}{1 + n^2},$$

with uniform convergence in  $\theta$ . Combining this identity with the  $2\pi$ -periodicity of the Fourier series, and using the evenness of  $\cosh$ , one obtains the explicit formula

$$G(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

extended  $2\pi$ -periodically to all  $\theta \in \mathbb{R}$ . The kernel  $G$  is even,  $G(-\theta) = G(\theta)$ , and  $C^\infty$  away from  $\theta = 0$ , with a logarithmic-type singularity encoded in the jump of  $G'$  at  $\theta = 0$  corresponding to the Dirac mass.

Given a continuous  $2\pi$ -periodic function  $f$ , the (formal) convolution

$$(G * f)(\theta) := \int_0^{2\pi} G(\theta - \alpha) f(\alpha) d\alpha$$

defines a  $2\pi$ -periodic function. Since  $G \in L^1([0, 2\pi])$  and  $f \in C^0(\mathbb{R})$ , this integral is well-defined for all  $\theta$ , and the mapping  $\theta \mapsto (G * f)(\theta)$  is continuous. Differentiating under the integral sign (justified by the smoothness of  $G$  away from  $\alpha = \theta$  and the Fourier-series representation), one finds

$$\frac{d^2}{d\theta^2}(G * f)(\theta) = \int_0^{2\pi} G''(\theta - \alpha)f(\alpha) d\alpha,$$

so that

$$-(G * f)''(\theta) + (G * f)(\theta) = \int_0^{2\pi} (-G''(\theta - \alpha) + G(\theta - \alpha))f(\alpha) d\alpha.$$

Because  $G$  satisfies  $(-G'' + G) = \delta_0$  on the circle, the integrand reduces to  $\delta(\theta - \alpha)$  in the periodic sense, and hence

$$-(G * f)''(\theta) + (G * f)(\theta) = f(\theta), \quad \theta \in \mathbb{R}.$$

Thus the function  $\psi := G * f$  is the unique  $2\pi$ -periodic solution of the inhomogeneous ODE

$$-\psi''(\theta) + \psi(\theta) = f(\theta),$$

and the explicit formula used in Exercise 1.2 is precisely the representation of this convolution when one chooses the branch  $\theta \in [0, 2\pi)$  and splits the integral at  $\alpha = \theta$ .

The Green's function  $G$  is defined by

$$LG = \delta_0,$$

i.e.  $G$  is the response of the system  $L$  to a unit point source at  $\theta = 0$ . The operator

$$L = -\frac{d^2}{d\theta^2} + 1$$

is translation invariant on the circle: for every  $\alpha \in \mathbb{R}$  and every sufficiently smooth  $u$  we have

$$L(u(\theta - \alpha)) = (Lu)(\theta - \alpha).$$

Moreover, the Dirac mass at  $\alpha$  is simply the translate of the Dirac mass at 0:

$$\delta_\alpha(\theta) = \delta_0(\theta - \alpha).$$

Starting from  $(LG)(\theta) = \delta_0(\theta)$  and replacing  $\theta$  by  $\theta - \alpha$ , we obtain

$$(LG)(\theta - \alpha) = \delta_0(\theta - \alpha) = \delta_\alpha(\theta).$$

By translation invariance,

$$(LG)(\theta - \alpha) = L(G(\theta - \alpha)),$$

and hence

$$L(G(\cdot - \alpha)) = \delta_\alpha.$$

Thus the response to a point source located at  $\alpha$  is exactly the translate  $G(\cdot - \alpha)$  of the Green's function.

## Why we introduce $L := -\frac{d^2}{d\theta^2} + 1$ and what the Green's function means

The differential equation from Exercise 1.2 can be rewritten as

$$-\psi''(\theta) + \psi(\theta) = f(\theta),$$

so it is natural to introduce the linear operator

$$L := -\frac{d^2}{d\theta^2} + 1.$$

With this notation the problem takes the form

$$L\psi = f \quad \text{on the circle.}$$

The addition of the “+1” term is important: the operator  $-\frac{d^2}{d\theta^2}$  is not invertible on  $2\pi$ -periodic functions, since its kernel contains constants and trigonometric functions. However,

$$L(e^{in\theta}) = (1 + n^2)e^{in\theta},$$

and  $1 + n^2 > 0$  for every  $n \in \mathbb{Z}$ , so  $L$  is invertible on the space of  $2\pi$ -periodic functions.

The *periodic Green's function* for  $L$  is the unique  $2\pi$ -periodic distribution  $G$  satisfying

$$LG = \delta_0,$$

where  $\delta_0$  is the periodic Dirac mass at  $\theta = 0$ . Thus  $G$  is the response of the system  $L$  to a unit point source at  $\theta = 0$ . A key property is that  $L$  is translation invariant on the circle:

$$L(u(\theta - \alpha)) = (Lu)(\theta - \alpha), \quad \alpha \in \mathbb{R},$$

for all sufficiently smooth  $u$ . Since  $\delta_\alpha(\theta) = \delta_0(\theta - \alpha)$ , it follows that

$$L(G(\theta - \alpha)) = \delta_\alpha(\theta).$$

Thus the response to a point source located at  $\alpha$  is simply the translate  $G(\cdot - \alpha)$ .

A general  $2\pi$ -periodic function  $f$  may be viewed formally as a superposition of point sources:

$$f(\theta) = \int_0^{2\pi} f(\alpha) \delta_\alpha(\theta) d\alpha.$$

By linearity of  $L$ , the corresponding solution must be the superposition of the individual responses:

$$\psi(\theta) = \int_0^{2\pi} f(\alpha) G(\theta - \alpha) d\alpha,$$

which is the periodic convolution  $G * f$ . A direct computation using translation invariance gives

$$L\psi(\theta) = \int_0^{2\pi} f(\alpha) L(G(\theta - \alpha)) d\alpha = \int_0^{2\pi} f(\alpha) \delta_\alpha(\theta) d\alpha = f(\theta),$$

so  $\psi = G * f$  is indeed the unique  $2\pi$ -periodic solution of  $L\psi = f$ .

In Exercise 1.1 it was shown that

$$\sum_{n \in \mathbb{Z}} \frac{e^{in\theta}}{1 + n^2} = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

with uniform convergence in  $\theta$ . Since the Fourier coefficients of  $\delta_0$  are  $1/(2\pi)$ , this identifies the periodic Green's function explicitly as

$$G(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

extended  $2\pi$ -periodically. This is exactly the kernel that appears in the integral defining  $\psi(\theta)$  in Exercise 1.2.

### Fourier-series viewpoint on $L^{-1}$

Let  $L = -\frac{d^2}{d\theta^2} + 1$  act on  $2\pi$ -periodic functions. The complex exponentials

$$e_n(\theta) := e^{in\theta}, \quad n \in \mathbb{Z},$$

form an orthonormal basis of  $L^2_{\text{per}}([0, 2\pi])$ , and they are eigenfunctions of  $L$ :

$$Le_n = -\frac{d^2}{d\theta^2} e^{in\theta} + e^{in\theta} = (n^2 + 1)e^{in\theta} = (1 + n^2)e_n.$$

Thus  $L$  is *diagonal* in the Fourier basis, with eigenvalues  $\lambda_n = 1 + n^2 > 0$ .

Let  $f$  be a  $2\pi$ -periodic function with Fourier series

$$f(\theta) = \sum_{n \in \mathbb{Z}} \hat{f}_n e^{in\theta}, \quad \hat{f}_n = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} d\theta.$$

We look for a solution  $\psi$  of  $L\psi = f$  in the same form,

$$\psi(\theta) = \sum_{n \in \mathbb{Z}} \hat{\psi}_n e^{in\theta}.$$

Applying  $L$  termwise gives

$$L\psi = \sum_{n \in \mathbb{Z}} (1 + n^2) \hat{\psi}_n e^{in\theta}.$$

Hence the equation  $L\psi = f$  is equivalent to

$$(1 + n^2) \hat{\psi}_n = \hat{f}_n, \quad n \in \mathbb{Z},$$

so

$$\hat{\psi}_n = \frac{\hat{f}_n}{1 + n^2}.$$

Therefore

$$\psi(\theta) = L^{-1}f(\theta) = \sum_{n \in \mathbb{Z}} \frac{\hat{f}_n}{1 + n^2} e^{in\theta}.$$

This shows that the inverse operator  $L^{-1}$  multiplies each Fourier mode  $e^{in\theta}$  by the factor  $\frac{1}{1 + n^2}$ . In particular, the kernel  $G$  of  $L^{-1}$  (the periodic Green's function) has Fourier coefficients

$$\hat{G}_n = \frac{1}{2\pi(1 + n^2)}, \quad n \in \mathbb{Z},$$

so

$$G(\theta) = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} \frac{e^{in\theta}}{1 + n^2}.$$

Using the identity from Exercise 1.1,

$$\sum_{n \in \mathbb{Z}} \frac{e^{in\theta}}{1 + n^2} = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

we recover the explicit formula

$$G(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

extended  $2\pi$ -periodically. Thus the Green's function is nothing more than the integral kernel of  $L^{-1}$  when viewed in the Fourier basis.

### Eigenvalues of $L$ and diagonalisation in the Fourier basis

Let

$$L := -\frac{d^2}{d\theta^2} + 1$$

act on  $2\pi$ -periodic functions. Consider the Fourier basis

$$e_n(\theta) := e^{in\theta}, \quad n \in \mathbb{Z}.$$

We first compute the action of  $L$  on  $e_n$ . Differentiating,

$$\frac{d}{d\theta} e^{in\theta} = in e^{in\theta}, \quad \frac{d^2}{d\theta^2} e^{in\theta} = (in)^2 e^{in\theta} = -n^2 e^{in\theta}.$$

Hence

$$Le_n = -\frac{d^2}{d\theta^2} e^{in\theta} + e^{in\theta} = -(-n^2 e^{in\theta}) + e^{in\theta} = (1 + n^2) e^{in\theta}.$$

Thus  $e_n$  is an eigenfunction of  $L$  with eigenvalue

$$\lambda_n = 1 + n^2.$$

Because each basis vector  $e_n$  is an eigenfunction of  $L$ , the operator acts diagonally in this basis. That is, if we represent  $L$  as an infinite matrix with respect to the basis  $\{e_n\}_{n \in \mathbb{Z}}$ , then

$$Le_n = \lambda_n e_n \quad \Longleftrightarrow \quad (L)_{mn} = \lambda_n \delta_{mn},$$

so only the diagonal entries are nonzero. The Fourier basis therefore diagonalises  $L$ , with diagonal entries  $\{1 + n^2 : n \in \mathbb{Z}\}$ .

### Why $L$ is diagonal in the Fourier basis

Think of writing a linear operator as an infinite matrix with respect to a basis. If  $\{v_k\}$  is a basis and

$$Tv_k = \sum_j a_{jk} v_j,$$

then the matrix of  $T$  is the array  $(a_{jk})$ , where *column*  $k$  records how  $T$  acts on the  $k$ -th basis vector.

If each basis vector  $v_k$  is an eigenvector,

$$Tv_k = \lambda_k v_k,$$

then

$$Tv_k = \lambda_k v_k = \lambda_k v_k + 0 \cdot v_j \quad (j \neq k).$$

Therefore the matrix entries satisfy

$$a_{kk} = \lambda_k, \quad a_{jk} = 0 \quad (j \neq k).$$

Such a matrix has nonzero entries only on the diagonal; it is a *diagonal matrix*. Thus an operator is diagonal in a given basis precisely when every basis vector is an eigenvector.

For our operator

$$L := -\frac{d^2}{d\theta^2} + 1,$$

the relevant basis is the Fourier basis  $\{e_n\}_{n \in \mathbb{Z}}$  with  $e_n(\theta) = e^{in\theta}$ . We showed earlier that

$$Le_n = (1 + n^2) e_n,$$

so  $e_n$  is an eigenfunction of  $L$  with eigenvalue  $1 + n^2$ .

Hence the “matrix” of  $L$  in the Fourier basis is

$$\text{diag}(\dots, 1 + (-2)^2, 1 + (-1)^2, 1 + 0^2, 1 + 1^2, 1 + 2^2, \dots),$$

i.e. diagonal with entries  $1 + n^2$ .

This is the precise sense in which we say:

$L$  is diagonal in the Fourier basis, with eigenvalues  $\lambda_n = 1 + n^2$ .

Given a continuous  $2\pi$ -periodic function  $f$ , the (formal) convolution

$$(G * f)(\theta) := \int_0^{2\pi} G(\theta - \alpha) f(\alpha) d\alpha$$

appears naturally from linearity and translation invariance of  $L$ .

Indeed,  $G$  is defined by  $LG = \delta_0$ , so  $G$  is the response of the system  $L$  to a unit point source at 0. Since  $L$  is translation invariant on the circle, the response to a point source at  $\alpha$  is the translate  $G(\theta - \alpha)$ , satisfying

$$L(G(\theta - \alpha)) = \delta_\alpha(\theta).$$

A general right-hand side  $f$  may be viewed as a superposition of point sources,

$$f(\theta) = \int_0^{2\pi} f(\alpha) \delta_\alpha(\theta) d\alpha.$$

By linearity of  $L$ , the corresponding solution must be the superposition of the individual responses:

$$\psi(\theta) = \int_0^{2\pi} f(\alpha) G(\theta - \alpha) d\alpha = (G * f)(\theta).$$

Thus convolution with the Green's function  $G$  is exactly the operation that sends  $f$  to the unique  $2\pi$ -periodic solution  $\psi$  of  $L\psi = f$ .

## Green functions for simple one-dimensional operators

In this section we collect a few basic examples of Green functions for second-order operators in one dimension. Throughout, a Green function  $G$  for an operator  $L$  is a distributional solution of

$$LG = \delta_0,$$

with boundary conditions specified by the domain of  $L$  (whole line, finite interval, or circle). For a given right-hand side  $f$ , the solution of  $Lu = f$  is then given by the convolution  $u = G * f$  (or its appropriate periodic / bounded-interval analogue).

## 1. The operator $L_0 u = -u''$ on $\mathbb{R}$

We look for  $G_0$  such that

$$-G_0''(x) = \delta_0(x) \quad \text{on } \mathbb{R}.$$

Away from  $x = 0$  we have  $-G_0'' = 0$ , hence  $G_0$  is affine on  $(-\infty, 0)$  and on  $(0, \infty)$ :

$$G_0(x) = \begin{cases} a_+x + b_+, & x > 0, \\ a_-x + b_-, & x < 0. \end{cases}$$

We impose

- symmetry  $G_0(-x) = G_0(x)$  (by uniqueness and symmetry of the operator), which gives  $a_+ = -a_-$  and  $b_+ = b_- =: b$ ;
- continuity at 0:  $G_0(0^+) = G_0(0^-)$ , already encoded above;
- the distributional equation  $-G_0'' = \delta_0$ , which is equivalent to the jump condition  $G_0'(0^+) - G_0'(0^-) = -1$ .

Here  $G_0'(0^+) = a_+$  and  $G_0'(0^-) = a_-$ , so

$$a_+ - a_- = -1.$$

Together with  $a_+ = -a_-$  this yields  $2a_+ = -1$ , hence  $a_+ = -\frac{1}{2}$  and  $a_- = \frac{1}{2}$ . Thus

$$G_0(x) = -\frac{1}{2}|x|.$$

**Figure.** The graph of  $G_0$  is a “V” shaped function.

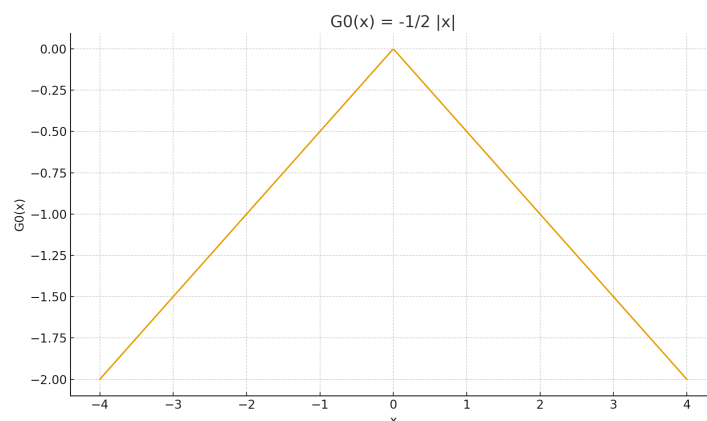


Figure 10: Green's function  $G_0(x) = -\frac{1}{2}|x|$  for the operator  $L_0 u = -u''$  on  $\mathbb{R}$ . The function is continuous and piecewise linear, and its second derivative is a Dirac mass at  $x = 0$  in the distributional sense.



## 2. The operator $L_1 u = -u'' + u$ on $\mathbb{R}$

We now solve

$$-G_1''(x) + G_1(x) = \delta_0(x) \quad \text{on } \mathbb{R}.$$

For  $x \neq 0$  this reduces to the homogeneous ODE

$$-G_1''(x) + G_1(x) = 0,$$

whose general solution is  $G_1(x) = Ae^x + Be^{-x}$ . To obtain a Green function that is bounded at  $\pm\infty$ , we choose

$$G_1(x) = \begin{cases} Ce^{-x}, & x > 0, \\ Ce^x, & x < 0, \end{cases}$$

so that  $G_1$  decays at  $\pm\infty$  and is even.

Continuity at 0 is automatic:

$$G_1(0^+) = C = G_1(0^-).$$

As before, the equation  $-G_1'' + G_1 = \delta_0$  is equivalent to the jump condition

$$G_1'(0^+) - G_1'(0^-) = -1.$$

Differentiating,

$$G_1'(x) = \begin{cases} -Ce^{-x}, & x > 0, \\ Ce^x, & x < 0, \end{cases} \Rightarrow G_1'(0^+) = -C, \quad G_1'(0^-) = C.$$

Hence

$$G_1'(0^+) - G_1'(0^-) = -C - C = -2C = -1,$$

so  $C = \frac{1}{2}$ . Therefore

$$G_1(x) = \frac{1}{2}e^{-|x|}.$$

**Figure.** The Green function is a symmetric, exponentially decaying bump.

## 3. The operator $L_\lambda u = -u'' + \lambda u$ on $\mathbb{R}$

More generally, for  $\lambda > 0$  we consider

$$-G_\lambda''(x) + \lambda G_\lambda(x) = \delta_0(x).$$

For  $x \neq 0$  the homogeneous equation is

$$-G_\lambda''(x) + \lambda G_\lambda(x) = 0 \iff G_\lambda''(x) = \lambda G_\lambda(x),$$

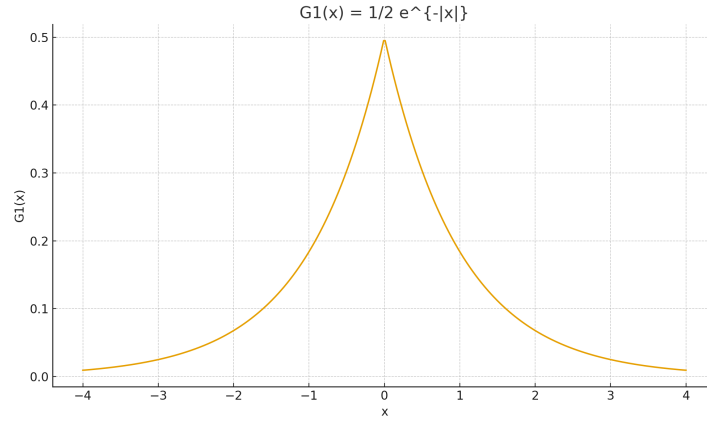


Figure 11: Green's function  $G_1(x) = \frac{1}{2}e^{-|x|}$  for  $L_1u = -u'' + u$  on  $\mathbb{R}$ . The exponential decay shows that high frequencies are strongly damped by  $L_1^{-1}$ .

with solutions  $G_\lambda(x) = Ae^{\sqrt{\lambda}x} + Be^{-\sqrt{\lambda}x}$ . As in the previous case, boundedness and evenness lead to

$$G_\lambda(x) = \begin{cases} Ce^{-\sqrt{\lambda}x}, & x > 0, \\ Ce^{\sqrt{\lambda}x}, & x < 0. \end{cases}$$

The jump condition for  $-G_\lambda'' + \lambda G_\lambda = \delta_0$  again gives

$$G_\lambda'(0^+) - G_\lambda'(0^-) = -1.$$

Differentiating,

$$G_\lambda'(x) = \begin{cases} -C\sqrt{\lambda}e^{-\sqrt{\lambda}x}, & x > 0, \\ C\sqrt{\lambda}e^{\sqrt{\lambda}x}, & x < 0, \end{cases}$$

so  $G_\lambda'(0^+) = -C\sqrt{\lambda}$  and  $G_\lambda'(0^-) = C\sqrt{\lambda}$ . Thus

$$G_\lambda'(0^+) - G_\lambda'(0^-) = -C\sqrt{\lambda} - C\sqrt{\lambda} = -2C\sqrt{\lambda} = -1 \implies C = \frac{1}{2\sqrt{\lambda}}.$$

Therefore

$$G_\lambda(x) = \frac{1}{2\sqrt{\lambda}} e^{-\sqrt{\lambda}|x|}.$$

#### 4. Periodic operator $L_{\text{per}}u = -u'' + u$ on the circle

We finally consider the operator

$$L_{\text{per}}u := -\frac{d^2u}{d\theta^2} + u$$

on  $2\pi$ -periodic functions. We seek a  $2\pi$ -periodic Green function  $G_{\text{per}}$  such that

$$-G_{\text{per}}''(\theta) + G_{\text{per}}(\theta) = \delta_0(\theta)$$

on the circle. One way to construct  $G_{\text{per}}$  is to *periodise* the line kernel  $G_1$ :

$$G_{\text{per}}(\theta) = \sum_{k \in \mathbb{Z}} G_1(\theta + 2\pi k) = \frac{1}{2} \sum_{k \in \mathbb{Z}} e^{-|\theta + 2\pi k|}.$$

This series converges absolutely and defines a  $2\pi$ -periodic function which satisfies  $(-G_{\text{per}}'' + G_{\text{per}}) = \delta_0$  in the periodic sense.

Using the Fourier-series computation of Exercise 1.1, one obtains the closed form

$$G_{\text{per}}(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}, \quad \theta \in [-\pi, \pi],$$

extended  $2\pi$ -periodically.

**Figure.** The periodic Green function is a smooth bump on the circle, symmetric around 0.

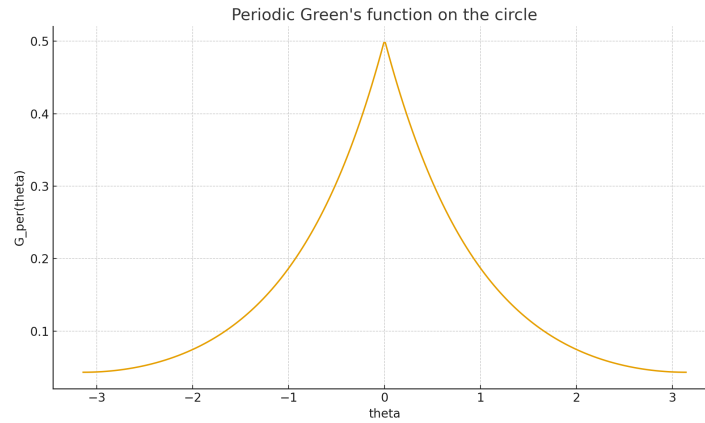


Figure 12: Periodic Green's function  $G_{\text{per}}(\theta) = \frac{\cosh(\pi - |\theta|)}{2 \sinh \pi}$  for  $L_{\text{per}}u = -u'' + u$  on the circle. It can be obtained either by periodising the line kernel  $G_1(x) = \frac{1}{2}e^{-|x|}$  or by inverting  $L_{\text{per}}$  in the Fourier basis.

## 5. Summary table

| Operator                                      | Domain / BC               | Green's function $G$ with $LG = \delta_0$                                                        |
|-----------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------|
| $L_0 = -\frac{d^2}{dx^2}$                     | $\mathbb{R}$              | $G_0(x) = -\frac{1}{2} x $                                                                       |
| $L_1 = -\frac{d^2}{dx^2} + 1$                 | $\mathbb{R}$              | $G_1(x) = \frac{1}{2}e^{- x }$                                                                   |
| $L_\lambda = -\frac{d^2}{dx^2} + \lambda$     | $\mathbb{R}, \lambda > 0$ | $G_\lambda(x) = \frac{1}{2\sqrt{\lambda}}e^{-\sqrt{\lambda} x }$                                 |
| $L_{\text{per}} = -\frac{d^2}{d\theta^2} + 1$ | Circle, $2\pi$ -periodic  | $G_{\text{per}}(\theta) = \frac{\cosh(\pi -  \theta )}{2 \sinh \pi}$ on $[-\pi, \pi]$ , periodic |

## 6. Convolution representation

Given a continuous function  $f$  on the domain of  $L$ , the solution of

$$Lu = f$$

with the appropriate boundary conditions is obtained by convolving  $f$  with the corresponding Green function. For example, on  $\mathbb{R}$ ,

$$u(x) = (G_\lambda * f)(x) := \int_{\mathbb{R}} G_\lambda(x - y)f(y) dy,$$

and on the circle

$$u(\theta) = \int_0^{2\pi} G_{\text{per}}(\theta - \alpha)f(\alpha) d\alpha.$$

In this way the Green function is the integral kernel of the inverse operator  $L^{-1}$ .

## Dirichlet and Neumann Green functions on $(0, 1)$

We consider the second-order operator  $L_0 u = -u''$  on the interval  $(0, 1)$  with different boundary conditions.

### 1. Dirichlet Green function

We first impose Dirichlet boundary conditions

$$u(0) = u(1) = 0.$$

A Dirichlet Green function  $G_D(x, \xi)$  is a function such that

$$-\partial_x^2 G_D(x, \xi) = \delta(x - \xi) \quad \text{for } 0 < x < 1,$$

with

$$G_D(0, \xi) = G_D(1, \xi) = 0.$$

For  $x \neq \xi$  we have  $-G_D'' = 0$ , hence  $G_D$  is affine on each side:

$$G_D(x, \xi) = \begin{cases} a_-x + b_-, & 0 \leq x < \xi, \\ a_+x + b_+, & \xi < x \leq 1. \end{cases}$$

From  $G_D(0, \xi) = 0$  we get  $b_- = 0$ , and from  $G_D(1, \xi) = 0$  we obtain  $a_+ + b_+ = 0$ , i.e.  $b_+ = -a_+$ . Continuity at  $x = \xi$  gives

$$a_- \xi = a_+ \xi + b_+ = a_+ (\xi - 1),$$

and the jump condition

$$G_{D,x}(\xi^+, \xi) - G_{D,x}(\xi^-, \xi) = -1$$

(from  $-G_D'' = \delta$ ) reads  $a_+ - a_- = -1$ . Solving these equations yields

$$a_- = 1 - \xi, \quad a_+ = -\xi,$$

so that

$$G_D(x, \xi) = \begin{cases} x(1 - \xi), & 0 \leq x \leq \xi, \\ \xi(1 - x), & \xi \leq x \leq 1. \end{cases}$$

This kernel is a “tent” function with peak at  $x = \xi$  and zeros at both endpoints.

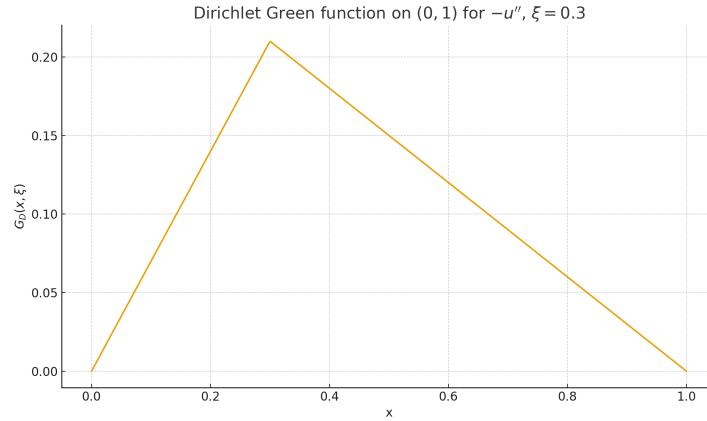


Figure 13: Dirichlet Green function  $G_D(x, \xi)$  on  $(0, 1)$  for  $L_0 u = -u''$ , plotted for  $\xi = 0.3$ . The function vanishes at  $x = 0$  and  $x = 1$  and is piecewise linear with a kink at  $x = \xi$ .

Given  $f \in C([0, 1])$ , the unique solution of  $-u'' = f$  with  $u(0) = u(1) = 0$  is

$$u(x) = \int_0^1 G_D(x, \xi) f(\xi) d\xi.$$

## 2. Neumann Green function

We now impose Neumann boundary conditions

$$u'(0) = u'(1) = 0.$$

In this case  $L_0 u = -u''$  has constants in its kernel, so the equation  $-u'' = f$  is solvable only if

$$\int_0^1 f(x) dx = 0,$$

and the solution is unique only up to an additive constant. To define a Green function we seek  $G_N(x, \xi)$  satisfying

$$-\partial_x^2 G_N(x, \xi) = \delta(x - \xi) - 1$$

on  $(0, 1)$ , with Neumann boundary conditions

$$G_{N,x}(0, \xi) = G_{N,x}(1, \xi) = 0,$$

and the normalisation

$$\int_0^1 G_N(x, \xi) dx = 0 \quad \text{for each fixed } \xi.$$

For  $x \neq \xi$  the equation reduces to  $G_N''(x, \xi) = 1$ , so  $G_N$  is quadratic on each side:

$$G_N(x, \xi) = \begin{cases} \frac{1}{2}x^2 + a_-x + b_-, & 0 \leq x < \xi, \\ \frac{1}{2}x^2 + a_+x + b_+, & \xi < x \leq 1. \end{cases}$$

The Neumann conditions give  $G_{N,x}(0, \xi) = 0$  and  $G_{N,x}(1, \xi) = 0$ , that is

$$G_{N,x}(x, \xi) = \begin{cases} x + a_-, & x < \xi, \\ x + a_+, & x > \xi, \end{cases} \Rightarrow a_- = 0, \quad a_+ = -1.$$

Continuity at  $x = \xi$  implies  $b_- = b_+ - \xi$ , and the jump condition coming from  $-G_N'' = \delta - 1$ ,

$$G_{N,x}(\xi^+, \xi) - G_{N,x}(\xi^-, \xi) = -1,$$

is automatically satisfied with  $a_- = 0$ ,  $a_+ = -1$ . The normalisation  $\int_0^1 G_N(x, \xi) dx = 0$  then determines  $b_+$  and  $b_-$ ; a short computation yields

$$b_+ = \frac{\xi^2}{2} + \frac{1}{3}, \quad b_- = \frac{\xi^2}{2} - \xi + \frac{1}{3}.$$

Thus

$$G_N(x, \xi) = \begin{cases} \frac{1}{2}x^2 + \frac{1}{2}\xi^2 - \xi + \frac{1}{3}, & 0 \leq x \leq \xi, \\ \frac{1}{2}x^2 - x + \frac{1}{2}\xi^2 + \frac{1}{3}, & \xi \leq x \leq 1. \end{cases}$$

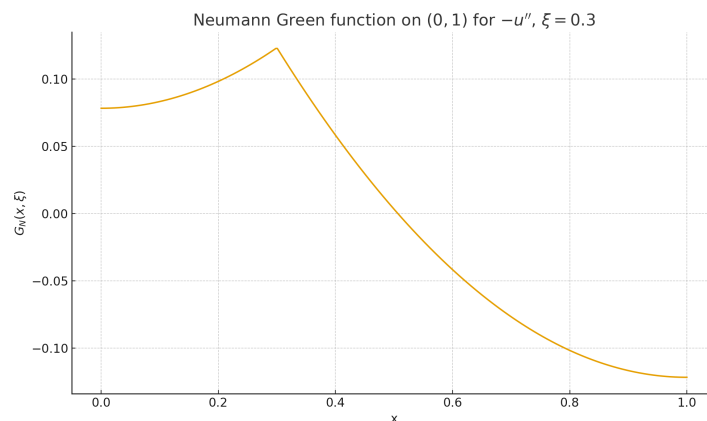


Figure 14: Neumann Green function  $G_N(x, \xi)$  on  $(0, 1)$  for  $L_0 u = -u''$ , plotted for  $\xi = 0.3$ . The derivative vanishes at  $x = 0$  and  $x = 1$ , and  $G_N$  is piecewise quadratic with a slope jump of  $-1$  at  $x = \xi$ . The normalisation  $\int_0^1 G_N(x, \xi) dx = 0$  fixes the additive constant.

If  $f$  satisfies  $\int_0^1 f(x) dx = 0$ , then

$$u(x) = \int_0^1 G_N(x, \xi) f(\xi) d\xi$$

solves  $-u'' = f$  with Neumann boundary conditions  $u'(0) = u'(1) = 0$ , and is determined up to an additive constant.

## Helmholtz and damped one-dimensional operators

### 1. Helmholtz operator on $\mathbb{R}$

Consider the Helmholtz operator

$$L_H u := -u'' - k^2 u, \quad k > 0,$$

on the real line. A Green function  $G_H$  satisfies

$$-G_H''(x) - k^2 G_H(x) = \delta_0(x) \quad \text{on } \mathbb{R}.$$

For  $x \neq 0$  the homogeneous equation  $G_H'' + k^2 G_H = 0$  has solutions  $A \cos(kx) + B \sin(kx)$ . We look for an even Green function and try

$$G_H(x) = -\frac{1}{2k} \sin(k|x|).$$

For  $x > 0$  we have  $G_H(x) = -\frac{1}{2k} \sin(kx)$  and for  $x < 0$ ,  $G_H(x) = \frac{1}{2k} \sin(kx)$ . In both cases  $G_H'' + k^2 G_H = 0$ , so the equation holds away from  $x = 0$ .

The derivative is

$$G_H'(x) = \begin{cases} -\frac{1}{2} \cos(kx), & x > 0, \\ \frac{1}{2} \cos(kx), & x < 0, \end{cases}$$

hence

$$G_H'(0^+) = -\frac{1}{2}, \quad G_H'(0^-) = \frac{1}{2},$$

and

$$G_H'(0^+) - G_H'(0^-) = -1.$$

This jump condition is equivalent to  $-G_H'' = \delta_0$  in the distribution sense, while the  $-k^2 G_H$  term integrates to zero across  $x = 0$ . Thus

$$G_H(x) = -\frac{1}{2k} \sin(k|x|)$$

is a Green function for  $L_H$  on  $\mathbb{R}$ . It is oscillatory and does not decay at infinity.

### 2. Damped operator on $\mathbb{R}$

Now consider the non-self-adjoint operator

$$L_D u := -u'' + cu', \quad c > 0.$$

We look for a one-sided (retarded) Green function  $G_D$  solving

$$-G_D''(x) + cG_D'(x) = \delta_0(x)$$

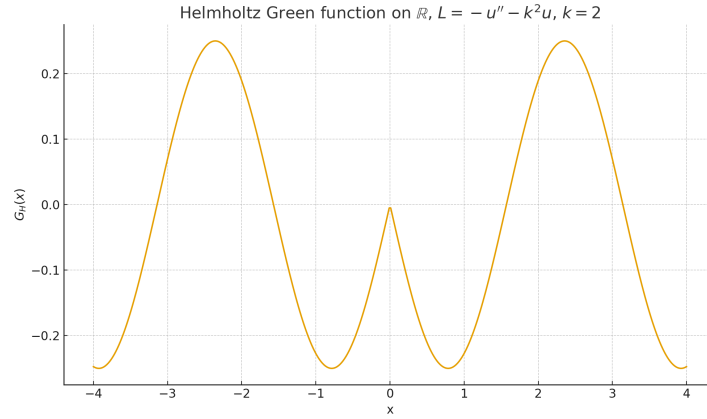


Figure 15: Helmholtz Green function  $G_H(x) = -\frac{1}{2k} \sin(k|x|)$  for  $L_H u = -u'' - k^2 u$ , plotted for  $k = 2$ . The kernel is oscillatory and bounded.

and vanishing for  $x > 0$ .

For  $x \neq 0$  we solve  $-G_D'' + cG_D' = 0$ , i.e.  $G_D'' - cG_D' = 0$ . Writing  $v = G_D'$  gives  $v' - cv = 0$  so  $v(x) = Ae^{cx}$  and

$$G_D(x) = \frac{A}{c} e^{cx} + B.$$

We prescribe

$$G_D(x) = \begin{cases} \alpha + \beta e^{cx}, & x < 0, \\ 0, & x > 0, \end{cases}$$

and impose continuity at 0:

$$\alpha + \beta = 0 \quad \Rightarrow \quad \alpha = -\beta.$$

The jump condition from  $-G_D'' + cG_D' = \delta_0$  reads

$$G_D'(0^+) - G_D'(0^-) = -1.$$

Here  $G_D'(x) = 0$  for  $x > 0$  and  $G_D'(x) = c\beta e^{cx}$  for  $x < 0$ , so  $G_D'(0^-) = c\beta$  and hence  $-c\beta = -1$ , i.e.  $\beta = 1/c$  and  $\alpha = -1/c$ . Thus

$$G_D(x) = \begin{cases} -\frac{1}{c} + \frac{1}{c} e^{cx}, & x < 0, \\ 0, & x > 0. \end{cases}$$

This kernel is continuous at 0, has a slope jump of  $-1$ , and vanishes for  $x > 0$ , so it is a retarded Green function for  $L_D$ .

## Green function for $-\Delta$ in $\mathbb{R}^2$

In two dimensions the Laplacian is

$$\Delta u = u_{xx} + u_{yy}.$$



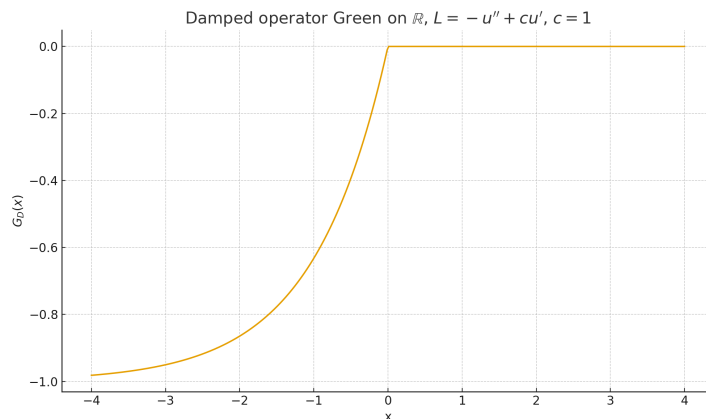


Figure 16: One-sided Green function  $G_D$  for the damped operator  $L_D u = -u'' + cu'$  on  $\mathbb{R}$ , plotted for  $c = 1$ . The kernel vanishes for  $x > 0$  and decays exponentially as  $x \rightarrow -\infty$ .

We seek a Green function  $G : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}$  such that

$$-\Delta G(x) = \delta_0(x)$$

in the sense of distributions. By rotational symmetry we look for a radial function  $G(x) = g(r)$  with  $r = |x|$ . For a radial function  $u(r)$  in  $\mathbb{R}^2$  one has

$$\Delta u = u''(r) + \frac{1}{r}u'(r).$$

Thus away from the origin  $-\Delta G = 0$  becomes

$$g''(r) + \frac{1}{r}g'(r) = 0.$$

Writing  $v(r) = g'(r)$ , this is  $v'(r) + \frac{1}{r}v(r) = 0$ , or equivalently  $(rv(r))' = 0$ , so  $v(r) = C/r$  and hence

$$g(r) = C \log r + D.$$

To fix the constant  $C$  we integrate  $-\Delta G$  over a disc  $B_R$  and apply the divergence theorem:

$$\int_{B_R} -\Delta G \, dx = \int_{\partial B_R} -\frac{\partial G}{\partial n} \, dS.$$

If  $G(x) = A \log |x|$ , then  $\partial G / \partial r = A/r$  and on  $\partial B_R$  we obtain

$$-\frac{\partial G}{\partial n} = -\frac{A}{R}.$$

Since  $dS = R \, d\theta$  on the circle of radius  $R$ ,

$$\int_{\partial B_R} -\frac{\partial G}{\partial n} \, dS = \int_0^{2\pi} \left(-\frac{A}{R}\right) R \, d\theta = -2\pi A.$$

We require this flux to equal 1 (because  $-\Delta G = \delta_0$ ), hence  $-2\pi A = 1$  and  $A = -1/(2\pi)$ . Therefore

$$G(x) = -\frac{1}{2\pi} \log |x|$$

is a Green function for the Laplacian in  $\mathbb{R}^2$  (unique up to an additive constant).

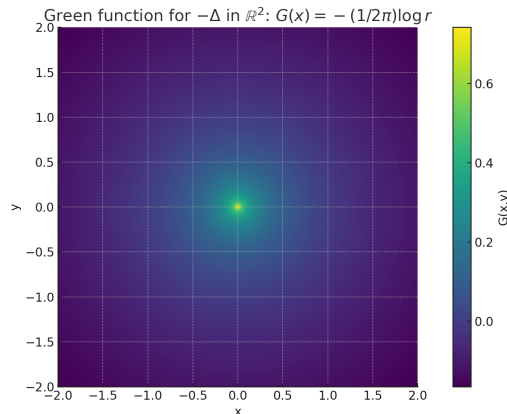


Figure 17: Green function  $G(x) = -\frac{1}{2\pi} \log r$  for  $-\Delta$  in  $\mathbb{R}^2$  (plotted on  $[-2, 2]^2$  with a small regularisation near  $r = 0$ ). The level sets are concentric circles and  $G$  has a logarithmic singularity at the origin.

## Green functions on the torus and sphere

### 1. Flat torus $\mathbb{T}^2$

Identify the flat torus  $\mathbb{T}^2$  with  $\mathbb{R}^2/(2\pi\mathbb{Z})^2$ . The eigenfunctions of the Laplacian are

$$e_{m,n}(x, y) = e^{i(mx+ny)}, \quad (m, n) \in \mathbb{Z}^2,$$

with eigenvalues

$$-\Delta e_{m,n} = (m^2 + n^2)e_{m,n}.$$

The constant mode  $(m, n) = (0, 0)$  has eigenvalue 0, so  $-\Delta$  is not invertible on all of  $L^2(\mathbb{T}^2)$ . To define a Green function  $G_{\mathbb{T}}$  one imposes

$$-\Delta G_{\mathbb{T}}(x, y) = \delta_0(x, y) - \frac{1}{(2\pi)^2},$$

together with the normalisation  $\int_{\mathbb{T}^2} G_{\mathbb{T}} = 0$ . Expanding in the Fourier basis and inverting mode by mode yields

$$G_{\mathbb{T}}(x, y) = \frac{1}{(2\pi)^2} \sum_{(m,n) \in \mathbb{Z}^2 \setminus \{(0,0)\}} \frac{e^{i(mx+ny)}}{m^2 + n^2}.$$

This series represents the periodisation of the planar Green function  $-(1/2\pi) \log |x|$  and has a logarithmic singularity at the origin.

### 2. Round sphere $\mathbb{S}^2$

On the unit sphere  $\mathbb{S}^2 \subset \mathbb{R}^3$  the eigenfunctions of the Laplace–Beltrami operator are the spherical harmonics  $Y_\ell^m$ ,  $\ell = 0, 1, 2, \dots$  and  $|m| \leq \ell$ , with eigenvalues

$$-\Delta_{\mathbb{S}^2} Y_\ell^m = \ell(\ell + 1) Y_\ell^m.$$

Again the constant mode ( $\ell = 0$ ) lies in the kernel. A Green function  $G_{\mathbb{S}^2}(p, q)$  is defined by

$$-\Delta_{\mathbb{S}^2} G_{\mathbb{S}^2}(p, q) = \delta_q(p) - \frac{1}{4\pi},$$

together with  $\int_{\mathbb{S}^2} G_{\mathbb{S}^2}(p, q) dp = 0$  for each fixed  $q$ . Using the completeness of the spherical harmonics one obtains the eigenfunction expansion

$$G_{\mathbb{S}^2}(p, q) = \sum_{\ell=1}^{\infty} \sum_{m=-\ell}^{\ell} \frac{1}{\ell(\ell+1)} Y_{\ell}^m(p) \overline{Y_{\ell}^m(q)}.$$

By rotational invariance,  $G_{\mathbb{S}^2}(p, q)$  depends only on the geodesic distance  $\gamma$  between  $p$  and  $q$  (equivalently on  $\cos \gamma = p \cdot q$ ), and one can write it as an explicit function of  $\gamma$  involving a logarithmic singularity as  $\gamma \rightarrow 0$ . Locally, the behaviour of  $G_{\mathbb{S}^2}$  near the diagonal  $p = q$  is therefore similar to the planar Green function  $-(1/2\pi) \log |x|$ .

### Exercise 1.3 — Maximum principle for $-\psi'' + \psi$ on the circle

Let  $\phi \in C^2(\mathbb{R})$  be  $2\pi$ -periodic, i.e.  $\phi(\theta + 2\pi) = \phi(\theta)$  for all  $\theta \in \mathbb{R}$ , and suppose that

$$-\phi''(\theta) + \phi(\theta) = 0, \quad \theta \in \mathbb{R}.$$

- (a) Show that if  $\phi$  attains a maximum at some  $\theta_0 \in \mathbb{R}$ , then  $\phi(\theta_0) \leq 0$ .
- (b) Show that if  $\phi$  attains a minimum at some  $\theta_0 \in \mathbb{R}$ , then  $\phi(\theta_0) \geq 0$ .
- (c) Deduce that  $\phi \equiv 0$ .
- (d) Let  $f \in C^0(\mathbb{R})$  be a given  $2\pi$ -periodic function. Conclude that there is at most one  $\psi \in C^2(\mathbb{R})$  such that  $\psi(\theta + 2\pi) = \psi(\theta)$  for all  $\theta$  and

$$-\psi''(\theta) + \psi(\theta) = f(\theta), \quad \theta \in \mathbb{R}.$$

Let  $\phi \in C^2(\mathbb{R})$  be  $2\pi$ -periodic, i.e.  $\phi(\theta + 2\pi) = \phi(\theta)$  for all  $\theta \in \mathbb{R}$ , and suppose that

$$-\phi''(\theta) + \phi(\theta) = 0, \quad \theta \in \mathbb{R}.$$

Equivalently,  $\phi''(\theta) = \phi(\theta)$ .

#### (a) Behaviour at a maximum

Suppose that  $\phi$  attains a maximum at some  $\theta_0 \in \mathbb{R}$ . By elementary calculus,

$$\phi'(\theta_0) = 0, \quad \phi''(\theta_0) \leq 0.$$

Evaluating the differential equation at  $\theta_0$  gives

$$0 = -\phi''(\theta_0) + \phi(\theta_0) \implies \phi(\theta_0) = \phi''(\theta_0).$$

Since  $\phi''(\theta_0) \leq 0$ , it follows that

$$\phi(\theta_0) \leq 0.$$

Thus any maximum value of  $\phi$  must be non-positive.

## (b) Behaviour at a minimum

Suppose that  $\phi$  attains a minimum at some  $\theta_0 \in \mathbb{R}$ . Then

$$\phi'(\theta_0) = 0, \quad \phi''(\theta_0) \geq 0.$$

Once more, evaluating the equation at  $\theta_0$  yields

$$0 = -\phi''(\theta_0) + \phi(\theta_0) \implies \phi(\theta_0) = \phi''(\theta_0) \geq 0.$$

Thus any minimum value of  $\phi$  must be non-negative.

## (c) Conclusion: $\phi \equiv 0$

Because  $\phi$  is continuous and  $2\pi$ -periodic, it attains both a maximum and a minimum on  $\mathbb{R}$ . Let

$$M := \max_{\theta \in \mathbb{R}} \phi(\theta), \quad m := \min_{\theta \in \mathbb{R}} \phi(\theta).$$

From part (a),  $M \leq 0$ , and from part (b),  $m \geq 0$ . Since always  $m \leq M$ , we obtain

$$m \leq M \leq 0 \quad \text{and} \quad m \geq 0.$$

Hence  $m = 0$  and  $M = 0$ . A continuous function whose maximum and minimum are both equal to 0 must be identically zero, so

$$\phi(\theta) \equiv 0 \quad \text{for all } \theta \in \mathbb{R}.$$

## (d) Uniqueness for $-\psi'' + \psi = f$

Let  $f \in C^0(\mathbb{R})$  be a given  $2\pi$ -periodic function and suppose that  $\psi_1, \psi_2 \in C^2(\mathbb{R})$  are  $2\pi$ -periodic solutions of

$$-\psi''(\theta) + \psi(\theta) = f(\theta), \quad \theta \in \mathbb{R}.$$

Consider their difference

$$\phi(\theta) := \psi_1(\theta) - \psi_2(\theta).$$

Then  $\phi$  is  $2\pi$ -periodic and satisfies

$$-\phi''(\theta) + \phi(\theta) = (-\psi_1''(\theta) + \psi_1(\theta)) - (-\psi_2''(\theta) + \psi_2(\theta)) = f(\theta) - f(\theta) = 0.$$

By part (c) we must have  $\phi \equiv 0$ , hence  $\psi_1 = \psi_2$ . Therefore there is at most one  $\psi \in C^2(\mathbb{R})$  which is  $2\pi$ -periodic and solves  $-\psi'' + \psi = f$ .

**Remark (Why “on the circle”?).** A  $2\pi$ -periodic function  $\phi \in C^2(\mathbb{R})$  naturally defines a function on the unit circle  $S^1$  via

$$\Phi(e^{i\theta}) := \phi(\theta).$$

This is well defined because  $\phi(\theta + 2\pi) = \phi(\theta)$ , so the value depends only on the class of  $\theta$  in  $\mathbb{R}/(2\pi\mathbb{Z}) \simeq S^1$ . Conversely, any function on  $S^1$  can be regarded as a  $2\pi$ -periodic function of  $\theta$ . In this identification the operator

$$L = -\frac{d^2}{d\theta^2} + 1$$

is precisely  $-\Delta_{S^1} + 1$ , the Laplacian plus a mass term on the circle. Thus Exercise 1.3 can be viewed as a maximum principle for the operator  $-\Delta_{S^1} + 1$  on the compact manifold  $S^1$ .

**Remark: periodic solutions of  $-\phi'' = 0$  vs.  $-\phi'' + \phi = 0$ .** It is important to distinguish between the operators  $-\phi''$  and  $-\phi'' + \phi$  when working on the circle  $S^1 \cong \mathbb{R}/(2\pi\mathbb{Z})$ .

**1. The equation  $-\phi'' = 0$ .** Solving  $-\phi''(\theta) = 0$  gives

$$\phi(\theta) = a\theta + b.$$

If  $\phi$  is  $2\pi$ -periodic, i.e.  $\phi(\theta + 2\pi) = \phi(\theta)$ , then

$$a(\theta + 2\pi) + b = a\theta + b \implies a = 0.$$

Hence the only  $2\pi$ -periodic solutions are *constants*. Therefore the kernel of  $-\phi''$  on  $S^1$  consists precisely of the constant functions:

$$\ker(-\phi'') = \{\text{constants}\}.$$

**2. The equation  $-\phi'' + \phi = 0$ .** Solving  $\phi''(\theta) = \phi(\theta)$  gives

$$\phi(\theta) = Ae^\theta + Be^{-\theta}.$$

There is no nonzero  $2\pi$ -periodic solution of this form. In Exercise 1.3(a)–(c) we proved a maximum principle showing that any  $2\pi$ -periodic  $C^2$  solution of

$$-\phi''(\theta) + \phi(\theta) = 0$$

must satisfy  $\phi \equiv 0$ . Thus the kernel of  $-\phi'' + \phi$  on  $S^1$  is *trivial*:

$$\ker(-\phi'' + \phi) = \{0\}.$$

**3. Consequence for uniqueness in (d).** Suppose  $\psi_1, \psi_2$  are  $2\pi$ -periodic  $C^2$  solutions of

$$-\psi''(\theta) + \psi(\theta) = f(\theta).$$

Then their difference  $\phi = \psi_1 - \psi_2$  satisfies

$$-\phi''(\theta) + \phi(\theta) = 0$$

and is  $2\pi$ -periodic. By the result above,  $\phi \equiv 0$ , hence  $\psi_1 = \psi_2$ . Therefore the inhomogeneous equation  $-\psi'' + \psi = f$  admits *at most one*  $2\pi$ -periodic solution.

**Summary.**

| Operator         | General solution           | $2\pi$ -periodic solution |
|------------------|----------------------------|---------------------------|
| $-\phi''$        | $a\theta + b$              | constants ( $a = 0$ )     |
| $-\phi'' + \phi$ | $Ae^\theta + Be^{-\theta}$ | only zero                 |

This distinction explains why part (d) has uniqueness but not constant solutions.

## Exercise 1.4

## Exercise 1.4

For  $t \in \mathbb{R}$  let

$$\chi(t) := \begin{cases} 0, & t \leq 0, \\ e^{-1/t}, & t > 0. \end{cases}$$

- (a) Show that  $\chi \in C^\infty(\mathbb{R})$ .
- (b) Show that there exists a function  $\psi \in C_0^\infty(\mathbb{R}^n)$  such that:
  - (i)  $0 \leq \psi(x) \leq 1$  for all  $x \in \mathbb{R}^n$ ;
  - (ii)  $\text{supp } \psi \subset B_2(0)$ ;
  - (iii)  $\psi(x) = 1$  for all  $|x| \leq 1$ .

*Hint:* First construct a smooth function  $\tilde{\chi} \in C^\infty(\mathbb{R})$  such that

$$\tilde{\chi}(x) = 0 \quad \text{for } x < -1, \quad \tilde{\chi}(x) = 1 \quad \text{for } x > 1.$$

For  $t \in \mathbb{R}$  let

$$\chi(t) := \begin{cases} 0, & t \leq 0, \\ e^{-1/t}, & t > 0. \end{cases}$$

**(a) Smoothness of  $\chi$** 

For  $t < 0$  the function  $\chi$  is identically zero, hence smooth. For  $t > 0$  we have  $\chi(t) = e^{-1/t}$ , which is smooth as a composition of smooth functions. The only nontrivial point is  $t = 0$ . A straightforward induction shows that for each  $k \in \mathbb{N}$  there is a polynomial  $P_k$  such that, for  $t > 0$ ,

$$\chi^{(k)}(t) = P_k(1/t) e^{-1/t}.$$

Moreover, for every  $m \geq 0$  one has

$$\lim_{t \rightarrow 0^+} t^m e^{-1/t} = 0,$$

so  $P_k(1/t)e^{-1/t} \rightarrow 0$  as  $t \rightarrow 0^+$ . On the other hand, for  $t < 0$  all derivatives of  $\chi$  vanish identically. Therefore for each  $k$  we can define  $\chi^{(k)}(0) := 0$  and obtain a continuous extension of  $\chi^{(k)}$  to all of  $\mathbb{R}$ . It follows that  $\chi \in C^\infty(\mathbb{R})$ .

**(b) Construction of a bump function  $\psi$  on  $\mathbb{R}^n$** 

**Step 1: a one-dimensional smooth step.** Define

$$\eta(x) := \frac{\chi(x)}{\chi(x) + \chi(1-x)}, \quad x \in \mathbb{R}.$$

The denominator is strictly positive for all  $x$  (at least one of  $\chi(x)$  or  $\chi(1-x)$  is nonzero), and both numerator and denominator are smooth by part (a), so  $\eta \in C^\infty(\mathbb{R})$ . By inspection,

$$\eta(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x \geq 1. \end{cases}$$

Now set

$$\tilde{\chi}(x) := \eta\left(\frac{x+1}{2}\right).$$

Then  $\tilde{\chi} \in C^\infty(\mathbb{R})$  and

$$\tilde{\chi}(x) = 0 \quad \text{for } x \leq -1, \quad \tilde{\chi}(x) = 1 \quad \text{for } x \geq 1,$$

as required in the hint.

**Step 2: a radial cutoff on  $\mathbb{R}^n$ .** Let  $r = |x|$  and define

$$h(r) := \tilde{\chi}(2-r), \quad r \geq 0.$$

Then  $h$  is smooth on  $[0, \infty)$  and satisfies

$$h(r) = \begin{cases} 1, & 0 \leq r \leq 1, \\ 0, & r \geq 2. \end{cases}$$

We now define a radial function  $\psi$  on  $\mathbb{R}^n$  by

$$\psi(x) := h(|x|) = \tilde{\chi}(2 - |x|).$$

Since  $h$  is smooth and depends only on  $|x|$ , standard arguments show that  $\psi \in C^\infty(\mathbb{R}^n)$ . Moreover,  $0 \leq \tilde{\chi} \leq 1$  implies  $0 \leq \psi \leq 1$ . If  $|x| \leq 1$  then  $2 - |x| \geq 1$  so  $\tilde{\chi}(2 - |x|) = 1$  and hence  $\psi(x) = 1$ . If  $|x| \geq 2$  then  $2 - |x| \leq 0$  so  $\tilde{\chi}(2 - |x|) = 0$  and hence  $\psi(x) = 0$ . Thus  $\text{supp } \psi \subset B_2(0)$  and  $\psi(x) = 1$  for  $|x| \leq 1$ .

We conclude that  $\psi \in C_0^\infty(\mathbb{R}^n)$  and

$$0 \leq \psi \leq 1, \quad \text{supp } \psi \subset B_2(0), \quad \psi(x) = 1 \text{ for } |x| \leq 1,$$

as required.

**Theory box: mollifiers and partitions of unity.** A *mollifier* in  $\mathbb{R}^n$  is a function  $\rho \in C_0^\infty(\mathbb{R}^n)$  such that

$$\rho(x) \geq 0 \quad \text{and} \quad \int_{\mathbb{R}^n} \rho(x) dx = 1.$$

For  $\varepsilon > 0$  we define the rescaled mollifier

$$\rho_\varepsilon(x) := \frac{1}{\varepsilon^n} \rho\left(\frac{x}{\varepsilon}\right).$$

If  $f \in L_{\text{loc}}^1(\mathbb{R}^n)$ , the mollified function  $f_\varepsilon := \rho_\varepsilon * f$  is smooth, and as  $\varepsilon \rightarrow 0$  one has  $f_\varepsilon \rightarrow f$  in various senses (e.g. in  $L_{\text{loc}}^p$ , or uniformly when  $f$  is continuous). Normalised bump functions of the type constructed in Exercise 1.4 are the basic building blocks for such mollifiers.

A *partition of unity* on an open set  $U \subset \mathbb{R}^n$  is a family  $\{\varphi_j\}_{j \in J} \subset C^\infty(U)$  such that

1.  $0 \leq \varphi_j \leq 1$  for all  $j$ ;
2. the family is locally finite (for each  $x \in U$  there is a neighbourhood where only finitely many  $\varphi_j$  are nonzero);
3.  $\sum_{j \in J} \varphi_j(x) = 1$  for all  $x \in U$ .

Given an open cover  $U = \bigcup_j U_j$ , one can construct a partition of unity  $\{\varphi_j\}$  subordinate to this cover (i.e.  $\text{supp } \varphi_j \subset U_j$  for each  $j$ ). The bump function  $\psi$  from Exercise 1.4, with  $\psi = 1$  on  $B_1(0)$  and  $\text{supp } \psi \subset B_2(0)$ , is a model for such building blocks: by translating and rescaling  $\psi$  one obtains the pieces of a partition of unity. This construction is fundamental in analysis on manifolds, where partitions of unity allow one to glue local objects into global ones.

We use the standard Euclidean norm

$$|x| = \sqrt{x_1^2 + \cdots + x_n^2}, \quad x \in \mathbb{R}^n,$$



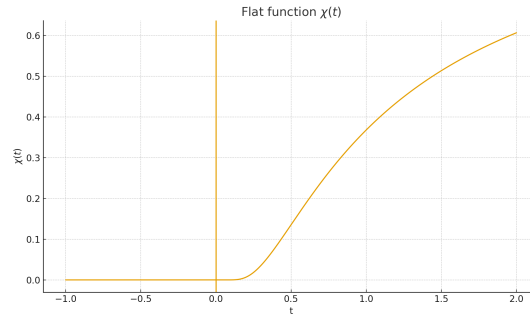


Figure 18: The flat function  $\chi(t)$ : 0 for  $t \leq 0$ , rapidly increasing for  $t > 0$ , with all derivatives vanishing at  $t = 0$ .

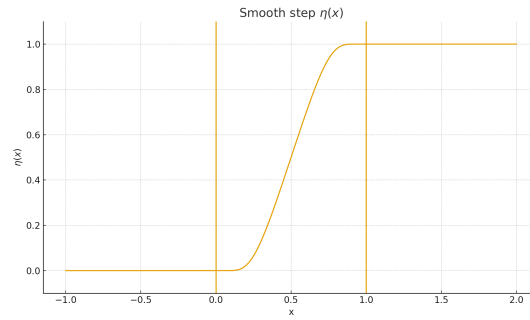


Figure 19: The smooth step  $\eta(x) = \chi(x)/(\chi(x) + \chi(1-x))$ :  $\eta(x) = 0$  for  $x \leq 0$  and  $\eta(x) = 1$  for  $x \geq 1$ , with a smooth transition in between.

and denote by

$$B_r(0) := \{x \in \mathbb{R}^n : |x| < r\}$$

the open ball of radius  $r$  centred at the origin. Thus  $B_1(0)$  is the unit ball and  $B_2(0)$  is the ball of radius 2. The conditions

$$\psi(x) = 1 \text{ for } |x| \leq 1, \quad \text{supp } \psi \subset B_2(0),$$

mean that  $\psi$  is identically 1 on the closed unit ball  $\overline{B_1(0)}$  and vanishes outside the ball  $B_2(0)$ .

## Exercise 1.5

- (a) Suppose  $\phi \in \mathcal{E}(\mathbb{R}^n)$ . Let  $\{x_l\}_{l=1}^\infty \subset \mathbb{R}^n$  be a sequence with  $x_l \rightarrow 0$ . Show that

$$\tau_{x_l} \phi \rightarrow \phi, \quad \text{as } l \rightarrow \infty,$$

in  $\mathcal{E}(\mathbb{R}^n)$ , where  $\tau_x$  is the translation operator defined in equation (1.2).

- (b) Suppose  $\phi \in \mathcal{E}(\mathbb{R}^n)$ . Show that

$$\Delta_i^h \phi \rightarrow D_i \phi, \quad \text{as } h \rightarrow 0,$$

in  $\mathcal{E}(\mathbb{R}^n)$ , where  $\Delta_i^h$  is the difference quotient defined in Example 2.

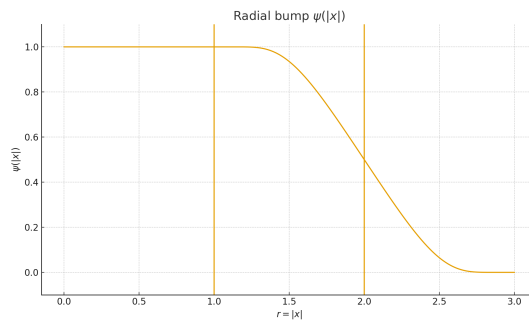


Figure 20: The radial bump  $\psi(|x|)$  obtained from  $\tilde{\chi}$ :  $\psi(r) = 1$  for  $0 \leq r \leq 1$  and  $\psi(r) = 0$  for  $r \geq 2$ , giving a smooth cutoff supported in  $B_2(0)$ .

*Translation operator.* For  $x \in \mathbb{R}^n$  and  $\phi : \mathbb{R}^n \rightarrow \mathbb{C}$  we define the translation operator  $\tau_x$  by

$$(\tau_x \phi)(y) := \phi(y - x), \quad y \in \mathbb{R}^n.$$

*Difference quotient in the  $i$ -th direction.* Let  $e_i$  denote the  $i$ -th coordinate vector in  $\mathbb{R}^n$ . For  $h \neq 0$  and  $\phi : \mathbb{R}^n \rightarrow \mathbb{C}$  we define

$$(\Delta_i^h \phi)(x) := \frac{\phi(x + he_i) - \phi(x)}{h}, \quad x \in \mathbb{R}^n.$$

Formally,  $\Delta_i^h \phi$  approximates the partial derivative  $D_i \phi = \partial_{x_i} \phi$  as  $h \rightarrow 0$ .

## Solution to Exercise 1.5

We use the standard Fréchet topology on  $\mathcal{E}(\mathbb{R}^n)$  given by the seminorms

$$p_{K,m}(\phi) := \sup_{\substack{x \in K \\ |\alpha| \leq m}} |D^\alpha \phi(x)|,$$

where  $K \subset \mathbb{R}^n$  is compact and  $m \in \mathbb{N}$ .

### (a) Continuity of translations

Let  $\phi \in \mathcal{E}(\mathbb{R}^n)$  and  $x_l \rightarrow 0$  in  $\mathbb{R}^n$ . Recall that  $(\tau_{x_l} \phi)(y) = \phi(y - x_l)$ . We must show that  $\tau_{x_l} \phi \rightarrow \phi$  in  $\mathcal{E}(\mathbb{R}^n)$ , i.e. that for every compact  $K$  and every  $m \in \mathbb{N}$ ,

$$p_{K,m}(\tau_{x_l} \phi - \phi) \rightarrow 0 \quad \text{as } l \rightarrow \infty.$$

Fix such  $K$  and  $m$ , and let  $|\alpha| \leq m$ . Then

$$D^\alpha(\tau_{x_l} \phi)(y) = D^\alpha \phi(y - x_l),$$

so

$$\sup_{y \in K} |D^\alpha(\tau_{x_l} \phi)(y) - D^\alpha \phi(y)| = \sup_{y \in K} |D^\alpha \phi(y - x_l) - D^\alpha \phi(y)|.$$

Choose a compact set  $K'$  such that  $K \cup (K - x_l) \subset K'$  for all  $l$  sufficiently large (for instance  $K' = K + \overline{B_1(0)}$  if  $|x_l| \leq 1$  eventually). The function  $D^\alpha \phi$  is continuous on the compact set  $K'$  and hence uniformly continuous there. Thus for any  $\varepsilon > 0$  there exists  $\delta > 0$  such that  $|z - z'| < \delta$  implies  $|D^\alpha \phi(z) - D^\alpha \phi(z')| < \varepsilon$  for all  $z, z' \in K'$ . Since  $x_l \rightarrow 0$ , for  $l$  large enough we have  $|x_l| < \delta$ , and then for all  $y \in K$

$$|D^\alpha \phi(y - x_l) - D^\alpha \phi(y)| < \varepsilon.$$

Taking the supremum over  $y \in K$  and  $|\alpha| \leq m$  shows that  $p_{K,m}(\tau_{x_l} \phi - \phi) \leq \varepsilon$  for all large  $l$ . Hence  $p_{K,m}(\tau_{x_l} \phi - \phi) \rightarrow 0$  and  $\tau_{x_l} \phi \rightarrow \phi$  in  $\mathcal{E}(\mathbb{R}^n)$ .

### (b) Difference quotients converge to the derivative

Let  $\phi \in \mathcal{E}(\mathbb{R}^n)$  and define, for  $h \neq 0$ ,

$$\Delta_i^h \phi(x) := \frac{\phi(x + he_i) - \phi(x)}{h},$$

where  $e_i$  is the  $i$ -th coordinate vector. We must show that  $\Delta_i^h \phi \rightarrow D_i \phi$  in  $\mathcal{E}(\mathbb{R}^n)$  as  $h \rightarrow 0$ .

Fix a compact  $K \subset \mathbb{R}^n$  and  $m \in \mathbb{N}$ . For a multi-index  $\alpha$  with  $|\alpha| \leq m$  we compute

$$D^\alpha(\Delta_i^h \phi)(x) = \frac{D^\alpha \phi(x + he_i) - D^\alpha \phi(x)}{h},$$

and hence

$$D^\alpha(\Delta_i^h \phi - D_i \phi)(x) = \frac{D^\alpha \phi(x + he_i) - D^\alpha \phi(x)}{h} - D_i D^\alpha \phi(x).$$

Set  $g(x) := D^\alpha \phi(x)$ , so  $g \in C^\infty$  and the expression becomes

$$\frac{g(x + he_i) - g(x)}{h} - D_i g(x).$$

Let  $K'$  be a compact set containing  $K$  and  $K + he_i$  for all  $x \in K$  and  $|h|$  small (e.g.  $K' = K + \overline{B_1(0)}$ ). On  $K'$  the function  $D_i g$  is continuous and therefore uniformly continuous. By the one-dimensional mean-value theorem, for each  $x \in K$  there exists  $\theta = \theta(x, h) \in (0, 1)$  such that

$$\frac{g(x + he_i) - g(x)}{h} = D_i g(x + \theta he_i),$$

so

$$\left| \frac{g(x + he_i) - g(x)}{h} - D_i g(x) \right| = |D_i g(x + \theta he_i) - D_i g(x)|.$$

Given  $\varepsilon > 0$ , uniform continuity of  $D_i g$  on  $K'$  yields  $\delta > 0$  such that  $|z - z'| < \delta$  implies  $|D_i g(z) - D_i g(z')| < \varepsilon$  for all  $z, z' \in K'$ . For  $|h| < \delta$  we have  $|x + \theta he_i - x| \leq |h| < \delta$ , so the right hand side is  $< \varepsilon$  for all  $x \in K$ . Thus

$$\sup_{x \in K} |D^\alpha(\Delta_i^h \phi - D_i \phi)(x)| < \varepsilon$$

for all  $|\alpha| \leq m$  and  $|h|$  sufficiently small. Hence  $p_{K,m}(\Delta_i^h \phi - D_i \phi) \rightarrow 0$  as  $h \rightarrow 0$ , and therefore  $\Delta_i^h \phi \rightarrow D_i \phi$  in  $\mathcal{E}(\mathbb{R}^n)$ .

**Example (Exercise 1.5 for  $\phi(x) = e^{-x^2}$  in  $\mathbb{R}$ ).** Let  $\phi(x) = e^{-x^2}$ . For a sequence  $x_l \rightarrow 0$  the translation  $\tau_{x_l}\phi$  is given by

$$(\tau_{x_l}\phi)(y) = \phi(y - x_l) = e^{-(y-x_l)^2}.$$

**(a) Translation example.** For each fixed  $y \in \mathbb{R}$  we have

$$e^{-(y-x_l)^2} \longrightarrow e^{-y^2} = \phi(y) \quad \text{as } l \rightarrow \infty.$$

The convergence is uniform on every compact interval  $K = [-R, R]$  since

$$(y - x_l)^2 \rightarrow y^2 \quad \text{uniformly on } [-R, R],$$

and the exponential function is uniformly continuous on bounded sets. Thus

$$\sup_{y \in [-R, R]} |(\tau_{x_l}\phi)(y) - \phi(y)| \rightarrow 0.$$

The same argument applies to all derivatives. For example,

$$\phi'(x) = -2xe^{-x^2}, \quad (\tau_{x_l}\phi)'(y) = -2(y - x_l)e^{-(y-x_l)^2},$$

and one checks that

$$(\tau_{x_l}\phi)'(y) \longrightarrow \phi'(y) \quad \text{uniformly on } [-R, R].$$

Hence  $\tau_{x_l}\phi \rightarrow \phi$  in  $\mathcal{E}(\mathbb{R})$ .

**(b) Difference quotient example.** In one dimension the difference quotient is

$$(\Delta^h\phi)(x) = \frac{\phi(x+h) - \phi(x)}{h} = \frac{e^{-(x+h)^2} - e^{-x^2}}{h}.$$

We know

$$\phi'(x) = -2xe^{-x^2}.$$

By the mean value theorem, for each  $x$  and each small  $h$ , there exists  $\theta = \theta(x, h) \in (0, 1)$  such that

$$\frac{e^{-(x+h)^2} - e^{-x^2}}{h} = \phi'(x + \theta h).$$

Thus

$$(\Delta^h\phi)(x) - \phi'(x) = \phi'(x + \theta h) - \phi'(x).$$

Fix a compact interval  $K = [-R, R]$ . For  $|h|$  sufficiently small, all points  $x + \theta h$  with  $x \in K$  lie in a slightly larger compact interval, say  $[-R-1, R+1]$ . Since  $\phi'$  is continuous

on this compact set, it is uniformly continuous there. Hence for any  $\varepsilon > 0$  there exists  $\delta > 0$  such that whenever  $|h| < \delta$ ,

$$\sup_{x \in [-R, R]} |\phi'(x + \theta(x, h)h) - \phi'(x)| < \varepsilon.$$

Therefore

$$\sup_{x \in [-R, R]} |(\Delta^h \phi)(x) - \phi'(x)| \longrightarrow 0 \quad \text{as } h \rightarrow 0.$$

The same argument applies to all derivatives of  $\Delta^h \phi$ , since  $d^k/dx^k$  applied to the difference quotient produces the same formula with  $\phi'$  replaced by a higher derivative. Consequently,

$$\Delta^h \phi \rightarrow \phi' \quad \text{in } \mathcal{E}(\mathbb{R}).$$

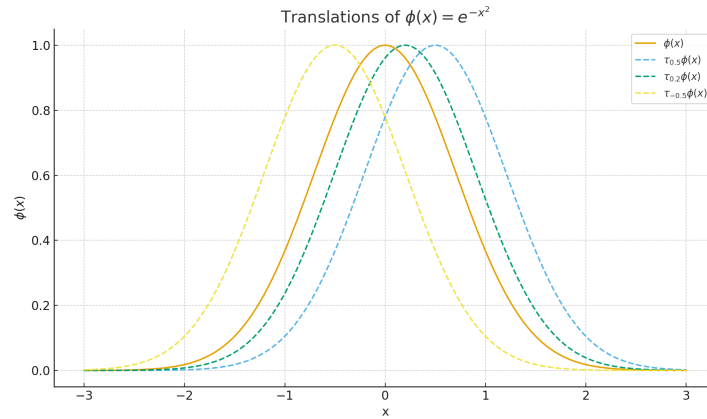


Figure 21: Translations  $\tau_a \phi$  of  $\phi(x) = e^{-x^2}$  for several values of  $a$ .

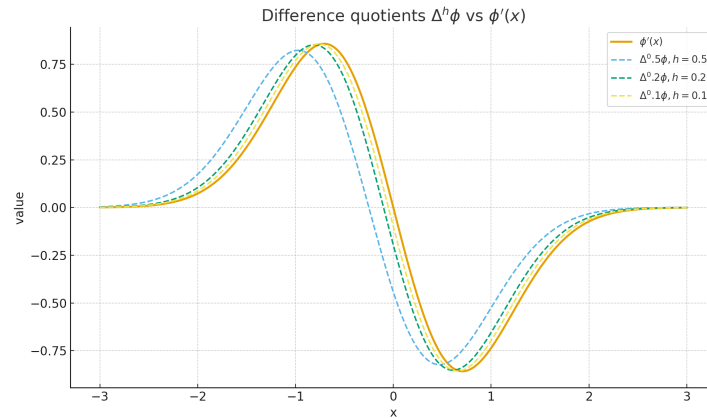


Figure 22: Difference quotients  $\Delta^h \phi$  for  $h = 0.5, 0.2, 0.1$  compared with the derivative  $\phi'(x)$ .

**Theorem 0.1** (Continuity of translations and difference quotients in  $\mathcal{E}(\mathbb{R}^n)$ ). *Let  $\mathcal{E}(\mathbb{R}^n)$  denote the space of smooth functions on  $\mathbb{R}^n$  endowed with the usual Fréchet topology given by the seminorms*

$$p_{K,m}(\phi) := \sup_{\substack{x \in K \\ |\alpha| \leq m}} |D^\alpha \phi(x)|,$$

where  $K \subset \mathbb{R}^n$  is compact and  $m \in \mathbb{N}$ .

- (a) For each fixed  $x \in \mathbb{R}^n$  the translation operator  $\tau_x : \mathcal{E}(\mathbb{R}^n) \rightarrow \mathcal{E}(\mathbb{R}^n)$ , defined by  $(\tau_x \phi)(y) = \phi(y - x)$ , is continuous. Moreover, if  $x_l \rightarrow 0$  in  $\mathbb{R}^n$  and  $\phi \in \mathcal{E}$ , then

$$\tau_{x_l} \phi \longrightarrow \phi \quad \text{in } \mathcal{E}(\mathbb{R}^n).$$

- (b) For each  $i \in \{1, \dots, n\}$  and  $h \neq 0$  the difference quotient operator

$$(\Delta_i^h \phi)(x) := \frac{\phi(x + he_i) - \phi(x)}{h}$$

is a continuous linear map on  $\mathcal{E}(\mathbb{R}^n)$ , and for every  $\phi \in \mathcal{E}(\mathbb{R}^n)$  one has

$$\Delta_i^h \phi \longrightarrow D_i \phi \quad \text{in } \mathcal{E}(\mathbb{R}^n) \quad \text{as } h \rightarrow 0.$$

*Proof.* Fix a compact  $K \subset \mathbb{R}^n$  and  $m \in \mathbb{N}$ .

- (a) *Translations.* For  $|\alpha| \leq m$  we have

$$D^\alpha(\tau_x \phi)(y) = D^\alpha \phi(y - x).$$

If  $x_l \rightarrow 0$ , choose a compact set  $K'$  containing  $K$  and  $K - x_l$  for all  $l$  large. Then  $D^\alpha \phi$  is uniformly continuous on  $K'$ , so given  $\varepsilon > 0$  there exists  $\delta > 0$  such that  $|z - z'| < \delta \Rightarrow |D^\alpha \phi(z) - D^\alpha \phi(z')| < \varepsilon$  for all  $z, z' \in K'$ . For  $l$  large enough we have  $|x_l| < \delta$  and therefore

$$\sup_{y \in K} |D^\alpha \phi(y - x_l) - D^\alpha \phi(y)| \leq \varepsilon.$$

Taking the supremum over  $|\alpha| \leq m$  yields  $p_{K,m}(\tau_{x_l} \phi - \phi) \rightarrow 0$ , i.e.  $\tau_{x_l} \phi \rightarrow \phi$  in  $\mathcal{E}(\mathbb{R}^n)$ .

- (b) *Difference quotients.* Let  $g(x) := D^\alpha \phi(x)$  for some  $|\alpha| \leq m$ . Then

$$D^\alpha(\Delta_i^h \phi)(x) = \frac{g(x + he_i) - g(x)}{h}.$$

By the mean value theorem,

$$\frac{g(x + he_i) - g(x)}{h} = D_i g(x + \theta he_i)$$

for some  $\theta = \theta(x, h) \in (0, 1)$ . Choosing a compact  $K'$  containing  $K$  and  $K + he_i$  for all  $|h|$  small, we see that  $D_i g$  is uniformly continuous on  $K'$ . Hence for every  $\varepsilon > 0$  there exists  $\delta > 0$  such that for  $|h| < \delta$ ,

$$\sup_{x \in K} |D^\alpha(\Delta_i^h \phi)(x) - D_i D^\alpha \phi(x)| = \sup_{x \in K} |D_i g(x + \theta he_i) - D_i g(x)| < \varepsilon.$$

Taking the supremum over  $|\alpha| \leq m$  shows that  $p_{K,m}(\Delta_i^h \phi - D_i \phi) \rightarrow 0$  as  $h \rightarrow 0$ , which is precisely  $\Delta_i^h \phi \rightarrow D_i \phi$  in  $\mathcal{E}(\mathbb{R}^n)$ .  $\square$

## Exercise 1.6

- (a) Show that  $\mathcal{S}$  is a vector subspace of  $\mathcal{E}(\mathbb{R}^n)$ . Show that if  $\{\phi_j\}_{j=1}^\infty$  is a sequence of rapidly decreasing functions which tends to zero in  $\mathcal{S}$ , then  $\phi_j \rightarrow 0$  in  $\mathcal{E}(\mathbb{R}^n)$ .
- (b) Show that  $\mathcal{D}(\mathbb{R}^n)$  is a vector subspace of  $\mathcal{S}$ . Show that if  $\{\phi_j\}_{j=1}^\infty$  is a sequence of compactly supported functions which tends to zero in  $\mathcal{D}(\mathbb{R}^n)$ , then  $\phi_j \rightarrow 0$  in  $\mathcal{S}$ .
- (c) Give an example of a sequence  $\{\phi_j\}_{j=1}^\infty \subset C_0^\infty(\mathbb{R}^n)$  such that
- (i)  $\phi_j \rightarrow 0$  in  $\mathcal{S}$ , but  $\{\phi_j\}$  has no limit in  $\mathcal{D}(\mathbb{R}^n)$ ;
  - (ii)  $\phi_j \rightarrow 0$  in  $\mathcal{E}(\mathbb{R}^n)$ , but  $\{\phi_j\}$  has no limit in  $\mathcal{S}$ .

## Exercise 1.7

- (a) Suppose  $\phi \in \mathcal{S}$ . Let  $\{x_l\}_{l=1}^\infty \subset \mathbb{R}^n$  be a sequence with  $x_l \rightarrow 0$ . Show that

$$\tau_{x_l}\phi \rightarrow \phi, \quad \text{as } l \rightarrow \infty,$$

in  $\mathcal{S}$ , where  $\tau_x$  is the translation operator defined in equation (1.2).

- (b) Suppose  $\phi \in \mathcal{S}$ , show that

$$\Delta_i^h \phi \rightarrow D_i \phi, \quad \text{as } h \rightarrow 0,$$

in  $\mathcal{S}$ , where  $\Delta_i^h$  is the difference quotient defined in Example 2.

## Exercise 1.6

- (a) Show that  $\mathcal{S}(\mathbb{R}^n)$  is a vector subspace of  $\mathcal{E}(\mathbb{R}^n)$ . Show that if  $\{\phi_j\}_{j=1}^\infty$  is a sequence of rapidly decreasing functions which tends to zero in  $\mathcal{S}$ , then  $\phi_j \rightarrow 0$  in  $\mathcal{E}(\mathbb{R}^n)$ .

**Solution.** Every  $\phi \in \mathcal{S}$  is smooth and satisfies

$$q_{\alpha,\beta}(\phi) = \sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta \phi(x)| < \infty, \quad \forall \alpha, \beta,$$

so in particular each derivative is bounded on every compact set. Thus  $\mathcal{S} \subset \mathcal{E}$  and is clearly a vector subspace.

If  $\phi_j \rightarrow 0$  in  $\mathcal{S}$ , then

$$q_{\alpha,\beta}(\phi_j) \rightarrow 0 \quad \forall \alpha, \beta.$$

Fix a compact  $K$  and  $m \in \mathbb{N}$ . On  $K$  we have  $|x^\alpha| \leq C_K$ , hence

$$p_{K,m}(\phi_j) = \sup_{x \in K, |\beta| \leq m} |D^\beta \phi_j(x)| \leq \max_{|\beta| \leq m} q_{0,\beta}(\phi_j) \rightarrow 0.$$

Thus  $\phi_j \rightarrow 0$  in  $\mathcal{E}$ .

- (b) Show that  $\mathcal{D}(\mathbb{R}^n)$  is a vector subspace of  $\mathcal{S}(\mathbb{R}^n)$ . Show that if  $\{\phi_j\}$  tends to 0 in  $\mathcal{D}(\mathbb{R}^n)$ , then  $\phi_j \rightarrow 0$  in  $\mathcal{S}$ .

**Solution.** If  $\phi \in C_0^\infty(\mathbb{R}^n)$  and  $\text{supp } \phi \subset B_R(0)$ , then

$$q_{\alpha,\beta}(\phi) = \sup_{x \in B_R(0)} |x^\alpha D^\beta \phi(x)| \leq R^{|\alpha|} \sup_{x \in B_R(0)} |D^\beta \phi(x)| < \infty,$$

so  $\phi \in \mathcal{S}$ . Linear combinations preserve smoothness and compact support, so  $\mathcal{D} \subset \mathcal{S}$  as a subspace.

If  $\phi_j \rightarrow 0$  in  $\mathcal{D}$ , then for some compact  $K$  we have  $\text{supp } \phi_j \subset K$  and

$$\sup_{x \in K, |\beta| \leq m} |D^\beta \phi_j(x)| \rightarrow 0 \quad \forall m.$$

Hence

$$q_{\alpha,\beta}(\phi_j) = \sup_{x \in K} |x^\alpha D^\beta \phi_j(x)| \leq C_K \sup_{x \in K} |D^\beta \phi_j(x)| \rightarrow 0.$$

Thus  $\phi_j \rightarrow 0$  in  $\mathcal{S}$ .

- (c) Give examples of sequences  $\{\phi_j\} \subset C_0^\infty(\mathbb{R}^n)$  such that

- (i)  $\phi_j \rightarrow 0$  in  $\mathcal{S}$ , but the sequence has no limit in  $\mathcal{D}(\mathbb{R}^n)$ ;
- (ii)  $\phi_j \rightarrow 0$  in  $\mathcal{E}(\mathbb{R}^n)$ , but the sequence has no limit in  $\mathcal{S}$ .

**Solution.**

Let  $\psi \in C_0^\infty(\mathbb{R}^n)$ ,  $\psi \not\equiv 0$ , with  $\text{supp } \psi \subset B_1(0)$ .

(i) Converges in  $\mathcal{S}$  but not in  $\mathcal{D}$ . Define

$$\phi_j(x) = e^{-j^2} \psi(x - je_1).$$

Then  $\text{supp } \phi_j \subset B_1(je_1)$  escapes to infinity. For any  $\alpha, \beta$ ,

$$|x^\alpha D^\beta \phi_j(x)| \leq C_{\alpha,\beta} (1+j)^{|\alpha|} e^{-j^2} \rightarrow 0.$$

Thus  $\phi_j \rightarrow 0$  in  $\mathcal{S}$ . However, no convergent sequence in  $\mathcal{D}$  may have supports escaping to infinity, so  $\{\phi_j\}$  has no limit in  $\mathcal{D}$ .

(ii) Converges in  $\mathcal{E}$  but not in  $\mathcal{S}$ . Define

$$\phi_j(x) = \psi(x - je_1).$$

For any compact  $K$ ,  $\phi_j \equiv 0$  on  $K$  for large  $j$ , hence  $\phi_j \rightarrow 0$  in  $\mathcal{E}$ . However,

$$q_{\alpha,0}(\phi_j) = \sup_x |x^\alpha \phi_j(x)| \geq |j|^{|\alpha|} \max_{|y| \leq 1} |\psi(y)| \rightarrow \infty,$$

so the sequence does not converge in  $\mathcal{S}$ .



## Exercise 1.7

(a) Suppose  $\phi \in \mathcal{S}$ . Let  $\{x_l\} \subset \mathbb{R}^n$  with  $x_l \rightarrow 0$ . Show that

$$\tau_{x_l}\phi \rightarrow \phi \quad \text{in } \mathcal{S}.$$

**Solution.** Recall  $\tau_x\phi(\xi) = \phi(\xi - x)$ . Fix  $\alpha, \beta$ . Using

$$\xi^\alpha = (\xi - x_l + x_l)^\alpha = \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} (\xi - x_l)^\gamma x_l^{\alpha-\gamma},$$

we obtain

$$q_{\alpha,\beta}(\tau_{x_l}\phi - \phi) \leq \sum_{\gamma < \alpha} C_{\alpha,\gamma} |x_l|^{\alpha-\gamma} q_{\gamma,\beta}(\phi) + \sup_{\xi} |\xi^\alpha (D^\beta \phi(\xi - x_l) - D^\beta \phi(\xi))|.$$

The first sum  $\rightarrow 0$  because  $x_l \rightarrow 0$ . The second term  $\rightarrow 0$  because  $\xi^\alpha D^\beta \phi(\xi)$  is uniformly continuous (it is rapidly decaying). Thus  $q_{\alpha,\beta}(\tau_{x_l}\phi - \phi) \rightarrow 0$ , proving convergence in  $\mathcal{S}$ .

(b) Show that for  $\phi \in \mathcal{S}$ ,

$$\Delta_i^h \phi \rightarrow D_i \phi \quad \text{in } \mathcal{S}, \quad h \rightarrow 0,$$

where

$$\Delta_i^h \phi(x) = \frac{\phi(x + he_i) - \phi(x)}{h}.$$

**Solution.** Fix  $\alpha, \beta$ . Use the mean value theorem in the  $e_i$  direction:

$$\frac{D^\beta \phi(x + he_i) - D^\beta \phi(x)}{h} = D_i D^\beta \phi(x + \theta he_i)$$

for some  $\theta \in (0, 1)$ . Hence

$$x^\alpha D^\beta (\Delta_i^h \phi - D_i \phi)(x) = x^\alpha (D_i D^\beta \phi(x + \theta he_i) - D_i D^\beta \phi(x)).$$

The function  $x^\alpha D_i D^\beta \phi(x)$  is continuous and decays rapidly, hence uniformly continuous. Therefore the supremum over  $x$  tends to 0 as  $h \rightarrow 0$  and

$$q_{\alpha,\beta}(\Delta_i^h \phi - D_i \phi) \rightarrow 0.$$

Thus  $\Delta_i^h \phi \rightarrow D_i \phi$  in  $\mathcal{S}$ .

**Spaces of test functions and their seminorms.**

- *Smooth functions:*

$$\mathcal{E}(\mathbb{R}^n) := C^\infty(\mathbb{R}^n)$$

with the Fréchet topology generated by seminorms

$$p_{K,m}(\phi) := \sup_{\substack{x \in K \\ |\alpha| \leq m}} |D^\alpha \phi(x)|, \quad K \subset \mathbb{R}^n \text{ compact, } m \in \mathbb{N}.$$

- *Compactly supported smooth functions:*

$$\mathcal{D}(\mathbb{R}^n) := C_0^\infty(\mathbb{R}^n),$$

endowed with the LF topology: a sequence  $\phi_j \rightarrow 0$  in  $\mathcal{D}$  iff there exists a compact  $K$  such that  $\text{supp } \phi_j \subset K$  for all large  $j$  and

$$\sup_{x \in K, |\alpha| \leq m} |D^\alpha \phi_j(x)| \rightarrow 0 \quad \forall m \in \mathbb{N}.$$

- *Schwartz space:*

$$\mathcal{S}(\mathbb{R}^n) := \left\{ \phi \in C^\infty(\mathbb{R}^n) : q_{\alpha,\beta}(\phi) < \infty \quad \forall \alpha, \beta \right\},$$

where the seminorms are

$$q_{\alpha,\beta}(\phi) := \sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta \phi(x)|, \quad \alpha, \beta \in \mathbb{N}^n.$$

They measure simultaneous control of all derivatives *and* all polynomial weights.

**Continuous embeddings.**

$$\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \subset \mathcal{E}(\mathbb{R}^n),$$

and both inclusions are continuous:

- Every compactly supported smooth function is rapidly decreasing, so  $\mathcal{D} \subset \mathcal{S}$ .
- Every Schwartz function is smooth, so  $\mathcal{S} \subset \mathcal{E}$ .
- Convergence in  $\mathcal{D} \Rightarrow$  convergence in  $\mathcal{S}$  (Exercise 1.6(b)).
- Convergence in  $\mathcal{S} \Rightarrow$  convergence in  $\mathcal{E}$  (Exercise 1.6(a)).

**Examples of different convergences.** Fix  $\psi \in C_0^\infty(\mathbb{R})$ ,  $\psi \neq 0$ ,  $\text{supp } \psi \subset [-1, 1]$ .

- (1)  $\phi_j(x) = \frac{1}{j} \psi(x)$ : converges to 0 in  $\mathcal{D}, \mathcal{S}, \mathcal{E}$ .

| Space         | Elements                                 | Seminorms / topology                                                                      | Convergence of a sequence $\phi_j \rightarrow 0$                                                                                                                          |
|---------------|------------------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{E}$ | $C^\infty$ functions on $\mathbb{R}^n$   | Fréchet topology:<br>$p_{K,m}(\phi) = \sup_{x \in K,  \alpha  \leq m}  D^\alpha \phi(x) $ | For every compact $K$ and every $m$ : $p_{K,m}(\phi_j) \rightarrow 0$ , i.e. all derivatives converge to 0 <i>uniformly on each compact</i> .                             |
| $\mathcal{D}$ | Compactly supported $C^\infty$ functions | LF topology: inductive limit of $\mathcal{D}_K$ with fixed compact support $K$            | There exists a compact $K$ s.t. $\text{supp } \phi_j \subset K$ for all large $j$ , and $p_{K,m}(\phi_j) \rightarrow 0$ for all $m$ . Supports cannot escape to infinity. |
| $\mathcal{S}$ | Rapidly decreasing $C^\infty$ functions  | Schwartz seminorms:<br>$q_{\alpha,\beta}(\phi) = \sup_x  x^\alpha D^\beta \phi(x) $       | $q_{\alpha,\beta}(\phi_j) \rightarrow 0$ for every $\alpha, \beta$ , i.e. all derivatives tend to 0 faster than any inverse polynomial, uniformly on $\mathbb{R}^n$ .     |

Table 1: Three topologies on smooth functions:  $\mathcal{D} \subset \mathcal{S} \subset \mathcal{E}$ .

- (2)  $\phi_j(x) = e^{-j^2} \psi(x - j)$ : the supports escape to  $+\infty$ , but the factors  $e^{-j^2}$  force all Schwartz seminorms  $q_{\alpha,\beta}(\phi_j) \rightarrow 0$ . Hence  $\phi_j \rightarrow 0$  in  $\mathcal{S}$  but the sequence has no limit in  $\mathcal{D}$ .
- (3)  $\phi_j(x) = \psi(x - j)$ : for any compact  $K$ , one has  $\phi_j \equiv 0$  on  $K$  for all large  $j$ , so  $\phi_j \rightarrow 0$  in  $\mathcal{E}$ . However  $q_{1,0}(\phi_j) \sim j \rightarrow \infty$ , so there is no convergence in  $\mathcal{S}$ .
- (4)  $\phi_j(x) = e^{-(x-a_j)^2}$  with  $a_j \rightarrow 0$ : translations of a Gaussian. Then  $\phi_j \rightarrow \phi$  in  $\mathcal{S}$  (and hence in  $\mathcal{E}$ ).
- (5)  $\phi_h(x) = \frac{\phi(x+h) - \phi(x)}{h}$  with  $\phi(x) = e^{-x^2}$ : difference quotients; one checks that  $\phi_h \rightarrow \phi'$  in  $\mathcal{S}$  as  $h \rightarrow 0$ .

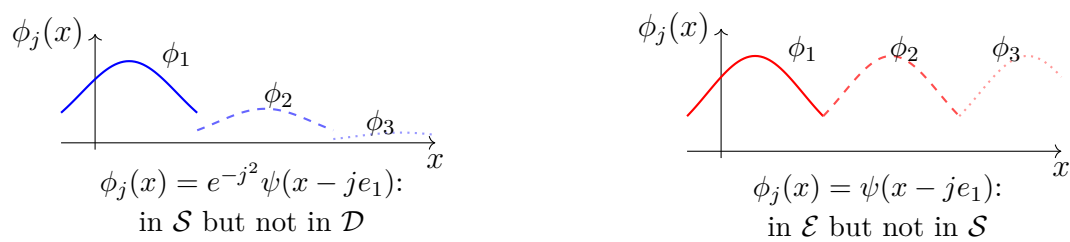


Figure 23: Schematic of the sequences in Exercise 1.6(c): left: scaled bumps converging in  $\mathcal{S}$  but not in  $\mathcal{D}$ ; right: bumps of fixed height converging in  $\mathcal{E}$  but not in  $\mathcal{S}$ .

## Exercise 2.1 (\*): Schwartz functions and integrability

Assume  $f, g, h \in \mathcal{S}(\mathbb{R}^n)$ .

(a) Show that for any multi-index  $\alpha$  and any  $1 \leq p < \infty$ , one has  $D^\alpha f \in L^p(\mathbb{R}^n)$ .

(b) Define

$$F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{C}, \quad F(x, y) = f(x) g(y - x).$$

Show that  $F \in L^1(\mathbb{R}^n \times \mathbb{R}^n)$ .

(c) For each  $x \in \mathbb{R}^n$ , set

$$G_x : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{C}, \quad G_x(y, z) = f(y) g(z) h(x - y - z).$$

Show that  $G_x \in L^1(\mathbb{R}^n \times \mathbb{R}^n)$  for every  $x$ .

### Solution

(a)  $D^\alpha f \in L^p$ . Since  $f \in \mathcal{S}(\mathbb{R}^n)$ , for each multi-index  $\alpha$  and each integer  $N > 0$  there exists  $C_{\alpha, N} > 0$  such that

$$|D^\alpha f(x)| \leq \frac{C_{\alpha, N}}{(1 + |x|)^N} \quad \forall x \in \mathbb{R}^n.$$

Fix  $1 \leq p < \infty$  and choose  $N$  with  $Np > n$ . Then

$$|D^\alpha f(x)|^p \leq \frac{C_{\alpha, N}^p}{(1 + |x|)^{Np}},$$

and the right-hand side is integrable on  $\mathbb{R}^n$ . Hence  $\|D^\alpha f\|_{L^p} < \infty$ , so  $D^\alpha f \in L^p(\mathbb{R}^n)$ .

(b)  $F \in L^1(\mathbb{R}^{2n})$ . We compute

$$\begin{aligned} \iint_{\mathbb{R}^n \times \mathbb{R}^n} |F(x, y)| \, dx \, dy &= \iint |f(x)g(y - x)| \, dx \, dy \\ &= \int_{\mathbb{R}^n} |f(x)| \left( \int_{\mathbb{R}^n} |g(y - x)| \, dy \right) dx \\ &\quad (\text{let } t = y - x \text{ so } dy = dt) \\ &= \int_{\mathbb{R}^n} |f(x)| \left( \int_{\mathbb{R}^n} |g(t)| \, dt \right) dx \\ &= \|g\|_{L^1(\mathbb{R}^n)} \|f\|_{L^1(\mathbb{R}^n)} < \infty, \end{aligned}$$

since  $f, g \in \mathcal{S} \subset L^1$ . Therefore  $F \in L^1(\mathbb{R}^{2n})$ .

(c)  $G_x \in L^1(\mathbb{R}^{2n})$  for each fixed  $x$ . Fix  $x \in \mathbb{R}^n$  and consider

$$I(x) := \iint_{\mathbb{R}^n \times \mathbb{R}^n} |G_x(y, z)| dy dz = \iint |f(y)g(z)h(x - y - z)| dy dz.$$

Make the change of variables  $(y, z) \mapsto (y, w)$  with  $w := y + z$ . Then  $z = w - y$ , the Jacobian is 1, and

$$\begin{aligned} I(x) &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(y)| |g(w - y)| |h(x - w)| dy dw \\ &= \int_{\mathbb{R}^n} |h(x - w)| \left( \int_{\mathbb{R}^n} |f(y)| |g(w - y)| dy \right) dw \\ &= \int_{\mathbb{R}^n} |h(x - w)| (|f| * |g|)(w) dw \\ &= (|f| * |g| * |h|)(x). \end{aligned}$$

Since  $f, g, h \in \mathcal{S} \subset L^1$ , the convolution  $|f| * |g| \in L^1$  and then  $(|f| * |g|) * |h| \in L^1$ . In particular,  $(|f| * |g| * |h|)(x) < \infty$  for every  $x$ , so  $I(x) < \infty$  and  $G_x \in L^1(\mathbb{R}^{2n})$ .

## Schwartz functions: definition, examples, integrability

**Definition 0.2** (Schwartz space). The Schwartz space  $\mathcal{S}(\mathbb{R}^n)$  consists of all functions  $f \in C^\infty(\mathbb{R}^n)$  such that

$$q_{\alpha, \beta}(f) := \sup_{x \in \mathbb{R}^n} |x^\alpha D^\beta f(x)| < \infty \quad \text{for all multi-indices } \alpha, \beta.$$

Equivalently, for every multi-index  $\beta$  and  $N \in \mathbb{N}$  there exists  $C_{N, \beta} > 0$  such that

$$|D^\beta f(x)| \leq \frac{C_{N, \beta}}{(1 + |x|)^N}, \quad x \in \mathbb{R}^n.$$

### Examples.

- The Gaussian  $f(x) = e^{-|x|^2}$  belongs to  $\mathcal{S}(\mathbb{R}^n)$ , since each derivative is a polynomial times  $e^{-|x|^2}$ .
- More generally, for any polynomial  $P$  we have  $P(x)e^{-|x|^2} \in \mathcal{S}(\mathbb{R}^n)$ .
- Every compactly supported smooth function is Schwartz, so  $\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n)$ .
- If  $f \in \mathcal{S}$ , then all derivatives  $D^\alpha f$ , translations  $x \mapsto f(x - a)$ , and polynomial multiples  $x \mapsto P(x)f(x)$  also lie in  $\mathcal{S}$ .

**Integrability properties.** Let  $f \in \mathcal{S}(\mathbb{R}^n)$ .

- For every multi-index  $\alpha$  and every  $1 \leq p < \infty$ ,

$$D^\alpha f \in L^p(\mathbb{R}^n).$$

Indeed, for each  $N$  we have  $|D^\alpha f(x)| \leq C_{\alpha,N}(1 + |x|)^{-N}$ , and choosing  $N$  with  $Np > n$  gives  $|D^\alpha f|^p \in L^1$ .

- For every multi-index  $\gamma$  and  $1 \leq p < \infty$ ,

$$x^\gamma D^\alpha f(x) \in L^p(\mathbb{R}^n),$$

since we may take  $N$  large enough to dominate the polynomial growth of  $x^\gamma$ .

- In particular  $f \in L^p(\mathbb{R}^n)$  for all  $1 \leq p \leq \infty$ , and the same holds for all derivatives  $D^\alpha f$ .
- If  $f, g \in \mathcal{S}$  then  $f, g \in L^1$  and the convolution  $f * g$  is well-defined and belongs to  $\mathcal{S}(\mathbb{R}^n)$ .

### Problem 3

Let  $A + B := \{a + b : a \in A, b \in B\}$  and  $B_r(x) = \{y \in \mathbb{R}^n : \|y - x\| < r\}$ ,  $\overline{B_r(x)} = \{y : \|y - x\| \leq r\}$ .

- (a) Prove the following identities for  $r, s > 0$  and  $x \in \mathbb{R}^n$ :

- i)  $B_r(x) + B_s(0) = B_{r+s}(x)$ ;
- ii)  $\overline{B_r(x)} + B_s(0) = B_{r+s}(x)$ ;
- iii)  $\overline{B_r(x)} + \overline{B_s(0)} = \overline{B_{r+s}(x)}$ .

- (b) Suppose that  $A, B \subset \mathbb{R}^n$ . Show that:

- b) If one of  $A$  or  $B$  is open, then so is  $A + B$ .
- b) If  $A$  and  $B$  are both bounded, then so is  $A + B$ .
- b) If  $A$  is closed and  $B$  is compact, then  $A + B$  is closed.
- b) If  $A$  and  $B$  are both compact, then so is  $A + B$ .

*Proof.* (a) (i) Let  $z \in B_r(x) + B_s(0)$ , so  $z = y + u$  with  $\|y - x\| < r$  and  $\|u\| < s$ . Then

$$\|z - x\| \leq \|y - x\| + \|u\| < r + s,$$

hence  $z \in B_{r+s}(x)$ .

Conversely, let  $z \in B_{r+s}(x)$ , so  $\|z - x\| < r + s$ . Set

$$y := x + \frac{r}{r+s}(z - x), \quad u := z - y = \frac{s}{r+s}(z - x).$$

Then

$$\|y - x\| = \frac{r}{r+s}\|z - x\| < r, \quad \|u\| = \frac{s}{r+s}\|z - x\| < s,$$

so  $y \in B_r(x)$ ,  $u \in B_s(0)$  and  $z = y + u \in B_r(x) + B_s(0)$ . Thus  $B_r(x) + B_s(0) = B_{r+s}(x)$ .

(ii) If  $z = y + u$  with  $\|y - x\| \leq r$  and  $\|u\| < s$ , then

$$\|z - x\| \leq \|y - x\| + \|u\| \leq r + \|u\| < r + s,$$

so  $z \in B_{r+s}(x)$  and  $\overline{B_r(x)} + B_s(0) \subset B_{r+s}(x)$ .

Conversely, if  $z \in B_{r+s}(x)$ , part (i) gives  $z = y + u$  with  $y \in B_r(x) \subset \overline{B_r(x)}$  and  $u \in B_s(0)$ , hence  $z \in \overline{B_r(x)} + B_s(0)$ . Therefore  $\overline{B_r(x)} + B_s(0) = B_{r+s}(x)$ .

(iii) If  $z = y + u$  with  $\|y - x\| \leq r$ ,  $\|u\| \leq s$ , then  $\|z - x\| \leq r + s$ , so  $\overline{B_r(x)} + \overline{B_s(0)} \subset \overline{B_{r+s}(x)}$ .

For the reverse inclusion, let  $\|z - x\| < r + s$ . Then  $z \in B_{r+s}(x)$  and by (i) we have  $z = y + u$  with  $y \in B_r(x) \subset \overline{B_r(x)}$  and  $u \in B_s(0) \subset \overline{B_s(0)}$ , so  $z \in \overline{B_r(x)} + \overline{B_s(0)}$ .

If  $\|z - x\| = r + s$ , set  $z_t = x + t(z - x)$ ,  $0 < t < 1$ . Then  $\|z_t - x\| = t(r + s) < r + s$ , hence  $z_t \in B_{r+s}(x)$  and thus  $z_t \in \overline{B_r(x)} + \overline{B_s(0)}$ . The set  $\overline{B_r(x)} + \overline{B_s(0)}$  is compact (sum of two compact sets), hence closed, so  $z = \lim_{t \rightarrow 1} z_t$  also belongs to it. Therefore  $\overline{B_{r+s}(x)} \subset \overline{B_r(x)} + \overline{B_s(0)}$  and

$$\overline{B_r(x)} + \overline{B_s(0)} = \overline{B_{r+s}(x)}.$$

(b) Assume  $A$  is open (the case where  $B$  is open is analogous). Let  $z \in A + B$ , so  $z = a + b$  with  $a \in A$ ,  $b \in B$ . Since  $A$  is open, there exists  $r > 0$  with  $B_r(a) \subset A$ . If  $\|y - z\| < r$  and we set  $a' := a + (y - z)$ , then  $\|a' - a\| = \|y - z\| < r$ , so  $a' \in B_r(a) \subset A$ , and  $y = a' + b \in A + B$ . Thus  $B_r(z) \subset A + B$  and  $A + B$  is open.

(c) Suppose  $A \subset B_M(0)$  and  $B \subset B_N(0)$  for some  $M, N > 0$ . If  $z = a + b \in A + B$ , then

$$\|z\| \leq \|a\| + \|b\| \leq M + N,$$

so  $A + B \subset B_{M+N}(0)$  and  $A + B$  is bounded.

(d) Assume  $A$  is closed and  $B$  is compact. Let  $(z_k)$  be a sequence in  $A + B$  with  $z_k \rightarrow z$ . Write  $z_k = a_k + b_k$  with  $a_k \in A$  and  $b_k \in B$ .

By compactness of  $B$ , there is a subsequence  $(b_{k_j})$  such that  $b_{k_j} \rightarrow b \in B$ . Then

$$a_{k_j} = z_{k_j} - b_{k_j} \longrightarrow z - b.$$

Since  $A$  is closed and  $a_{k_j} \in A$ , we have  $z - b \in A$ , hence  $z = (z - b) + b \in A + B$ . Therefore  $A + B$  is closed.

(e) If  $A$  and  $B$  are compact, then  $A \times B$  is compact and the map

$$f : A \times B \rightarrow \mathbb{R}^n, \quad f(a, b) = a + b$$

is continuous. Hence  $A + B = f(A \times B)$  is compact as the continuous image of a compact set.  $\square$



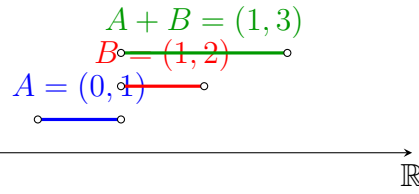


Figure 24: Minkowski sum of intervals on the real line.

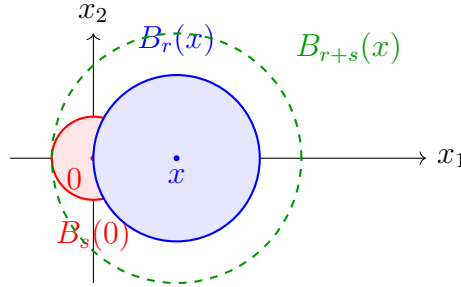
Figure 25: Minkowski sum of balls:  $B_r(x) + B_s(0) = B_{r+s}(x)$ .

Figure ?? gives a concrete picture for part (e): the Minkowski sum  $A + B$  of two compact sets is again compact. Here  $A$  is a closed triangle and  $B$  is a small closed ball; the sum is just a uniformly thickened version of  $A$ .

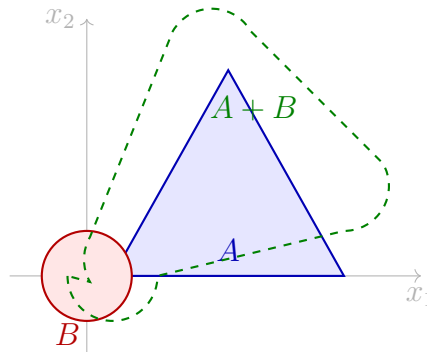


Figure 26: Minkowski sum of a compact set  $A$  (triangle) with a small ball  $B$ . Geometrically,  $A + B$  is obtained by “thickening”  $A$  by the radius of  $B$ . Since both  $A$  and  $B$  are compact, their sum  $A + B$  is compact as well.

**Example (openness of  $A + B$ ).** Let  $A = B_1(0) = \{x \in \mathbb{R}^2 : \|x\| < 1\}$  and let  $b \in \mathbb{R}^2$ . Define

$$B = \{0, b\}.$$

Then

$$A + B = \{a + p : a \in A, p \in B\} = A + \{0, b\} = A \cup (A + b).$$

In particular, for the two specific choices

$$b_1 = (2, 0), \quad b_2 = (0, 2),$$

we obtain

$$A + \{0, b_1\} = B_1(0) \cup B_1((2, 0)), \quad A + \{0, b_2\} = B_1(0) \cup B_1((0, 2)).$$

In each case  $A$  is open,  $B$  is finite, and  $A + B$  is the union of two open discs, hence open but not connected.

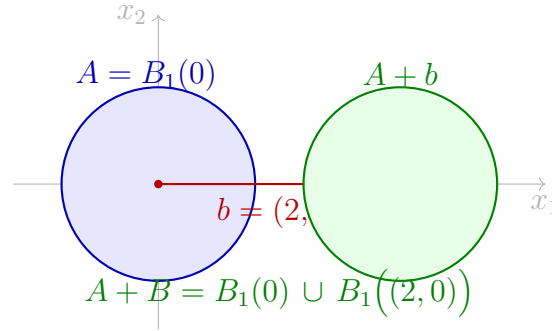


Figure 27: Example for part (b) with  $b = (2, 0)$ . The set  $A$  is the blue open disc,  $B = \{0, b\}$  consists of the two red points, and  $A + B$  is the union of the two open discs.

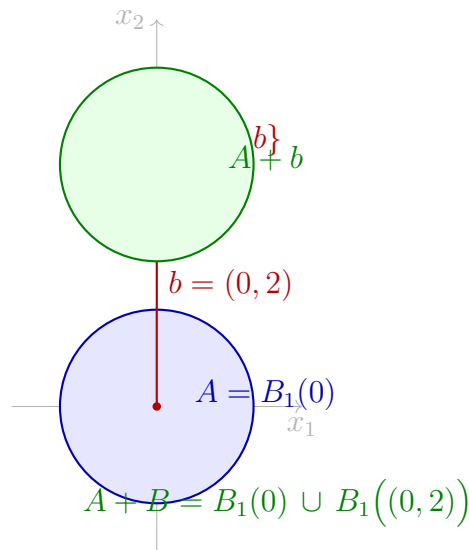


Figure 28: Example for part (b) with  $b = (0, 2)$ . Again  $A$  is the blue disc,  $B = \{0, b\}$  is the two-point set, and  $A + B$  is the union of two open discs, now vertically separated.

**Example (two-point set  $B$ ).** Let

$$A = B_1(0) = \{x \in \mathbb{R}^2 : \|x\| < 1\}$$

and choose two vectors

$$b_1 = (2, 0), \quad b_2 = (0, 2).$$

Define

$$B = \{b_1, b_2\}.$$

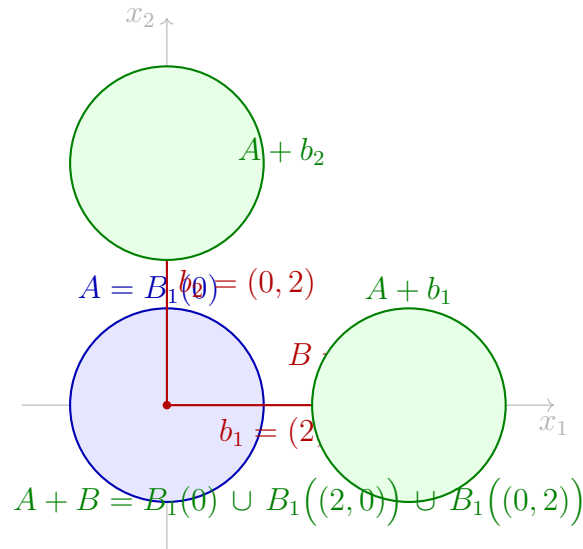


Figure 29: Example for part (b) with  $A = B_1(0)$  and  $B = \{0, b_1, b_2\}$ . The Minkowski sum  $A + B$  is the union of three open discs, hence open but not connected.

Then the Minkowski sum is

$$\begin{aligned}
 A + B &= \{a + p : a \in A, p \in B\} = A + \{b_1, b_2\} \\
 &= (A + b_1) \cup (A + b_2) \\
 &= B_1((2, 0)) \cup B_1((0, 2)).
 \end{aligned}$$

Thus  $A + B$  is the union of two open discs, hence open but not connected. Note that, in contrast with the case  $B = \{0, b_1, b_2\}$ , there is now no disc centred at the origin, because  $0 \notin B$ .

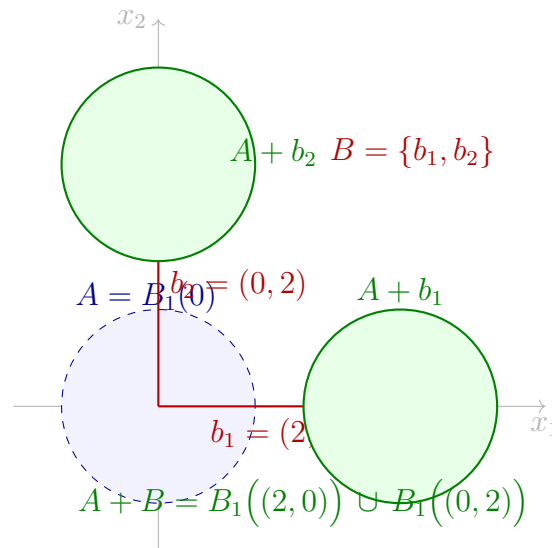


Figure 30: Example with  $A = B_1(0)$  and  $B = \{b_1, b_2\}$ , where  $b_1 = (2, 0)$  and  $b_2 = (0, 2)$ . The Minkowski sum  $A + B$  is the union of the two translated discs  $A + b_1$  and  $A + b_2$ , hence open but not connected.

## Exercise 2.4 (Convolution preserves smoothness)

Show that if  $f \in C_0^k(\mathbb{R}^n)$  and  $g \in C_0^\ell(\mathbb{R}^n)$  then  $f * g \in C_0^{k+\ell}(\mathbb{R}^n)$ . Conclude that  $\mathcal{D}(\mathbb{R}^n)$  is closed under convolution.

**Solution.** Let

$$(f * g)(x) := \int_{\mathbb{R}^n} f(x - y) g(y) dy.$$

**Step 1: Compact support.** Let  $K_f = \text{supp } f$  and  $K_g = \text{supp } g$ . If  $(f * g)(x) \neq 0$ , then there exists  $y$  such that  $f(x - y) \neq 0$  and  $g(y) \neq 0$ , so  $x - y \in K_f$  and  $y \in K_g$ . Hence

$$x = (x - y) + y \in K_f + K_g := \{a + b : a \in K_f, b \in K_g\}.$$

Thus

$$\text{supp}(f * g) \subset K_f + K_g.$$

Since  $K_f$  and  $K_g$  are compact, their Minkowski sum  $K_f + K_g$  is compact. Therefore  $f * g$  has compact support.

**Step 2: Differentiation under the integral.** Let  $\alpha$  be a multiindex. Since  $f$  has compact support, the integrand in

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y) g(y) dy$$

vanishes outside a fixed compact set, so we may differentiate under the integral:

$$\partial^\alpha (f * g)(x) = \int_{\mathbb{R}^n} \partial^\alpha f(x - y) g(y) dy = (\partial^\alpha f) * g.$$

Similarly,

$$\partial^\alpha (f * g) = f * (\partial^\alpha g).$$

Both expressions are continuous, since convolution of continuous compactly supported functions is continuous.

**Step 3: Smoothness of order  $k + \ell$ .** Let  $\alpha$  be a multiindex with  $|\alpha| \leq k + \ell$ . Choose multiindices  $\beta, \gamma$  with  $\alpha = \beta + \gamma$ ,  $|\beta| \leq k$ ,  $|\gamma| \leq \ell$ . Then:

$$\partial^\alpha (f * g) = \partial^\beta \partial^\gamma (f * g) = \partial^\beta (f * (\partial^\gamma g)) = (\partial^\beta f) * (\partial^\gamma g).$$

Here  $\partial^\beta f$  and  $\partial^\gamma g$  are continuous and compactly supported, so their convolution is continuous. Thus  $f * g \in C_0^{k+\ell}(\mathbb{R}^n)$ .

**Step 4: Closure of  $\mathcal{D}(\mathbb{R}^n)$  under convolution.** Recall:

$$\mathcal{D}(\mathbb{R}^n) = C_0^\infty(\mathbb{R}^n) = \bigcap_{m=0}^{\infty} C_0^m(\mathbb{R}^n).$$

If  $f, g \in \mathcal{D}(\mathbb{R}^n)$ , then for each  $m$  we have  $f, g \in C_0^m$ . Applying the previous result with  $k = \ell = m$  yields  $f * g \in C_0^{2m}$  for every  $m$ , so  $f * g \in C_0^r$  for all  $r$ . Hence

$$f * g \in C_0^\infty(\mathbb{R}^n) = \mathcal{D}(\mathbb{R}^n),$$

and  $\mathcal{D}(\mathbb{R}^n)$  is closed under convolution.

**Example: hat functions in  $\mathbb{R}$**

Take

$$f(x) = \max(1 - |x|, 0), \quad g(x) = \max(1 - |x|, 0).$$

- Both  $f$  and  $g$  are continuous, piecewise linear.
- They vanish outside the interval  $[-1, 1]$ .
- Hence  $f, g \in C_0^0(\mathbb{R})$  (continuous with compact support,  $k = 0$ ).

Their convolution is

$$(f * g)(x) = \int_{\mathbb{R}} f(x - y) g(y) dy,$$

which is a continuous “rounded tent” supported in the interval  $[-2, 2]$ .

- If  $|x| > 2$ , then the supports of  $y \mapsto f(x - y)$  and  $y \mapsto g(y)$  do not overlap, so  $(f * g)(x) = 0$ .
- If  $|x| \leq 2$ , the integrand is nonzero on an interval in  $y$ , giving  $(f * g)(x) > 0$ .

Thus

$$\text{supp } f = [-1, 1], \quad \text{supp } g = [-1, 1], \quad \text{supp}(f * g) \subset [-1, 1] + [-1, 1] = [-2, 2],$$

and  $f * g$  is “smoother” than  $f$  and  $g$  (it has no sharp corner at 0).

For higher regularity one can replace the hat by a standard smooth bump

$$\varphi(x) = \begin{cases} e^{-1/(1-x^2)}, & |x| < 1, \\ 0, & |x| \geq 1, \end{cases}$$

and take  $f = \varphi$ ,  $g = \varphi$ . Then  $f, g \in C_0^\infty(\mathbb{R})$ , and the result of Exercise 2.4 says that

$$f * g \in C_0^\infty(\mathbb{R}) \quad \text{with} \quad \text{supp}(f * g) \subset [-2, 2].$$

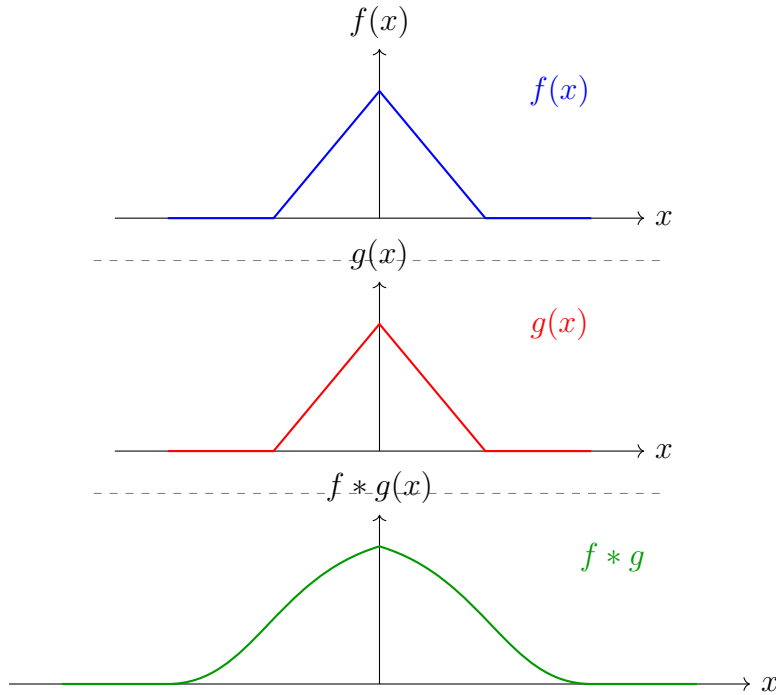


Figure 31: Example of convolution of compactly supported functions on  $\mathbb{R}$ . Two hat functions  $f, g$  with support in  $[-1, 1]$  produce a wider, smoother convolution  $f * g$  supported in  $[-2, 2]$ .

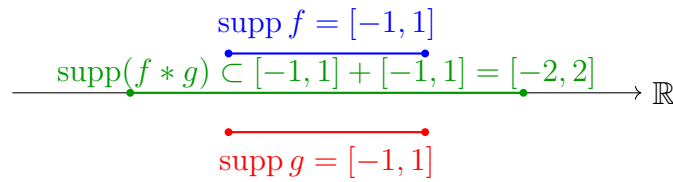


Figure 32: Support of the convolution  $f * g$  is contained in the Minkowski sum of the supports of  $f$  and  $g$ .

### (i) Supports of $f$ and $g$

Consider the hat functions

$$f(x) = \max(1 - |x|, 0), \quad g(x) = \max(1 - |x|, 0).$$

Both vanish outside the interval  $[-1, 1]$ , hence

$$\text{supp } f = \text{supp } g = [-1, 1].$$

### (ii) Support of the convolution $f * g$

The convolution is

$$(f * g)(x) = \int_{\mathbb{R}} f(x - y) g(y) dy.$$

The integrand is nonzero only when

$$x - y \in [-1, 1] \quad \text{and} \quad y \in [-1, 1],$$

which implies

$$x \in [-1, 1] + [-1, 1] = [-2, 2].$$

Thus

$$\text{supp}(f * g) \subset [-2, 2].$$

### (iii) Shape and smoothness of $f * g$

The hat functions  $f$  and  $g$  are continuous but not differentiable at  $x = 0$ . Their convolution  $f * g$  is smoother: it is continuous, piecewise quadratic, and has no corner at  $x = 0$ . Convolution therefore has a regularizing effect even when the input functions are only continuous.

### (iv) A $C^\infty$ example

Let

$$\varphi(x) = \begin{cases} e^{-1/(1-x^2)}, & |x| < 1, \\ 0, & |x| \geq 1. \end{cases}$$

Then  $\varphi \in C_0^\infty(\mathbb{R})$ . Setting  $f = \varphi$ ,  $g = \varphi$  gives

$$f, g \in C_0^\infty(\mathbb{R}) \implies f * g \in C_0^\infty(\mathbb{R}),$$

and

$$\text{supp}(f * g) \subset [-2, 2].$$

### (v) Interpretation via Minkowski sums

For compactly supported functions  $f$  and  $g$  one always has

$$\text{supp}(f * g) \subset \text{supp } f + \text{supp } g.$$

In the present case,

$$[-1, 1] + [-1, 1] = [-2, 2],$$

which geometrically explains the growth of the support under convolution.

## The Meaning and Applications of Convolution

Given two functions  $f$  and  $g$  on  $\mathbb{R}^n$ , their convolution is

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y) g(y) dy.$$

Conceptually, we *flip* one function, *slide* it across the other, and measure their weighted overlap. Convolution describes how two shapes interact under translation. Below we present the main interpretations.

### 1. Convolution as overlapping shapes

Convolution measures how much of the shape of  $f$  overlaps with the shape of  $g$  when  $g$  is shifted by  $x$ . If the overlap is strong,  $(f * g)(x)$  is large; if the functions barely overlap,  $(f * g)(x)$  is small; if there is no overlap, the convolution is zero. Thus  $(f * g)(x)$  describes the overlap as one function slides across the other.

### 2. Convolution as smoothing or averaging

If  $g$  is smooth, then  $f * g$  is smoother than  $f$ . Indeed, for each  $x$ ,

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y) g(y) dy = \text{a weighted average of } f \text{ near } x.$$

Thus convolution removes noise and irregularities in  $f$ . This is the basis of smoothing kernels, mollifiers, and the regularizing effects of the heat equation.

### 3. Convolution as propagation of influence (PDE)

If  $g$  is a Green's function, then convolution represents the solution of a partial differential equation. For instance,

$$u = f * G$$

solves the equation  $Lu = f$  for a differential operator  $L$ . Examples: heat kernels solve the heat equation; Poisson kernels give harmonic extensions; the fundamental solution of Laplace gives Newtonian potentials. In this sense convolution spreads or propagates the influence of  $f$  through the medium described by  $g$ .

### 4. Convolution in probability

If  $X$  and  $Y$  are independent random variables with densities  $f$  and  $g$ , then the density of  $X + Y$  is the convolution  $f * g$ . Thus convolution describes the distribution of sums of random variables. In particular, adding many independent random variables produces repeated convolutions, leading toward Gaussian limits (the Central Limit Theorem).



## 5. Convolution in signal processing and Fourier analysis

Convolution corresponds to filtering. If  $g$  is a “filter” (for instance a low-pass, high-pass, or band-pass filter), then  $f * g$  is the input signal  $f$  passed through that filter. In the Fourier domain,

$$\mathcal{F}(f * g) = \widehat{f} \widehat{g},$$

so convolution in physical space becomes multiplication in frequency space.

### Meaningful examples of convolution

**Example 1: Averaging window.** Let

$$g(t) = \frac{1}{2h} 1_{[-h, h]}(t).$$

Then

$$(f * g)(x) = \frac{1}{2h} \int_{x-h}^{x+h} f(t) dt,$$

which is just the *average* of  $f$  around  $x$  over an interval of length  $2h$ .

$\Rightarrow$  *Smoothing*: local averaging of the signal.

---

**Example 2: Gaussian smoothing.** Let

$$g_\varepsilon(y) = \frac{1}{\sqrt{\pi} \varepsilon} e^{-y^2/\varepsilon^2}.$$

Then  $f * g_\varepsilon$  is the *Gaussian-blurred* version of  $f$ .

$\Rightarrow$  *Effect*: removes noise, increases differentiability.

---

**Example 3: Derivative filters.** If  $g = f'$ , then formally

$$f * f' = \frac{d}{dx} (f * f).$$

More generally, for a multiindex  $\alpha$ ,

$$\partial^\alpha (f * g) = (\partial^\alpha f) * g = f * (\partial^\alpha g).$$

$\Rightarrow$  *Effect*: convolution propagates derivatives and can act as a differentiating filter.

---

**Example 4: Probability (sum of random variables).** Let  $f$  be the density of a random variable  $X$  and  $g$  the density of an independent random variable  $Y$ . Then the density of  $X + Y$  is  $f * g$ :

$$(f * g)(x) = \int_{\mathbb{R}} f(x - y) g(y) dy.$$

$\Rightarrow$  *Effect*: convolution describes the law of sums of independent random variables.

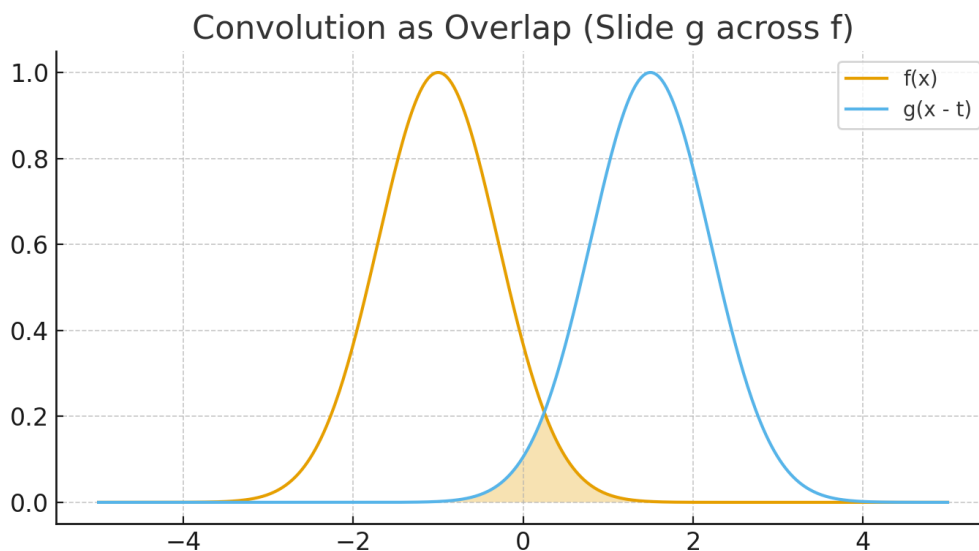


Figure 33: Convolution as overlap: the function  $g$  is shifted horizontally and its overlap with  $f$  determines the value of  $(f * g)(t)$ .

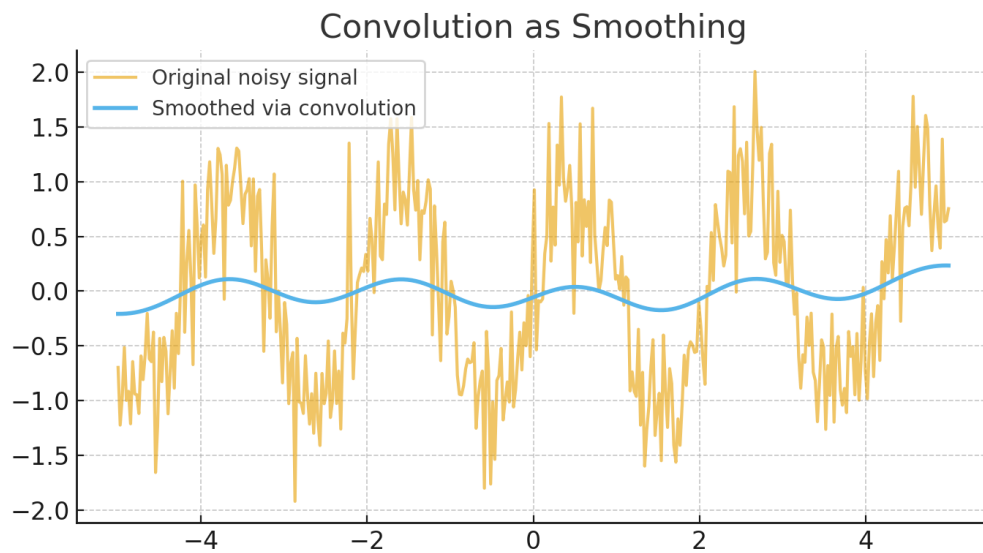


Figure 34: Convolution as smoothing: a noisy signal (blue) is convolved with a Gaussian kernel, producing a smoother function (orange).

## Exercise 2.5 (Young, Hölder and Minkowski inequalities)

Suppose that  $F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$  is a positive simple function.

- (a) Show that Minkowski's integral inequality holds in the case  $p = 1$ :

$$\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right| dx \leq \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |F(x, y)| dy dx.$$

- (b) Prove Young's inequality: if  $a, b \geq 0$  and  $p, q > 1$  satisfy  $p^{-1} + q^{-1} = 1$ , then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

*Hint:* Set  $t = p^{-1}$ , consider the function  $\log[ta^p + (1-t)b^q]$  and use the concavity of  $\log$ .

- (c) With  $p, q > 1$  such that  $p^{-1} + q^{-1} = 1$ , show that if  $\|f\|_{L^p} = 1$  and  $\|g\|_{L^q} = 1$  then

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq 1.$$

Deduce Hölder's inequality

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \|f\|_{L^p} \|g\|_{L^q}, \quad f \in L^p(\mathbb{R}^n), \quad g \in L^q(\mathbb{R}^n).$$

- (d) Set

$$G(y) := \left( \int_{\mathbb{R}^n} F(x, y) dx \right)^{p-1}.$$

- i) Show that if  $q = \frac{p}{p-1}$ , then

$$\|G\|_{L^q(\mathbb{R}^n)} = \left\| \int_{\mathbb{R}^n} F(x, \cdot) dx \right\|_{L^p(\mathbb{R}^n)}^{p-1}.$$

- ii) Show that

$$\left\| \int_{\mathbb{R}^n} F(x, \cdot) dx \right\|_{L^p(\mathbb{R}^n)}^p = \int_{\mathbb{R}^n} \left( \int_{\mathbb{R}^n} G(y) F(x, y) dy \right) dx.$$

- iii) Applying Hölder's inequality, deduce a Minkowski-type inequality for  $F$ .

- (e) Deduce Minkowski's integral inequality:

$$\left( \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right|^p dx \right)^{1/p} \leq \int_{\mathbb{R}^n} \left( \int_{\mathbb{R}^n} |F(x, y)|^p dx \right)^{1/p} dy,$$

which holds for any measurable function  $F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{C}$  and all  $1 \leq p < \infty$ .

## Deep Theory / Roadmap

This exercise reconstructs the chain of ideas leading from convexity to Young's inequality, Hölder's inequality, and finally to Minkowski's integral inequality. We outline the conceptual structure.

### Part (a): Minkowski's inequality for $p = 1$

Suppose  $F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$  is a positive simple function. We want to show

$$\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right| dx \leq \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |F(x, y)| dy dx.$$

**Solution.** Since  $F(x, y) \geq 0$  for all  $(x, y)$ , we have for each fixed  $x$ :

$$\int_{\mathbb{R}^n} F(x, y) dy \geq 0 \implies \left| \int_{\mathbb{R}^n} F(x, y) dy \right| = \int_{\mathbb{R}^n} F(x, y) dy.$$

Therefore

$$\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right| dx = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x, y) dy dx.$$

On the other hand  $F \geq 0$  implies  $|F| = F$ , hence

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |F(x, y)| dy dx = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x, y) dy dx.$$

Thus the two sides are equal, and in particular the desired inequality holds.

**Example.** Let

$$F(x, y) = 1_{[0,1]^n}(x) 1_{[0,1]^n}(y).$$

Then

$$\int_{\mathbb{R}^n} F(x, y) dy = 1_{[0,1]^n}(x),$$

so

$$\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right| dx = \int_{[0,1]^n} 1 dx = 1.$$

Moreover  $|F| = F$ , and

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |F(x, y)| dy dx = \int_{[0,1]^n} \int_{[0,1]^n} 1 dy dx = 1.$$

This concrete case illustrates that for  $p = 1$  and  $F \geq 0$  the inequality is actually an equality.

**Additional examples for part (a)**

The inequality

$$\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) dy \right| dx \leq \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |F(x, y)| dy dx$$

holds with equality whenever  $F \geq 0$ . Here are three regular examples.

**Example 1: Gaussian.** Let  $F(x, y) = e^{-x^2-y^2}$  on  $\mathbb{R}$ . Then

$$\int_{\mathbb{R}} F(x, y) dy = \sqrt{\pi} e^{-x^2}.$$

Hence

$$\int_{\mathbb{R}} \left| \int_{\mathbb{R}} F(x, y) dy \right| dx = \sqrt{\pi} \int_{\mathbb{R}} e^{-x^2} dx = \pi.$$

Since  $F \geq 0$ ,

$$\int_{\mathbb{R}} \int_{\mathbb{R}} |F(x, y)| dy dx = \pi.$$

**Example 2: Polynomial times Gaussian.** Let

$$F(x, y) = (1 + x^2)(1 + y^2)e^{-x^2-y^2}.$$

A direct computation gives

$$\int_{\mathbb{R}} (1 + t^2)e^{-t^2} dt = \frac{3\sqrt{\pi}}{2}.$$

Thus

$$\int_{\mathbb{R}} \left| \int_{\mathbb{R}} F(x, y) dy \right| dx = \left( \frac{3\sqrt{\pi}}{2} \right)^2 = \frac{9\pi}{4},$$

and the same value is obtained on the right-hand side.

**Example 3: Trigonometric example.** On  $[0, \pi]$  let

$$F(x, y) = \sin^2 x \sin^2 y.$$

Since  $\int_0^\pi \sin^2 t dt = \frac{\pi}{2}$ ,

$$\int_0^\pi \left| \int_0^\pi F(x, y) dy \right| dx = \frac{\pi^2}{4} = \int_0^\pi \int_0^\pi F(x, y) dy dx.$$

These examples illustrate how regular, smooth functions behave exactly like simple functions when  $p = 1$  and  $F \geq 0$ : the inequality becomes an equality.

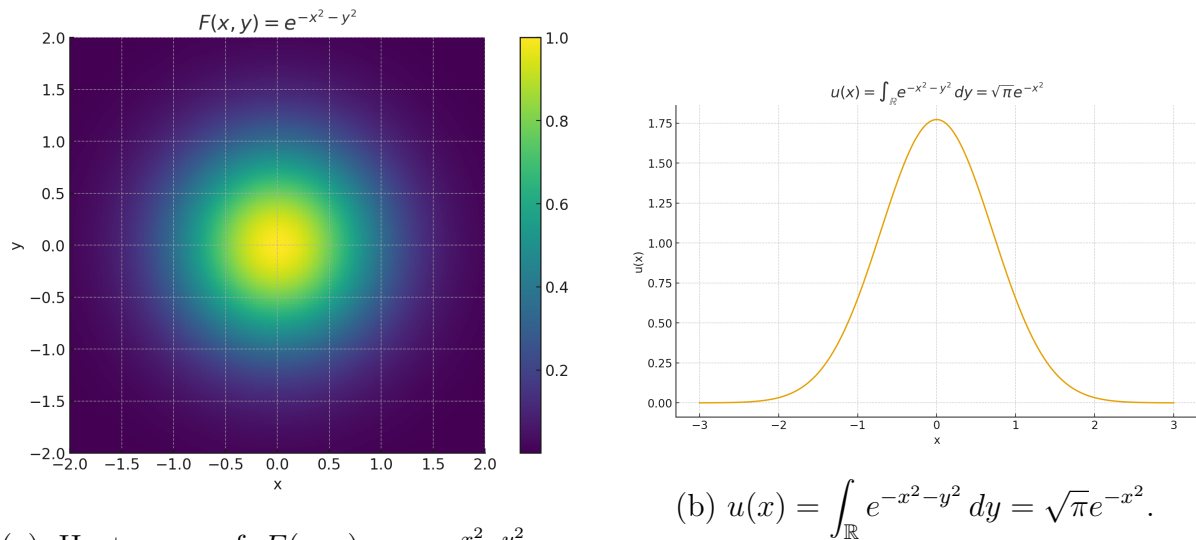


Figure 35: Gaussian example for Minkowski's integral inequality with  $p = 1$ : the area under  $u(x)$  equals the double integral of  $F(x, y)$ .

## How simple functions and regular functions are related

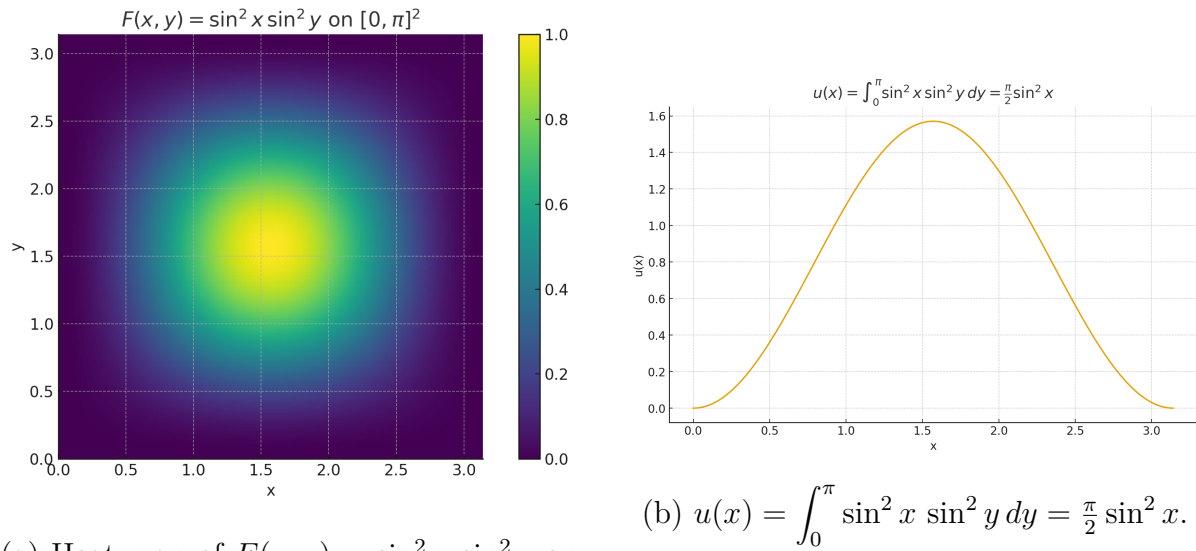
The Minkowski inequality (and many foundational results in integration theory) is first proved for *simple functions*, because these have a completely transparent structure:

- simple functions are finite linear combinations of indicator sets;
- all integrals reduce to finite sums;
- Tonelli–Fubini can be verified directly;
- no pathological behaviour can occur.

Once the inequality is established for simple functions, one extends it to general measurable functions (continuous,  $C^\infty$ , polynomials, Gaussians, trigonometric functions, etc.) by:

- approximation from below by increasing sequences of simple functions;
- the Monotone or Dominated Convergence Theorem;
- density of simple functions in  $L^p$ .

We now describe this relation step by step.



(a) Heat map of  $F(x, y) = \sin^2 x \sin^2 y$  on  $[0, \pi]^2$ .

Figure 36: Trigonometric example for Minkowski's inequality with  $p = 1$ : the integral of  $u(x)$  over  $[0, \pi]$  equals the double integral of  $F(x, y)$ .

### 1. Simple functions as the foundational case

A simple function has the form

$$F(x, y) = \sum_{k=1}^m a_k 1_{A_k}(x) 1_{B_k}(y), \quad a_k \geq 0.$$

For such  $F$ , all integrals become finite sums:

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x, y) \, dy \, dx = \sum_{k=1}^m a_k |A_k| |B_k|.$$

Thus Minkowski's inequality for  $p = 1$  is an equality for simple functions.

### 2. Approximation of general nonnegative measurable functions

Every nonnegative measurable function  $F$  on  $\mathbb{R}^n \times \mathbb{R}^n$  admits an increasing sequence of simple functions  $F_n$  such that

$$F_n(x, y) \uparrow F(x, y) \quad \text{for all } (x, y).$$

This construction is canonical and works for every  $F \geq 0$ .

### 3. The Monotone Convergence Theorem connects the two

By the Monotone Convergence Theorem,

$$\int F_n \rightarrow \int F, \quad \int \left| \int F_n(\cdot, y) \, dy \right| \rightarrow \int \left| \int F(\cdot, y) \, dy \right|.$$



Since Minkowski holds exactly for each  $F_n$ , taking limits shows it holds for  $F$  as well.

#### 4. Sign-changing regular functions

If  $F$  is a smooth or regular function that takes both signs, use the triangle inequality:

$$\left| \int_{\mathbb{R}^n} F(x, y) dy \right| \leq \int_{\mathbb{R}^n} |F(x, y)| dy.$$

Thus the sign-changing case reduces to the nonnegative case by replacing  $F$  with  $|F|$ .

#### 5. Regular functions (Gaussians, trigonometric, polynomials)

Regular functions such as

$$e^{-x^2-y^2}, \quad (1+x^2)(1+y^2)e^{-x^2-y^2}, \quad \sin^2 x \sin^2 y,$$

lie in every  $L^p$ , and satisfy

$$F_n \uparrow F \quad \text{or} \quad F_n \rightarrow F \text{ in } L^p.$$

Therefore all results established for simple functions extend to these regular functions by approximation.

**Summary.** Simple functions form the atomic building blocks. Every regular function is a limit of simple ones. Minkowski's inequality is verified first on the atoms, and then passed to the limit using monotone or dominated convergence.

#### (i) $L^p$ -spaces and Hölder conjugates.

- For  $1 \leq p < \infty$ , the space  $L^p(\mathbb{R}^n)$  consists of all measurable functions  $f$  such that

$$\int_{\mathbb{R}^n} |f(x)|^p dx < \infty.$$

- The Hölder conjugate exponent  $q$  of  $p$  is defined by

$$\frac{1}{p} + \frac{1}{q} = 1, \quad 1 < p < \infty.$$

- The pair  $(p, q)$  is fundamental: these are exactly the exponents for which Young's and Hölder's inequalities hold and for which the duality  $(L^p)^* \cong L^q$  is valid.

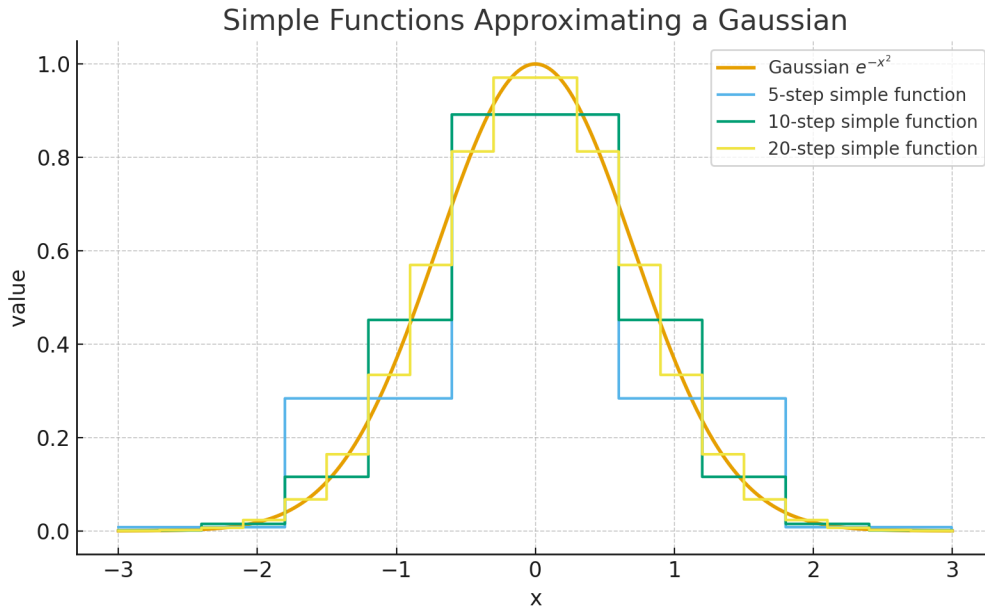


Figure 37: Approximation of the Gaussian  $e^{-x^2}$  by simple functions. The step functions with 5, 10, and 20 intervals converge monotonically to the Gaussian, illustrating how general integrable functions are obtained as limits of simple functions.

(ii) **Young's inequality as convex duality.**

- The function  $\phi(t) = e^t$  is convex and the logarithm is concave. The hint in part (b) suggests proving Young's inequality using concavity of  $\log$ .
- The inequality

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

is equivalent to the statement

$$\log(ab) \leq \log\left(\frac{a^p}{p} + \frac{b^q}{q}\right).$$

Because  $\log$  is concave, this inequality follows from Jensen's inequality applied to  $\log$ .

- More conceptually: the functions  $x \mapsto x^p/p$  and  $y \mapsto y^q/q$  are Legendre transforms of each other. Young's inequality expresses this duality.

**Young's inequality and concavity of  $\log$ .**

- The inequality

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

is equivalent (since  $\log$  is increasing) to the statement

$$\log(ab) \leq \log\left(\frac{a^p}{p} + \frac{b^q}{q}\right).$$

- Set  $t = p^{-1}$  and consider the function

$$\varphi(t) = \log[ta^p + (1-t)b^q].$$

Because  $\log$  is concave, Jensen's inequality yields

$$\log[ta^p + (1-t)b^q] \geq t \log(a^p) + (1-t) \log(b^q) = p t \log a + q(1-t) \log b.$$

Choosing  $t = 1/p$  (so  $1-t = 1/q$ ) gives

$$\log\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \log a + \log b = \log(ab),$$

which is precisely Young's inequality.

**Legendre-transform viewpoint.** Let

$$\Phi(x) = \frac{x^p}{p}, \quad x \geq 0.$$

Then  $\Phi$  is convex and its Legendre transform (convex conjugate) is

$$\Phi^*(y) = \sup_{x \geq 0} (xy - \Phi(x)) = \frac{y^q}{q}, \quad y \geq 0,$$

where  $q$  is the Hölder conjugate of  $p$ , i.e.  $\frac{1}{p} + \frac{1}{q} = 1$ . The Fenchel–Young inequality asserts that

$$xy \leq \Phi(x) + \Phi^*(y), \quad x, y \geq 0.$$

Substituting  $\Phi(x) = x^p/p$  and  $\Phi^*(y) = y^q/q$  yields

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q},$$

which is Young's inequality (with  $x = a$ ,  $y = b$ ). Thus Young's inequality expresses the Legendre duality between the functions  $x \mapsto x^p/p$  and  $y \mapsto y^q/q$ .

**(iii) Hölder's inequality.**

- Using Young's inequality, apply it pointwise to  $|f(x)|$  and  $|g(x)|$ :

$$|f(x)g(x)| \leq \frac{|f(x)|^p}{p} + \frac{|g(x)|^q}{q}.$$

- Integrating over  $\mathbb{R}^n$  gives

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \frac{1}{p} \int |f|^p + \frac{1}{q} \int |g|^q.$$

- If  $\|f\|_{L^p} = \|g\|_{L^q} = 1$ , the right side equals 1.
- Scaling  $f$  and  $g$  yields the general form of Hölder's inequality:

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \|f\|_{L^p} \|g\|_{L^q}.$$

- This identity shows that multiplication by  $g \in L^q$  defines a bounded linear functional on  $L^p$  with operator norm  $\|g\|_{L^q}$  — the starting point of  $L^p$  duality.

**(iv) Minkowski's inequality and  $L^p$ -norms.**

- The classical Minkowski inequality

$$\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}$$

is the triangle inequality for the  $L^p$ -norm. Hölder's inequality is the main tool in its proof.

- In this exercise we prove a *two-variable* version:

$$\left\| \int_{\mathbb{R}^n} F(\cdot, y) dy \right\|_{L^p(\mathbb{R}^n)} \leq \int_{\mathbb{R}^n} \|F(\cdot, y)\|_{L^p(\mathbb{R}^n)} dy.$$

- The key idea is to define

$$u(x) := \int_{\mathbb{R}^n} F(x, y) dy, \quad G(y) := |u(y)|^{p-1} \operatorname{sgn}(u(y)),$$

so that

$$\|u\|_{L^p}^p = \int_{\mathbb{R}^n} u(x)G(x) dx.$$

- Applying Hölder's inequality to the right-hand side gives

$$\|u\|_{L^p}^p \leq \|G\|_{L^q} \int_{\mathbb{R}^n} \left( \int_{\mathbb{R}^n} |F(x, y)|^p dx \right)^{1/p} dy,$$

from which Minkowski's integral inequality follows.

**(v) Simple functions and general measurable functions.**

- Parts (a)–(d) assume  $F$  is a *positive simple function*: a finite linear combination of characteristic functions. On these, one may freely permute and decompose integrals.
- For general nonnegative measurable  $F$ , approximate  $F$  from below by simple functions and use the Monotone Convergence Theorem.
- For complex-valued  $F$ , apply the inequality to  $|F|$ .
- This yields Minkowski's integral inequality in full generality for all  $1 \leq p < \infty$ .

**From Fenchel–Young to Hölder's inequality.** The Fenchel–Young inequality for the convex function  $\Phi(x) = x^p/p$  and its conjugate  $\Phi^*(y) = y^q/q$  reads

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}, \quad x, y \geq 0.$$

Applying this pointwise with  $x = |f(x)|$  and  $y = |g(x)|$  gives

$$|f(x)g(x)| \leq \frac{|f(x)|^p}{p} + \frac{|g(x)|^q}{q}, \quad x \in \mathbb{R}^n.$$

Integrating over  $\mathbb{R}^n$  yields

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \frac{1}{p} \int_{\mathbb{R}^n} |f(x)|^p dx + \frac{1}{q} \int_{\mathbb{R}^n} |g(x)|^q dx = \frac{\|f\|_{L^p}^p}{p} + \frac{\|g\|_{L^q}^q}{q}.$$

If  $\|f\|_{L^p}, \|g\|_{L^q} > 0$ , apply this to  $\tilde{f} = f/\|f\|_{L^p}$  and  $\tilde{g} = g/\|g\|_{L^q}$  to obtain

$$\int_{\mathbb{R}^n} |\tilde{f}(x)\tilde{g}(x)| dx \leq \frac{1}{p} + \frac{1}{q} = 1,$$

and hence

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \|f\|_{L^p} \|g\|_{L^q}.$$

This recovers Hölder's inequality directly from the Fenchel–Young inequality for  $\Phi(x) = x^p/p$ .

## Examples for Exercise 2.5

### (i) Example for Young's inequality

Take  $p = q = 2$  and  $a = 3, b = 4$ . Young's inequality states that

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Here

$$ab = 3 \cdot 4 = 12,$$

while

$$\frac{a^2}{2} + \frac{b^2}{2} = \frac{9}{2} + \frac{16}{2} = \frac{25}{2} = 12.5.$$

Thus

$$12 = ab \leq 12.5 = \frac{a^2}{2} + \frac{b^2}{2},$$

which illustrates the inequality for a specific choice of parameters.

### (ii) Example for Hölder's inequality

Let  $p = q = 2$  and define functions on  $\mathbb{R}$  by

$$f(x) = 1_{[0,1]}(x), \quad g(x) = x 1_{[0,1]}(x).$$

Then

$$f(x)g(x) = x 1_{[0,1]}(x),$$

and

$$\int_{\mathbb{R}} |f(x)g(x)| dx = \int_0^1 x dx = \frac{1}{2}.$$

The norms are

$$\|f\|_{L^2}^2 = \int_0^1 1^2 dx = 1 \implies \|f\|_{L^2} = 1,$$

and

$$\|g\|_{L^2}^2 = \int_0^1 x^2 dx = \frac{1}{3} \implies \|g\|_{L^2} = \frac{1}{\sqrt{3}}.$$

Thus

$$\int_{\mathbb{R}} |f(x)g(x)| dx = \frac{1}{2} \leq \frac{1}{\sqrt{3}} = \|f\|_{L^2} \|g\|_{L^2},$$

which confirms Hölder's inequality in this concrete case (with strict inequality).

### (iii) Example for Minkowski's integral inequality

Let  $F : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$  be given by

$$F(x, y) = 1_{[0,1]}(x) 1_{[0,1]}(y),$$

and take  $p = 2$ . Define

$$u(x) := \int_{\mathbb{R}} F(x, y) dy.$$

Then

$$u(x) = \int_{\mathbb{R}} 1_{[0,1]}(x) 1_{[0,1]}(y) dy = 1_{[0,1]}(x) \cdot 1,$$

so  $u(x) = 1$  for  $x \in [0, 1]$  and  $u(x) = 0$  otherwise. Hence

$$\left( \int_{\mathbb{R}} |u(x)|^2 dx \right)^{1/2} = \left( \int_0^1 1 dx \right)^{1/2} = 1.$$

On the other hand,

$$\int_{\mathbb{R}} \left( \int_{\mathbb{R}} |F(x, y)|^2 dx \right)^{1/2} dy$$

is computed as follows. Since  $F^2 = F$ ,

$$\int_{\mathbb{R}} |F(x, y)|^2 dx = \int_{\mathbb{R}} 1_{[0,1]}(x) 1_{[0,1]}(y) dx = 1_{[0,1]}(y) \cdot 1.$$

Thus

$$\left( \int_{\mathbb{R}} |F(x, y)|^2 dx \right)^{1/2} = 1_{[0,1]}(y),$$

and therefore

$$\int_{\mathbb{R}} \left( \int_{\mathbb{R}} |F(x, y)|^2 dx \right)^{1/2} dy = \int_0^1 1 dy = 1.$$

In this example both sides of Minkowski's integral inequality are equal to 1, so equality holds.

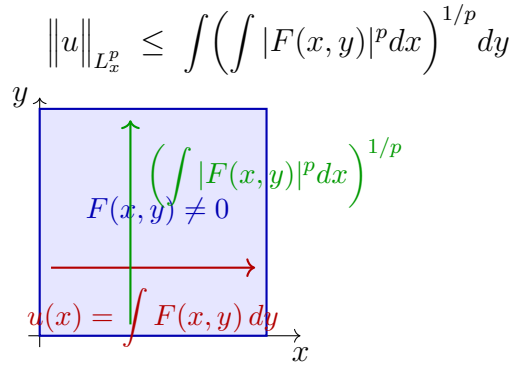


Figure 38: Geometric view of Minkowski's integral inequality. Horizontal integration in  $y$  gives  $u(x) = \int F(x, y) dy$ , whose  $L^p$ -norm appears on the left-hand side. For each fixed  $y$ , the vertical arrow represents the  $L^p$ -norm in  $x$  of the slice  $F(\cdot, y)$ ; these norms are then integrated over  $y$  on the right-hand side.

## Exercise 2.6. Distributions and their orders

- (a) Show that  $\delta_x$ , as defined in Example 7, is continuous and linear, hence a distribution. Find its order.
- (b) Show that  $T_f$ , as defined in Example 7, is continuous and linear, hence a distribution. Find its order.
- (c) By constructing a suitable sequence of smooth functions show that the order of  $\text{P.V.}\left(\frac{1}{x}\right)$  is one.

(a) **The Dirac mass  $\delta_x$ .** For fixed  $x \in \mathbb{R}^n$  the functional  $\delta_x$  is defined by

$$\delta_x(\varphi) := \varphi(x), \quad \varphi \in \mathcal{D}(\mathbb{R}^n).$$

It is clearly linear:

$$\delta_x(\alpha\varphi + \beta\psi) = (\alpha\varphi + \beta\psi)(x) = \alpha\varphi(x) + \beta\psi(x) = \alpha\delta_x(\varphi) + \beta\delta_x(\psi).$$

To show continuity and determine its order, let  $K \subset \mathbb{R}^n$  be compact and suppose  $\text{supp } \varphi \subset K$ . If  $x \notin K$  then  $\delta_x(\varphi) = 0$ . If  $x \in K$ , then

$$|\delta_x(\varphi)| = |\varphi(x)| \leq \sup_{y \in K} |\varphi(y)|.$$

Thus for each compact  $K$  we can choose  $C_K = 1$  and obtain

$$|\delta_x(\varphi)| \leq C_K \sup_K |\varphi|.$$

Hence  $\delta_x$  is a distribution of order 0.

**(b) The regular distribution  $T_f$ .** Let  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$  and define

$$T_f(\varphi) := \int_{\mathbb{R}^n} f(x) \varphi(x) dx.$$

Linearity is immediate from linearity of the integral. For continuity, fix a compact set  $K \subset \mathbb{R}^n$  and suppose  $\text{supp } \varphi \subset K$ . Then

$$|T_f(\varphi)| = \left| \int_K f(x) \varphi(x) dx \right| \leq \int_K |f(x)| |\varphi(x)| dx \leq \left( \int_K |f(x)| dx \right) \sup_K |\varphi(x)|.$$

Setting  $C_K := \int_K |f(x)| dx < \infty$ , we obtain

$$|T_f(\varphi)| \leq C_K \sup_K |\varphi|,$$

so  $T_f$  is a distribution of order 0.

**(c) The order of P.V.(1/x).** The principal value distribution P.V.(1/x) acts on  $\varphi \in \mathcal{D}(\mathbb{R})$  by

$$\langle \text{P.V.}(1/x), \varphi \rangle := \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx.$$

*Step 1:* P.V.(1/x) has order at most 1. One shows (by integration by parts on the truncated integrals and passing to the limit) that

$$\langle \text{P.V.}(1/x), \varphi \rangle = - \int_{\mathbb{R}} \log |x| \varphi'(x) dx.$$

Let  $K \subset \mathbb{R}$  be compact and  $\text{supp } \varphi \subset K$ . Then

$$\left| \langle \text{P.V.}(1/x), \varphi \rangle \right| \leq \int_K |\log |x|| |\varphi'(x)| dx \leq \left( \int_K |\log |x|| dx \right) \sup_K |\varphi'(x)|.$$

Thus there is a constant  $C_K$  such that

$$\left| \langle \text{P.V.}(1/x), \varphi \rangle \right| \leq C_K \sup_K |\varphi'|,$$

which means that P.V.(1/x) is a distribution of order  $\leq 1$ .

*Step 2:* P.V.(1/x) is not of order 0. Suppose, towards a contradiction, that it had order 0. Then for  $K = [-1, 1]$  there would exist  $C > 0$  such that for all  $\varphi \in \mathcal{D}(\mathbb{R})$  with  $\text{supp } \varphi \subset [-1, 1]$ ,

$$\left| \langle \text{P.V.}(1/x), \varphi \rangle \right| \leq C \sup_{[-1, 1]} |\varphi|. \quad (*)$$

Choose  $\eta \in C_c^\infty(\mathbb{R})$  with  $0 \leq \eta \leq 1$ ,  $\eta(x) = 0$  for  $x \leq 0$  and  $\eta(x) = 1$  for  $x \geq 1$ , and let  $\chi \in C_c^\infty((-1, 1))$  satisfy  $\chi \equiv 1$  on  $[-1/2, 1/2]$ . Define

$$\varphi_n(x) := \eta(nx) \chi(x).$$



Then  $\varphi_n \in C_c^\infty((-1, 1))$ ,  $0 \leq \varphi_n \leq 1$  and  $\sup |\varphi_n| = 1$  for all  $n$ . Moreover, for  $x \in [1/n, 1/2]$  we have  $\varphi_n(x) = 1$ .

For these test functions the principal value reduces to an improper integral:

$$\langle \text{P.V.}(1/x), \varphi_n \rangle = \int_0^1 \frac{\varphi_n(x)}{x} dx.$$

Hence

$$\langle \text{P.V.}(1/x), \varphi_n \rangle \geq \int_{1/n}^{1/2} \frac{1}{x} dx - \int_0^{1/n} \frac{|\varphi_n(x)|}{x} dx = \log n + \log\left(\frac{1}{2}\right) - R_n,$$

where  $R_n$  is a bounded remainder. Consequently

$$|\langle \text{P.V.}(1/x), \varphi_n \rangle| \rightarrow \infty \quad \text{as } n \rightarrow \infty,$$

while  $\sup_{[-1,1]} |\varphi_n| = 1$ . This contradicts (\*), so  $\text{P.V.}(1/x)$  cannot be of order 0.

Combining the two steps, we conclude that the order of  $\text{P.V.}(1/x)$  is exactly 1.

## Regular distributions, order, and singularity hierarchy

**Definition 0.3** (Regular distributions). Let  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ . The functional

$$T_f(\varphi) := \int_{\mathbb{R}^n} f(x) \varphi(x) dx, \quad \varphi \in \mathcal{D}(\mathbb{R}^n),$$

is called the *regular distribution* associated with  $f$ . Every regular distribution has order 0.

**Definition 0.4** (Order of a distribution). A distribution  $T \in \mathcal{D}'(\mathbb{R}^n)$  has *order*  $\leq m$  if for every compact  $K \subset \mathbb{R}^n$  there exists  $C_K > 0$  such that

$$|\langle T, \varphi \rangle| \leq C_K \sum_{|\alpha| \leq m} \sup_{x \in K} |\partial^\alpha \varphi(x)|, \quad \text{supp } \varphi \subset K.$$

The smallest such  $m$  is called the *order* of  $T$ .

## Examples and their orders

### 1. Dirac mass $\delta_x$ (order 0).

$$\delta_x(\varphi) := \varphi(x).$$

For compact  $K \ni x$ ,

$$|\delta_x(\varphi)| = |\varphi(x)| \leq \sup_K |\varphi|.$$

Thus  $\delta_x$  is linear, continuous, and of **order 0**.

**2. Regular distribution  $T_f$  (order 0).** If  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ ,

$$T_f(\varphi) = \int f(x)\varphi(x) dx.$$

If  $\text{supp } \varphi \subset K$ ,

$$|T_f(\varphi)| \leq \left( \int_K |f| \right) \sup_K |\varphi|.$$

Thus  $T_f$  is of **order 0**.

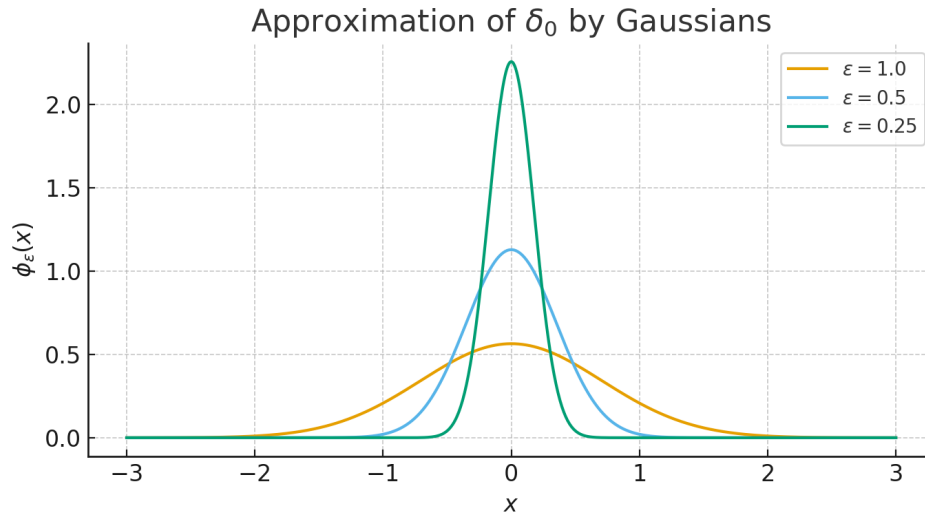


Figure 39: Approximation of the Dirac mass  $\delta_0$  by Gaussians  $\phi_\varepsilon(x) = \frac{1}{\sqrt{\pi\varepsilon}} e^{-x^2/\varepsilon^2}$  as  $\varepsilon \rightarrow 0$ . Each  $\phi_\varepsilon$  is a regular function, but the limit distribution has order 0.

**Example (b): regular distribution  $T_f$ .** Let  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$  and define

$$T_f(\varphi) = \int_{\mathbb{R}^n} f(x) \varphi(x) dx.$$

If  $\text{supp } \varphi \subset K$ , then

$$|T_f(\varphi)| \leq \int_K |f(x)| |\varphi(x)| dx \leq \left( \int_K |f(x)| dx \right) \sup_K |\varphi(x)|.$$

Hence  $T_f$  is linear, continuous, and of order 0.

**3. Principal value distribution P.V.(1/x) (order 1).**

$$\left\langle \text{P.V.}\left(\frac{1}{x}\right), \varphi \right\rangle := \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx.$$

One shows that

$$\langle \text{P.V.}(1/x), \varphi \rangle = - \int_{\mathbb{R}} \log |x| \varphi'(x) dx.$$

Thus

$$|\langle \text{P.V.}(1/x), \varphi \rangle| \leq C_K \sup_K |\varphi'|.$$

So P.V.(1/x) has **order**  $\leq 1$ . It is not of order 0, hence the order is **exactly 1**.

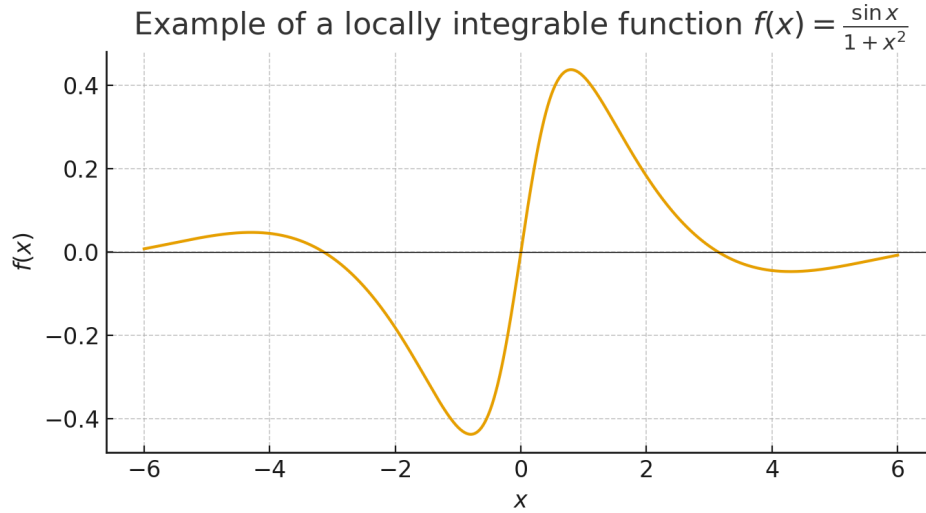


Figure 40: Example of a locally integrable function  $f(x) = \frac{\sin x}{1+x^2}$ . The associated regular distribution  $T_f$  has order 0: it only depends on the size of  $\varphi$ , not on its derivatives.

**4. Derivative of the Dirac mass  $\delta'_0$  (order 1).** By definition,

$$\langle \delta'_0, \varphi \rangle := -\varphi'(0).$$

Therefore, for compact  $K \ni 0$ ,

$$|\langle \delta'_0, \varphi \rangle| = |\varphi'(0)| \leq \sup_K |\varphi'|.$$

Hence  $\delta'_0$  has **order 1**.

**Singularity hierarchy: lower order means more regular**

functions  $\subset$  measures (order 0)  $\subset$  order 1 distributions  $\subset$  order 2 distributions  $\subset \dots$

- Order 0: only  $\sup |\varphi|$  matters (regular  $T_f$ ,  $\delta_x$ ).
- Order 1: depends on first derivatives of  $\varphi$  ( $\delta'$ , P.V.( $1/x$ )).
- Order  $m$ : depends on derivatives up to order  $m$  — increasingly singular.

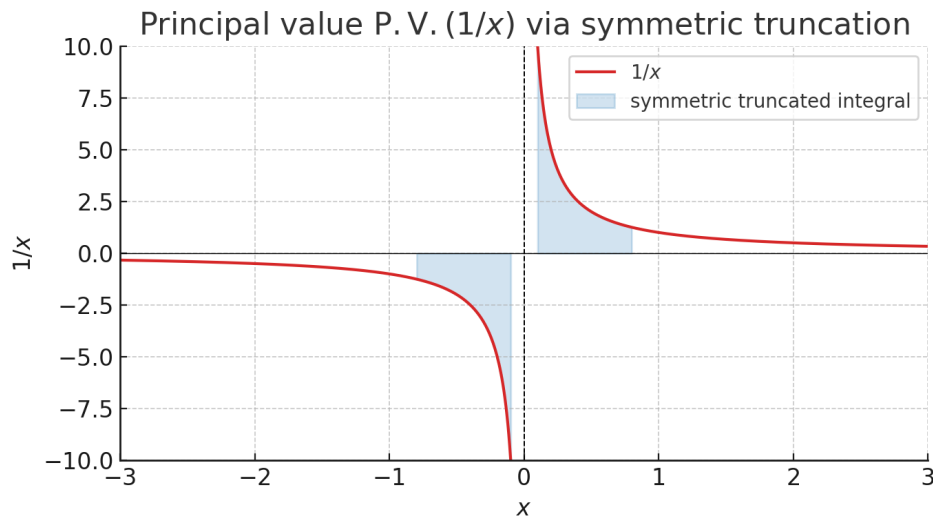


Figure 41: Graph of  $1/x$  with symmetric truncation around the origin, illustrating the principal value definition of  $P.V.(1/x)$ . The distribution cannot be represented by a locally integrable function, and it has order 1, depending on first derivatives of test functions.

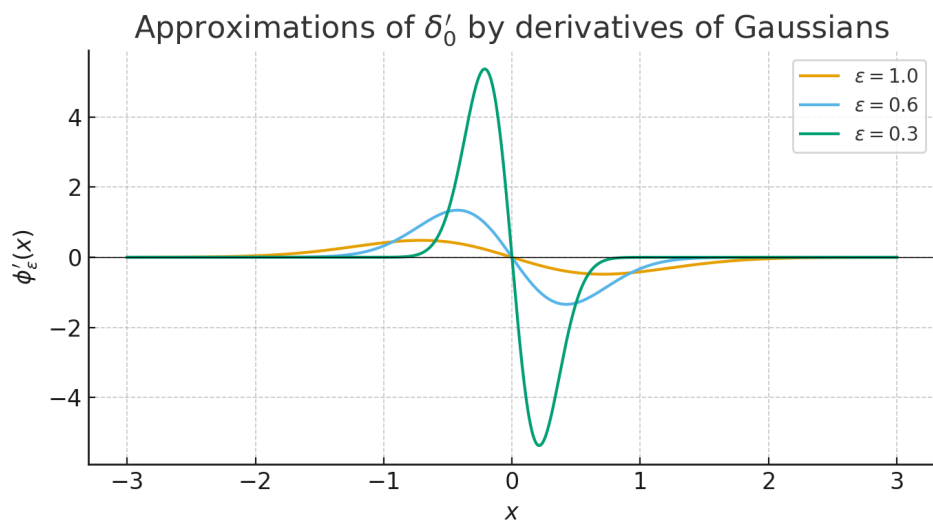


Figure 42: Approximations of the derivative of the Dirac mass  $\delta'_0$  by derivatives of Gaussians  $\phi'_\varepsilon(x) = -(2x/\varepsilon^2)\phi_\varepsilon(x)$ . The positive and negative lobes concentrate near 0, consistent with  $\langle \delta'_0, \varphi \rangle = -\varphi'(0)$ .

**Exercise 2.7. Local mollification in  $L^1_{\text{loc}}$** 

Suppose  $f \in L^1_{\text{loc}}(\Omega)$ . Let  $\phi_\varepsilon$  be the standard mollifier from Theorem 1.13 and, for  $x$  with  $d(x, \partial\Omega) \geq \varepsilon$ , define

$$f_\varepsilon(x) := T_f[T_x \check{\phi}_\varepsilon],$$

where  $\check{\phi}_\varepsilon(y) := \phi_\varepsilon(-y)$  and  $T_x \psi(y) := \psi(y - x)$ .

Show that for any compact  $K \subset \Omega$ ,

$$\|f_\varepsilon - f\|_{L^1(K)} \longrightarrow 0 \quad \text{as } \varepsilon \rightarrow 0.$$

*Hint:* Follow the proof of Theorem 2.2, but use part (b) of Theorem 1.13.

**Solution**

First note that for  $x$  with  $d(x, \partial\Omega) \geq \varepsilon$ ,

$$\begin{aligned} f_\varepsilon(x) &= T_f[T_x \check{\phi}_\varepsilon] = \int_{\Omega} f(y) (T_x \check{\phi}_\varepsilon)(y) dy \\ &= \int_{\Omega} f(y) \check{\phi}_\varepsilon(y - x) dy = \int_{\Omega} f(y) \phi_\varepsilon(x - y) dy. \end{aligned}$$

Thus  $f_\varepsilon(x) = (f * \phi_\varepsilon)(x)$  wherever  $B(x, \varepsilon) \subset \Omega$ .

Let  $K \subset \Omega$  be compact. There exists an open set  $U$  with

$$K \subset U, \quad \overline{U} \subset \Omega.$$

Choose  $\chi \in C_c^\infty(\Omega)$  such that  $\chi \equiv 1$  on  $U$ . Extend  $\chi f$  by 0 to a function in  $L^1(\mathbb{R}^n)$ .

For  $\varepsilon > 0$  sufficiently small we have  $\varepsilon < \text{dist}(K, \partial\Omega)$ , so that for all  $x \in K$ ,

$$f_\varepsilon(x) = \int_{\Omega} f(y) \phi_\varepsilon(x - y) dy = \int_{\Omega} \chi(y) f(y) \phi_\varepsilon(x - y) dy = ((\chi f) * \phi_\varepsilon)(x).$$

Hence, for such  $\varepsilon$ ,

$$\|f_\varepsilon - f\|_{L^1(K)} = \|(\chi f) * \phi_\varepsilon - \chi f\|_{L^1(K)} \leq \|(\chi f) * \phi_\varepsilon - \chi f\|_{L^1(\mathbb{R}^n)}.$$

By Theorem 1.13(b), since  $\chi f \in L^1(\mathbb{R}^n)$ , we have

$$\|(\chi f) * \phi_\varepsilon - \chi f\|_{L^1(\mathbb{R}^n)} \rightarrow 0 \quad \text{as } \varepsilon \rightarrow 0.$$

Therefore  $\|f_\varepsilon - f\|_{L^1(K)} \rightarrow 0$  as  $\varepsilon \rightarrow 0$ , which shows that  $f_\varepsilon \rightarrow f$  in  $L^1_{\text{loc}}(\Omega)$ .

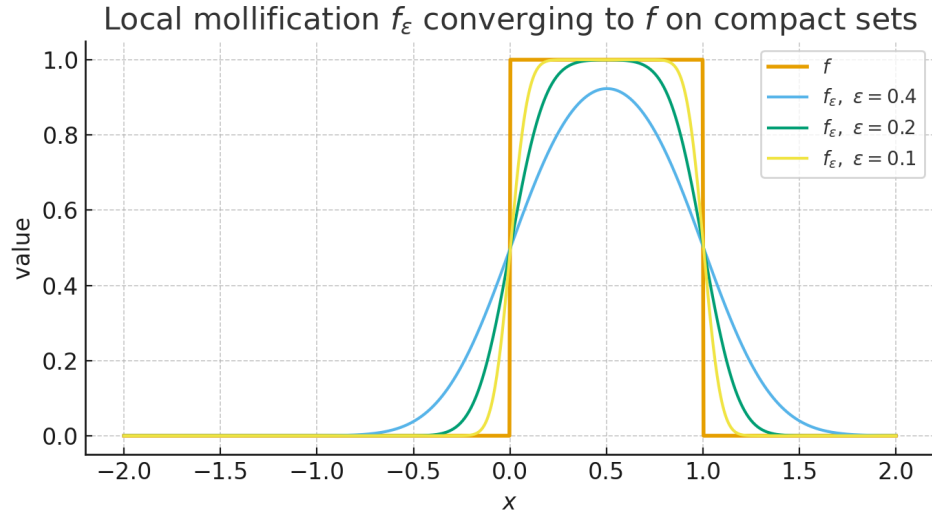


Figure 43: Example of local mollification. Here  $f(x) = 1$  on  $(0, 1)$  and 0 elsewhere, and  $f_\varepsilon = f * \phi_\varepsilon$  is plotted for several values of  $\varepsilon$ . On compact sets away from the boundary,  $f_\varepsilon$  converges to  $f$  in  $L^1$ .

## Mollifiers, test functions and tempered test functions

**Standard mollifiers as test functions.** A *standard mollifier* is built from a function  $\phi \in C_c^\infty(\mathbb{R}^n)$  such that

- $\phi \geq 0$ ,
- $\text{supp } \phi \subset B(0, 1)$ ,
- $\int_{\mathbb{R}^n} \phi(x) dx = 1$ .

For  $\varepsilon > 0$  we set

$$\phi_\varepsilon(x) := \frac{1}{\varepsilon^n} \phi\left(\frac{x}{\varepsilon}\right).$$

Each  $\phi_\varepsilon$  is smooth and compactly supported, hence

$$\phi_\varepsilon \in C_c^\infty(\mathbb{R}^n) = \mathcal{D}(\mathbb{R}^n).$$

Thus, in the framework of  $\mathcal{D}'(\mathbb{R}^n)$ , *standard mollifiers are particular test functions* which additionally satisfy the approximate identity properties

$$\phi_\varepsilon \geq 0, \quad \int \phi_\varepsilon = 1, \quad \phi_\varepsilon \rightarrow \delta_0 \text{ in } \mathcal{D}'.$$

**Gaussian mollifiers.** One often also uses the Gaussian kernels

$$\eta_\varepsilon(x) := \frac{1}{(2\pi\varepsilon^2)^{n/2}} e^{-\frac{|x|^2}{2\varepsilon^2}}, \quad \varepsilon > 0.$$

These functions are smooth and rapidly decaying, but not compactly supported. Hence

$$\eta_\varepsilon \notin \mathcal{D}(\mathbb{R}^n),$$

where compact support is required, but

$$\eta_\varepsilon \in \mathcal{S}(\mathbb{R}^n),$$

the Schwartz space of rapidly decreasing smooth functions. They also form an approximate identity, now in the sense of tempered distributions.

**Relation to tempered test functions and tempered distributions.** Recall that

$$\mathcal{D}(\mathbb{R}^n) = C_c^\infty(\mathbb{R}^n)$$

is the test function space for *distributions*  $\mathcal{D}'(\mathbb{R}^n)$ , while

$$\mathcal{S}(\mathbb{R}^n)$$

is the test function space for *tempered distributions*  $\mathcal{S}'(\mathbb{R}^n)$  (distributions of at most polynomial growth).

- Compactly supported mollifiers  $\phi_\varepsilon$  live in  $\mathcal{D}(\mathbb{R}^n)$  and are suitable for regularizing arbitrary distributions in  $\mathcal{D}'(\mathbb{R}^n)$  (at least away from the boundary in a domain).
- Gaussian-type mollifiers  $\eta_\varepsilon$  live in  $\mathcal{S}(\mathbb{R}^n)$  and are particularly well adapted to the theory of tempered distributions  $\mathcal{S}'(\mathbb{R}^n)$ , since convolution with  $\eta_\varepsilon$  preserves temperateness and interacts nicely with the Fourier transform.

In summary:

“standard mollifier” in Theorem 1.13  $\in \mathcal{D}$ ,      Gaussian mollifier  $\in \mathcal{S}$ ,

and both act as approximate identities in their respective distribution spaces  $\mathcal{D}'$  and  $\mathcal{S}'$ .

## Mollifiers acting on distributions and tempered distributions

**Lemma 0.5** (Mollifiers in  $\mathcal{D}'(\mathbb{R}^n)$ ). *Let  $(\phi_\varepsilon)_{\varepsilon>0} \subset \mathcal{D}(\mathbb{R}^n)$  be a standard approximate identity as in Theorem 1.13, and let  $T \in \mathcal{D}'(\mathbb{R}^n)$  be a distribution. Define the convolution  $\phi_\varepsilon * T$  by*

$$\langle \phi_\varepsilon * T, \psi \rangle := \langle T, \check{\phi}_\varepsilon * \psi \rangle, \quad \psi \in \mathcal{D}(\mathbb{R}^n),$$

*where  $\check{\phi}_\varepsilon(x) := \phi_\varepsilon(-x)$ . Then  $\phi_\varepsilon * T \in C^\infty(\mathbb{R}^n)$  for every  $\varepsilon > 0$ , and*

$$\phi_\varepsilon * T \longrightarrow T \quad \text{in } \mathcal{D}'(\mathbb{R}^n) \text{ as } \varepsilon \rightarrow 0.$$

*Proof.* First, for each fixed  $x$ , the function  $y \mapsto \phi_\varepsilon(x - y)$  is a test function, so

$$(\phi_\varepsilon * T)(x) := \langle T, \phi_\varepsilon(x - \cdot) \rangle$$

defines a smooth function of  $x$ . The distributional formulation is

$$\langle \phi_\varepsilon * T, \psi \rangle = \langle T, \check{\phi}_\varepsilon * \psi \rangle, \quad \psi \in \mathcal{D}(\mathbb{R}^n).$$

By Theorem 1.13(b),  $(\check{\phi}_\varepsilon)_{\varepsilon>0}$  is also an approximate identity, so for each  $\psi \in \mathcal{D}$  we have

$$\check{\phi}_\varepsilon * \psi \longrightarrow \psi \quad \text{in } \mathcal{D}(\mathbb{R}^n).$$

Since  $T$  is continuous on  $\mathcal{D}$ , it follows that

$$\langle \phi_\varepsilon * T, \psi \rangle = \langle T, \check{\phi}_\varepsilon * \psi \rangle \longrightarrow \langle T, \psi \rangle \quad (\varepsilon \rightarrow 0).$$

This is precisely the convergence  $\phi_\varepsilon * T \rightarrow T$  in  $\mathcal{D}'(\mathbb{R}^n)$ . □

**Lemma 0.6** (Mollifiers in  $\mathcal{S}'(\mathbb{R}^n)$ ). *Let  $(\eta_\varepsilon)_{\varepsilon>0} \subset \mathcal{S}(\mathbb{R}^n)$  be a Schwartz approximate identity (e.g. Gaussian mollifiers), and let  $U \in \mathcal{S}'(\mathbb{R}^n)$  be a tempered distribution. Define*

$$\langle \eta_\varepsilon * U, \psi \rangle := \langle U, \check{\eta}_\varepsilon * \psi \rangle, \quad \psi \in \mathcal{S}(\mathbb{R}^n),$$

*with  $\check{\eta}_\varepsilon(x) := \eta_\varepsilon(-x)$ . Then  $\eta_\varepsilon * U \in \mathcal{S}(\mathbb{R}^n)$  for each  $\varepsilon > 0$ , and*

$$\eta_\varepsilon * U \longrightarrow U \quad \text{in } \mathcal{S}'(\mathbb{R}^n) \text{ as } \varepsilon \rightarrow 0.$$

*Proof.* For each  $\varepsilon > 0$  and  $\psi \in \mathcal{S}$ , the convolution  $\check{\eta}_\varepsilon * \psi$  belongs to  $\mathcal{S}$  and, by the approximate identity property in the Schwartz topology,

$$\check{\eta}_\varepsilon * \psi \longrightarrow \psi \quad \text{in } \mathcal{S}(\mathbb{R}^n).$$

Since  $U$  is continuous on  $\mathcal{S}$ , we obtain

$$\langle \eta_\varepsilon * U, \psi \rangle = \langle U, \check{\eta}_\varepsilon * \psi \rangle \longrightarrow \langle U, \psi \rangle.$$

This is exactly the convergence  $\eta_\varepsilon * U \rightarrow U$  in  $\mathcal{S}'(\mathbb{R}^n)$ . □

### Fourier transform viewpoint for Gaussian mollifiers

Let  $\eta_\varepsilon \in \mathcal{S}(\mathbb{R}^n)$  be the Gaussian mollifier

$$\eta_\varepsilon(x) = \frac{1}{(2\pi\varepsilon^2)^{n/2}} e^{-\frac{|x|^2}{2\varepsilon^2}}.$$

Up to normalisation conventions, its Fourier transform is again a Gaussian:

$$\widehat{\eta_\varepsilon}(\xi) = e^{-\frac{\varepsilon^2|\xi|^2}{2}}.$$

In particular, for any tempered distribution  $U \in \mathcal{S}'(\mathbb{R}^n)$  we have

$$\mathcal{F}(\eta_\varepsilon * U) = \widehat{\eta_\varepsilon * U} = \widehat{\eta_\varepsilon} \widehat{U} = e^{-\frac{\varepsilon^2|\xi|^2}{2}} \widehat{U}(\xi).$$



Thus convolution with the Gaussian mollifier corresponds in the frequency domain to multiplication by the *low-pass filter*

$$m_\varepsilon(\xi) := e^{-\frac{\varepsilon^2|\xi|^2}{2}}.$$

For each fixed  $\varepsilon > 0$ , the factor  $m_\varepsilon(\xi)$  damps high frequencies strongly, so that  $\eta_\varepsilon * U$  is a very smooth (indeed Schwartz) function. As  $\varepsilon \rightarrow 0$ ,

$$m_\varepsilon(\xi) \longrightarrow 1 \quad \text{for all } \xi \in \mathbb{R}^n,$$

so  $\widehat{\eta_\varepsilon * U} \rightarrow \widehat{U}$  in  $\mathcal{S}'$ , which is equivalent to the convergence

$$\eta_\varepsilon * U \longrightarrow U \quad \text{in } \mathcal{S}'(\mathbb{R}^n).$$

In summary, Gaussian mollification can be viewed either as *smoothing in physical space* (convolution with  $\eta_\varepsilon$ ) or as *frequency cut-off* (multiplication by  $e^{-\varepsilon^2|\xi|^2/2}$ ) in the Fourier domain.

**Example A: regular tempered distribution  $T_f$ .** Let  $f(x) = \frac{\sin x}{1+x^2}$ . Then  $f \in L^1_{\text{loc}}(\mathbb{R})$  and even  $f \in \mathcal{S}(\mathbb{R})$ , so the functional

$$\langle T_f, \varphi \rangle := \int_{\mathbb{R}} f(x) \varphi(x) dx, \quad \varphi \in \mathcal{D}(\mathbb{R}),$$

defines a regular distribution of order 0. Since  $f$  is rapidly decreasing,  $T_f$  is a tempered distribution as well ( $T_f \in \mathcal{S}'(\mathbb{R})$ ).

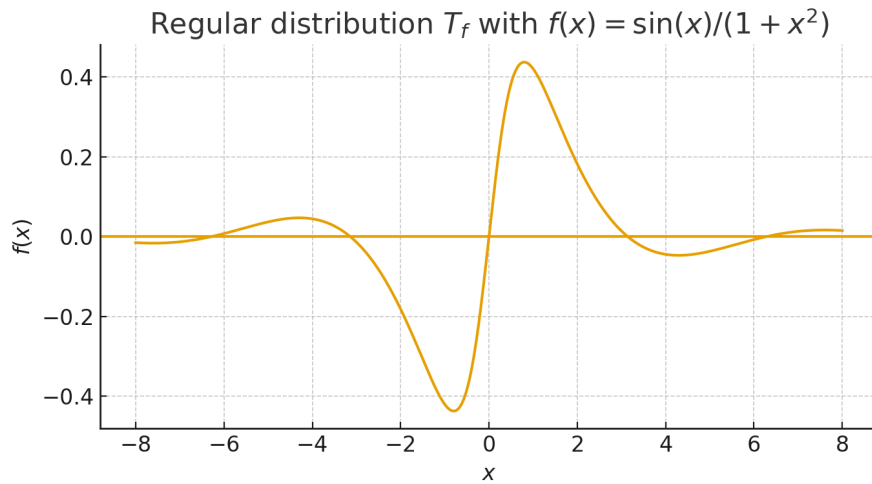


Figure 44: Graph of  $f(x) = \sin(x)/(1+x^2)$ , inducing the regular tempered distribution  $T_f$ .

**Concrete action of a regular distribution.** Let

$$f(x) = \frac{\sin x}{1 + x^2},$$

and let  $\varphi \in C_c^\infty(\mathbb{R})$  be a smooth bump supported in  $(-2, 2)$  (for instance  $\varphi(x) = C \exp(-1/(1 - (x/2)^2))$  for  $|x| < 2$  and 0 otherwise, with  $C$  chosen so that  $\int \varphi = 1$ ). The regular distribution  $T_f$  acts by

$$\langle T_f, \varphi \rangle = \int_{\mathbb{R}} f(x) \varphi(x) dx.$$

The integrand is the pointwise product  $f(x)\varphi(x)$ .

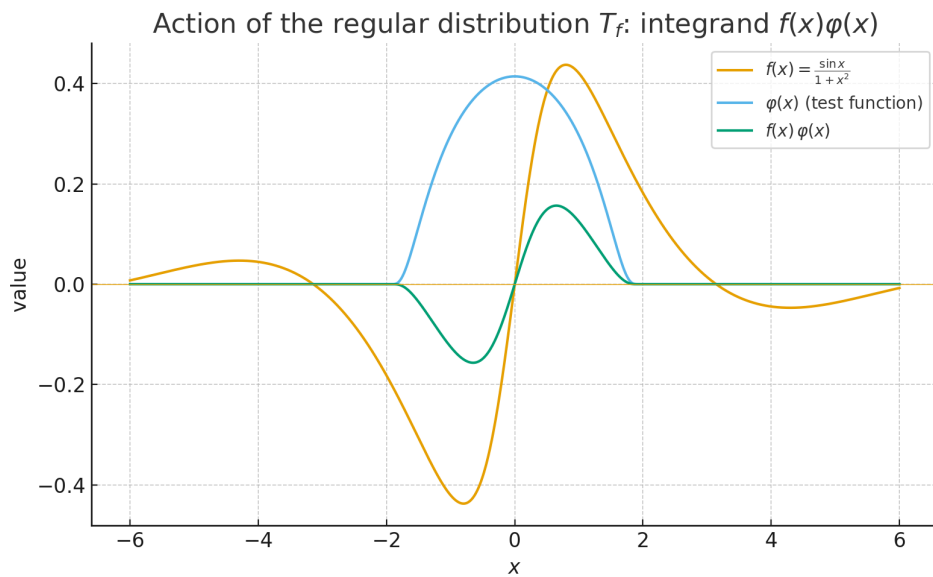


Figure 45: The function  $f(x) = \sin(x)/(1 + x^2)$ , a smooth compactly supported test function  $\varphi(x)$ , and their product  $f(x)\varphi(x)$ , which is the integrand in  $\langle T_f, \varphi \rangle = \int f(x)\varphi(x) dx$ .

**Example B: Heaviside function and Dirac mass.** Define the Heaviside function

$$H(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

It induces a regular distribution  $T_H(\varphi) = \int_{\mathbb{R}} H(x)\varphi(x) dx$  of order 0. Using integration by parts one checks that

$$\langle H', \varphi \rangle := -\langle H, \varphi' \rangle = \varphi(0) = \langle \delta_0, \varphi \rangle,$$

so in  $\mathcal{D}'(\mathbb{R})$  we have  $H' = \delta_0$ . Both  $T_H$  and  $\delta_0$  are tempered distributions.

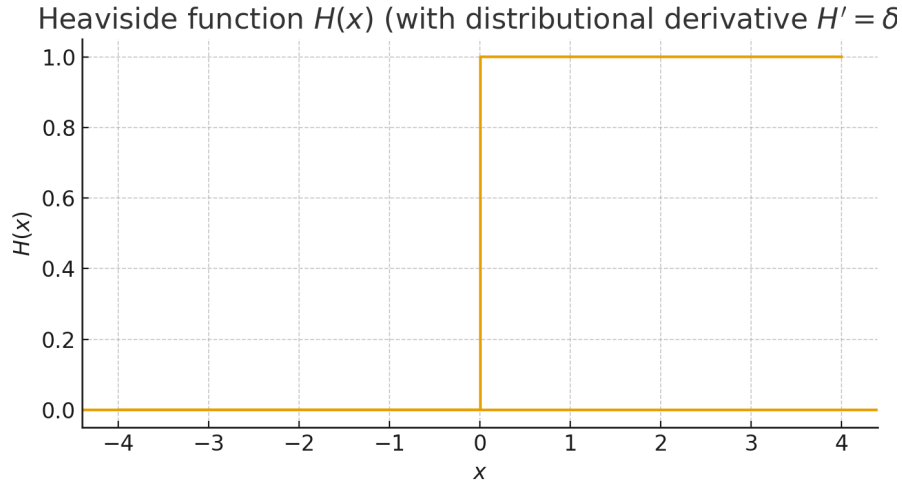


Figure 46: Heaviside function  $H(x)$ . Its distributional derivative is the Dirac mass  $\delta_0$ .

**Example C: principal value  $\text{P.V.}(1/x)$ .** The distribution

$$\langle \text{P.V.}(1/x), \varphi \rangle := \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx$$

is not given by a locally integrable function (since  $1/x$  is not integrable near 0), hence it is not regular. However, the growth of  $1/x$  at infinity is only of order  $|x|^{-1}$ , so the functional is continuous on the Schwartz space  $\mathcal{S}(\mathbb{R})$ , and thus

$$\text{P.V.}(1/x) \in \mathcal{S}'(\mathbb{R})$$

is a tempered distribution.

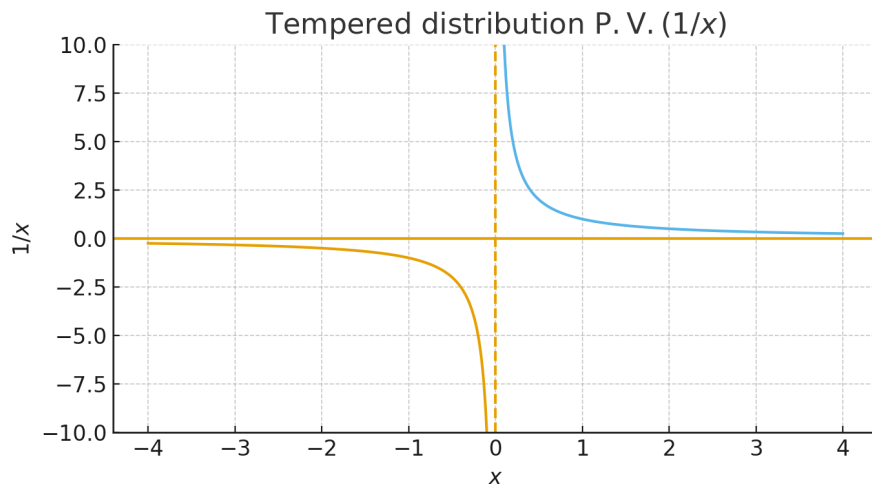


Figure 47: Graph of  $1/x$  with a dashed vertical line at 0, indicating the singularity underlying the tempered distribution  $\text{P.V.}(1/x)$ .

| Object                                     | Regular?                     | Tempered?            | Order |
|--------------------------------------------|------------------------------|----------------------|-------|
| $f \in L^1_{\text{loc}}$ , moderate growth | yes ( $T_f$ )                | yes                  | 0     |
| $H$ (Heaviside)                            | yes                          | yes                  | 0     |
| $\delta_0$                                 | no (not $L^1_{\text{loc}}$ ) | yes                  | 0     |
| $\delta'_0$                                | no                           | yes                  | 1     |
| P.V.( $1/x$ )                              | no                           | yes                  | 1     |
| $e^{x^2}$ (as $T_f$ )                      | yes                          | no (too fast growth) | 0     |

**Example D: Dirac mass  $\delta_0$ .** The Dirac distribution acts by evaluation:

$$\langle \delta_0, \varphi \rangle = \varphi(0).$$

The following figure shows a test function  $\varphi$  together with the highlighted value  $\varphi(0)$ .

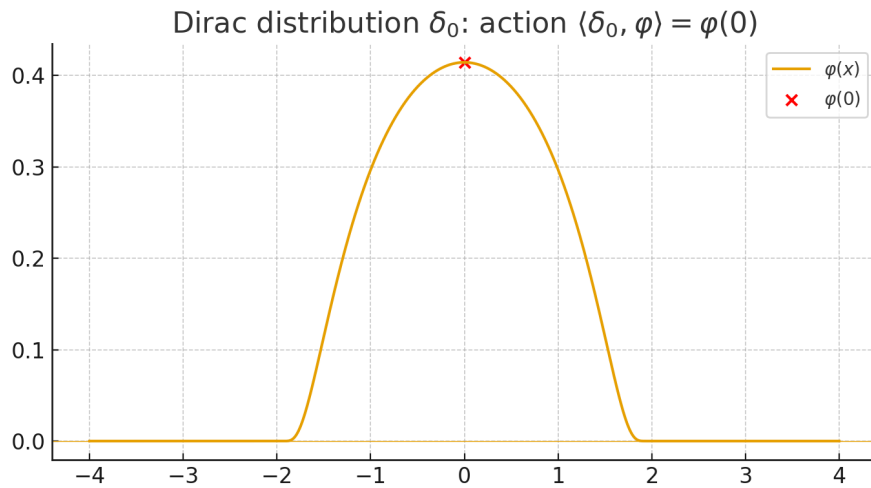


Figure 48: Test function  $\varphi$  and the point  $\varphi(0)$  corresponding to the action of  $\delta_0$ .

**Example E: derivative of Dirac  $\delta'_0$ .** The action of the derivative of the Dirac mass is

$$\langle \delta'_0, \varphi \rangle = -\varphi'(0).$$

The next figure displays  $\varphi$  and  $\varphi'$  and highlights  $\varphi'(0)$ .

**Example F: principal value P.V.( $1/x$ ).** The distribution P.V.( $1/x$ ) is defined by

$$\langle \text{P.V.}(1/x), \varphi \rangle = \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx.$$

The graph below shows  $\varphi(x)$  and the truncated integrand  $\varphi(x)/x$  for  $|x| > \varepsilon$ .

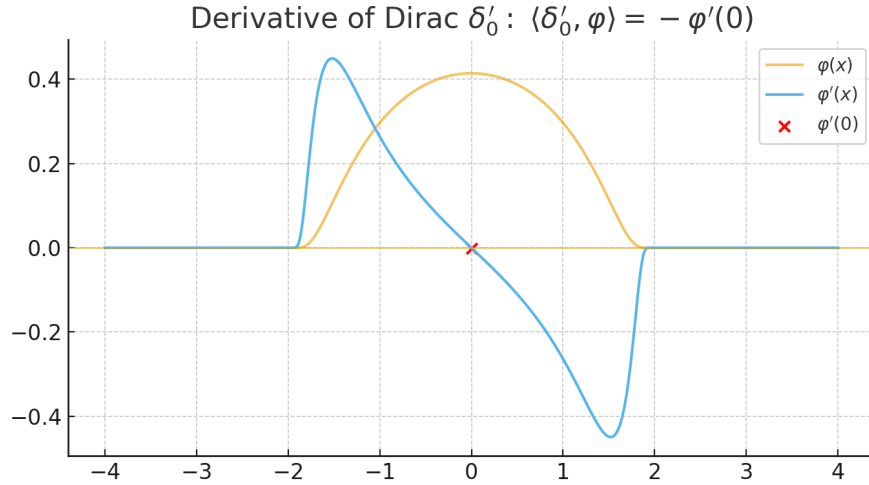


Figure 49: Test function  $\varphi$ , its derivative  $\varphi'$ , and the highlighted value  $\varphi'(0)$  corresponding to the action of  $\delta'_0$ .

### Exercise 2.8. Algebra and differentiation of regular distributions

(a) Show that if  $f_1, f_2 \in C^0(\Omega)$  and  $a \in C^\infty(\Omega)$ , then

$$aT_{f_1} + T_{f_2} = T_{af_1+f_2}.$$

(b) Show that if  $f \in C^k(\Omega)$  then

$$D^\alpha T_f = T_{D^\alpha f} \quad \text{for all multiindices } \alpha \text{ with } |\alpha| \leq k.$$

Deduce that  $\iota \circ D^\alpha = D^\alpha \circ \iota$ , where  $\iota : C^k(\Omega) \rightarrow \mathcal{D}'(\Omega)$  is the natural embedding  $f \mapsto T_f$ .

(c) Deduce that if  $f \in C^k(\Omega)$  then

$$\sum_{|\alpha| \leq k} a_\alpha D^\alpha T_f = T_{Lf},$$

where

$$Lf := \sum_{|\alpha| \leq k} a_\alpha D^\alpha f$$

and  $a_\alpha \in C^\infty(\Omega)$ .

Let  $f \in C^0(\Omega)$  and

$$T_f(\varphi) := \int_{\Omega} f(x) \varphi(x) dx.$$

For  $a \in C^\infty(\Omega)$  and  $T \in \mathcal{D}'(\Omega)$  define  $(aT)(\varphi) := T(a\varphi)$ . For a multiindex  $\alpha$ ,

$$\langle D^\alpha T, \varphi \rangle := (-1)^{|\alpha|} \langle T, D^\alpha \varphi \rangle.$$

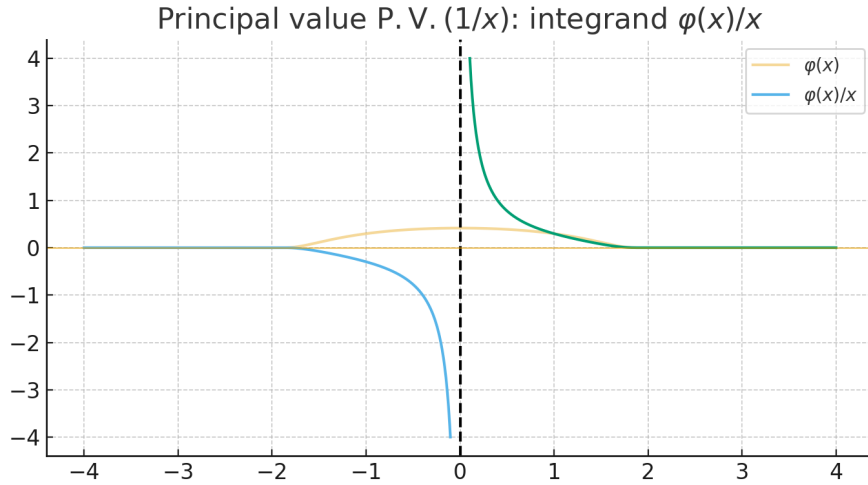


Figure 50: Test function  $\varphi(x)$  and the integrand  $\varphi(x)/x$  used in the principal value distribution.

(a) If  $f_1, f_2 \in C^0(\Omega)$  and  $a \in C^\infty(\Omega)$ , show

$$aT_{f_1} + T_{f_2} = T_{af_1+f_2}.$$

**solution**

For  $\varphi \in \mathcal{D}(\Omega)$ ,

$$\begin{aligned} \langle aT_{f_1} + T_{f_2}, \varphi \rangle &= \langle T_{f_1}, a\varphi \rangle + \langle T_{f_2}, \varphi \rangle \\ &= \int_{\Omega} f_1(x)a(x)\varphi(x) dx + \int_{\Omega} f_2(x)\varphi(x) dx \\ &= \int_{\Omega} (af_1 + f_2)(x)\varphi(x) dx = \langle T_{af_1+f_2}, \varphi \rangle. \end{aligned}$$

Thus  $aT_{f_1} + T_{f_2} = T_{af_1+f_2}$ .

(b) If  $f \in C^k(\Omega)$ , show that for  $|\alpha| \leq k$ ,

$$D^\alpha T_f = T_{D^\alpha f}.$$

Deduce that  $\iota \circ D^\alpha = D^\alpha \circ \iota$  for the embedding  $\iota : C^k(\Omega) \rightarrow \mathcal{D}'(\Omega)$ ,  $\iota(f) = T_f$ .

**solution**

For  $\varphi \in \mathcal{D}(\Omega)$ ,

$$\langle D^\alpha T_f, \varphi \rangle = (-1)^{|\alpha|} \langle T_f, D^\alpha \varphi \rangle = (-1)^{|\alpha|} \int_{\Omega} f D^\alpha \varphi.$$

Since  $f \in C^k(\Omega)$  and  $\varphi$  has compact support, we may integrate by parts  $|\alpha|$  times, with no boundary term:

$$(-1)^{|\alpha|} \int_{\Omega} f D^{\alpha} \varphi = \int_{\Omega} D^{\alpha} f \varphi = \langle T_{D^{\alpha} f}, \varphi \rangle.$$

Hence  $D^{\alpha} T_f = T_{D^{\alpha} f}$ .

For the embedding  $\iota$ , we have

$$(\iota \circ D^{\alpha})(f) = T_{D^{\alpha} f} = D^{\alpha} T_f = (D^{\alpha} \circ \iota)(f),$$

so  $\iota \circ D^{\alpha} = D^{\alpha} \circ \iota$  on  $C^k(\Omega)$ .

(c) Let  $f \in C^k(\Omega)$  and

$$Lf := \sum_{|\alpha| \leq k} a_{\alpha} D^{\alpha} f, \quad a_{\alpha} \in C^{\infty}(\Omega).$$

Show that

$$\sum_{|\alpha| \leq k} a_{\alpha} D^{\alpha} T_f = T_{Lf}.$$

**solution**

By part (b),  $D^{\alpha} T_f = T_{D^{\alpha} f}$ . By part (a),  $a_{\alpha} T_{D^{\alpha} f} = T_{a_{\alpha} D^{\alpha} f}$ , hence

$$\sum_{|\alpha| \leq k} a_{\alpha} D^{\alpha} T_f = \sum_{|\alpha| \leq k} T_{a_{\alpha} D^{\alpha} f} = T_{\sum_{|\alpha| \leq k} a_{\alpha} D^{\alpha} f} = T_{Lf}.$$

## Exercise 2.9. Convolution of regular and general distributions

(a) Show that for  $f, g \in C_0^{\infty}(\mathbb{R}^n)$ ,

$$T_{f * g} = T_f * T_g.$$

(b) Show that convolution is linear in both of its arguments, i.e. if  $u_i \in \mathcal{D}'(\mathbb{R}^n)$  and  $u_3, u_4$  have compact support, then for any  $a \in \mathbb{C}$ ,

$$(u_1 + au_2) * u_3 = u_1 * u_3 + a u_2 * u_3,$$

and

$$u_1 * (u_3 + au_4) = u_1 * u_3 + a u_1 * u_4.$$

## Solution

(a) Let  $f, g \in C_0^\infty(\mathbb{R}^n)$ . We show that  $T_{f*g} = T_f * T_g$ .

For any  $\varphi \in \mathcal{D}(\mathbb{R}^n)$ ,

$$\begin{aligned}
 \langle T_f * T_g, \varphi \rangle &= \left\langle T_f, x \mapsto \langle T_g, \varphi(x + \cdot) \rangle \right\rangle \\
 &= \int_{\mathbb{R}^n} f(x) \left( \int_{\mathbb{R}^n} g(y) \varphi(x + y) dy \right) dx \\
 &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} f(x) g(y) \varphi(x + y) dy dx \\
 &= \int_{\mathbb{R}^n} (f * g)(z) \varphi(z) dz \\
 &= \langle T_{f*g}, \varphi \rangle.
 \end{aligned}$$

Thus

$$T_f * T_g = T_{f*g}.$$

(b) Let  $u_1, u_2, u_3, u_4 \in \mathcal{D}'(\mathbb{R}^n)$ , with  $u_3, u_4$  of compact support. For any  $\varphi \in \mathcal{D}(\mathbb{R}^n)$ ,

$$\langle u_1 * u_3, \varphi \rangle = \left\langle u_1, x \mapsto \langle u_3, \varphi(x + \cdot) \rangle \right\rangle.$$

Then

$$\begin{aligned}
 \langle (u_1 + au_2) * u_3, \varphi \rangle &= \left\langle u_1 + au_2, x \mapsto \langle u_3, \varphi(x + \cdot) \rangle \right\rangle \\
 &= \langle u_1 * u_3, \varphi \rangle + a \langle u_2 * u_3, \varphi \rangle.
 \end{aligned}$$

Hence

$$(u_1 + au_2) * u_3 = u_1 * u_3 + a u_2 * u_3.$$

The same computation shows that

$$u_1 * (u_3 + au_4) = u_1 * u_3 + a u_1 * u_4.$$

□

## Remarks on convolution of distributions and regularisation

(1) **Why is compact support needed?** For functions  $f, g$  the convolution

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y) g(y) dy$$

is defined by an ordinary integral. For distributions  $u, v \in \mathcal{D}'(\mathbb{R}^n)$  one instead defines

$$\langle u * v, \varphi \rangle := \left\langle u_x, \left\langle v_y, \varphi(x + y) \right\rangle \right\rangle, \quad \varphi \in \mathcal{D}(\mathbb{R}^n),$$



whenever this makes sense. The inner pairing

$$x \mapsto \langle v_y, \varphi(x + y) \rangle$$

must be a test function in  $x$  (smooth with compact support), so that the outer pairing with  $u$  is defined. This is guaranteed, for example, if  $v$  has compact support. Without such a hypothesis there is no reason for  $x \mapsto \langle v_y, \varphi(x + y) \rangle$  to belong to  $\mathcal{D}(\mathbb{R}^n)$ , and the convolution  $u * v$  may fail to be well-defined. This is why, in the general theory, one usually assumes that at least one of the distributions has compact support (or stronger decay conditions) when defining  $u * v$ .

**(2) Convolution with test functions as regularisation.** If  $\phi \in C_0^\infty(\mathbb{R}^n)$  and  $T \in \mathcal{D}'(\mathbb{R}^n)$ , the convolution

$$(T * \phi)(x) := \langle T, \phi(x - \cdot) \rangle$$

is always well-defined and belongs to  $C^\infty(\mathbb{R}^n)$ . In fact, for every multiindex  $\alpha$ ,

$$D^\alpha(T * \phi) = T * (D^\alpha \phi).$$

If  $(\phi_\varepsilon)_{\varepsilon>0}$  is a mollifier (approximate identity), one moreover has

$$T * \phi_\varepsilon \longrightarrow T \quad \text{in } \mathcal{D}'(\mathbb{R}^n) \text{ as } \varepsilon \rightarrow 0.$$

Thus convolution with test functions provides a canonical way to *regularise* distributions:

$$T \in \mathcal{D}'(\mathbb{R}^n) \xrightarrow{* \phi_\varepsilon} T * \phi_\varepsilon \in C^\infty(\mathbb{R}^n), \quad T * \phi_\varepsilon \rightarrow T.$$

In the tempered setting, the same picture holds with Schwartz mollifiers  $\eta_\varepsilon \in \mathcal{S}(\mathbb{R}^n)$  and tempered distributions  $U \in \mathcal{S}'(\mathbb{R}^n)$ .

**Convolution with test functions.** If  $\phi \in C_0^\infty(\mathbb{R}^n)$  and  $T \in \mathcal{D}'(\mathbb{R}^n)$ , the convolution  $T * \phi$  is defined by

$$(T * \phi)(x) := \langle T, \phi(x - \cdot) \rangle, \quad x \in \mathbb{R}^n.$$

Here  $\phi(x - \cdot)$  denotes the test function of the dummy variable  $y$  given by  $y \mapsto \phi(x - y)$ . Thus we first fix  $x$ , then consider the test function  $y \mapsto \phi(x - y)$ , and finally let  $T$  act on it.

*Examples.*

1. If  $T = \delta_0$ , then

$$(\delta_0 * \phi)(x) = \langle \delta_0, \phi(x - \cdot) \rangle = \phi(x - 0) = \phi(x).$$

2. More generally, for the Dirac mass at  $a \in \mathbb{R}^n$ ,

$$(\delta_a * \phi)(x) = \langle \delta_a, \phi(x - \cdot) \rangle = \phi(x - a).$$

3. If  $T = T_f$  is the regular distribution induced by  $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ , then

$$(T_f * \phi)(x) = \int_{\mathbb{R}^n} f(y) \phi(x - y) dy,$$

which is the classical convolution of  $f$  and  $\phi$ .

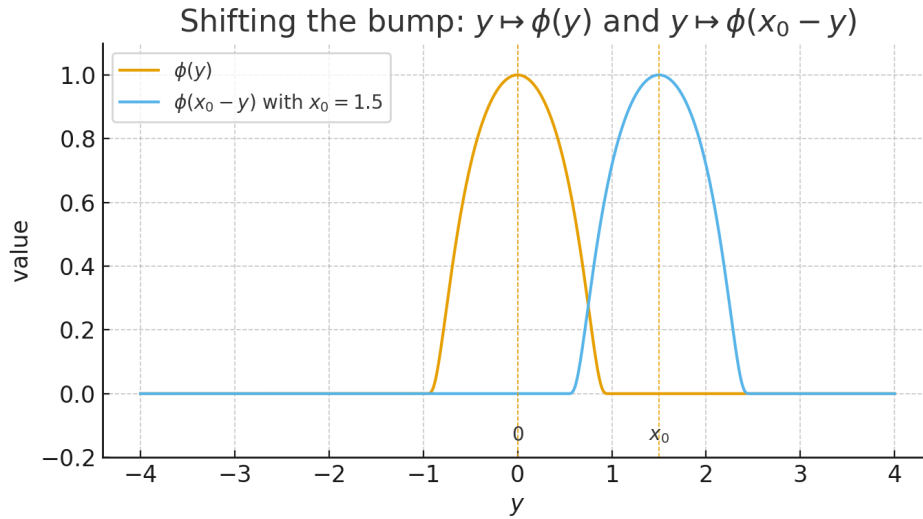


Figure 51: Geometric meaning of  $\phi(x - \cdot)$ . The blue curve is the bump  $y \mapsto \phi(y)$ , centred at 0. For a fixed  $x_0$ , the orange curve is  $y \mapsto \phi(x_0 - y)$ : the same bump, reflected and shifted so that it is attached to the point  $x_0$ . For each  $x$ , the test function  $y \mapsto \phi(x - y)$  is what we feed into the distribution  $T$  in the definition  $(T * \phi)(x) = \langle T, \phi(x - \cdot) \rangle$ .

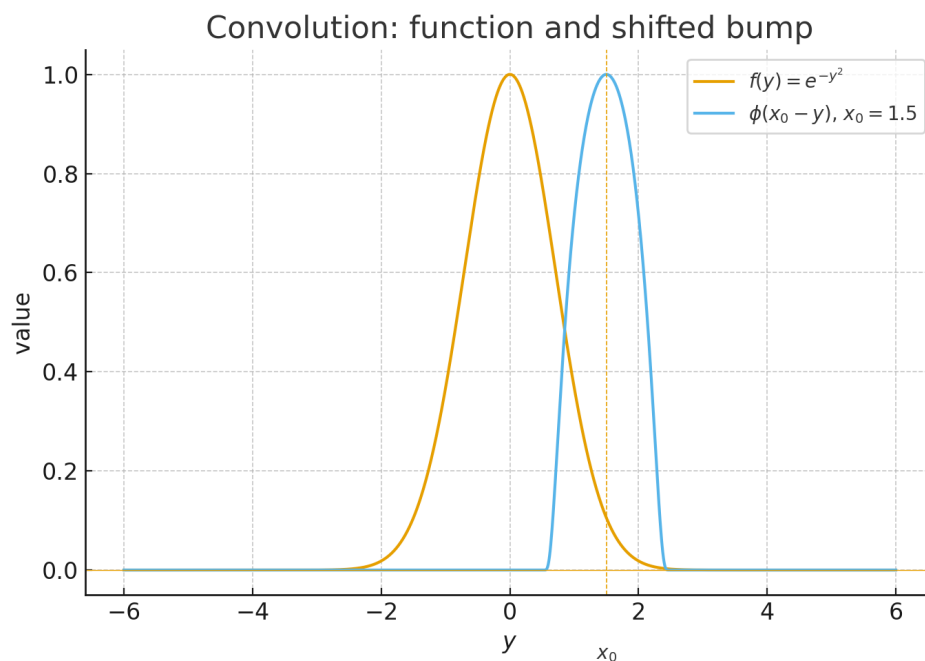


Figure 52: Convolution with a bump: the function  $f(y) = e^{-y^2}$  (blue) and the shifted test function  $y \mapsto \phi(x_0 - y)$  (orange) for a fixed  $x_0$ . As  $x_0$  varies, the bump slides along the  $y$ -axis.

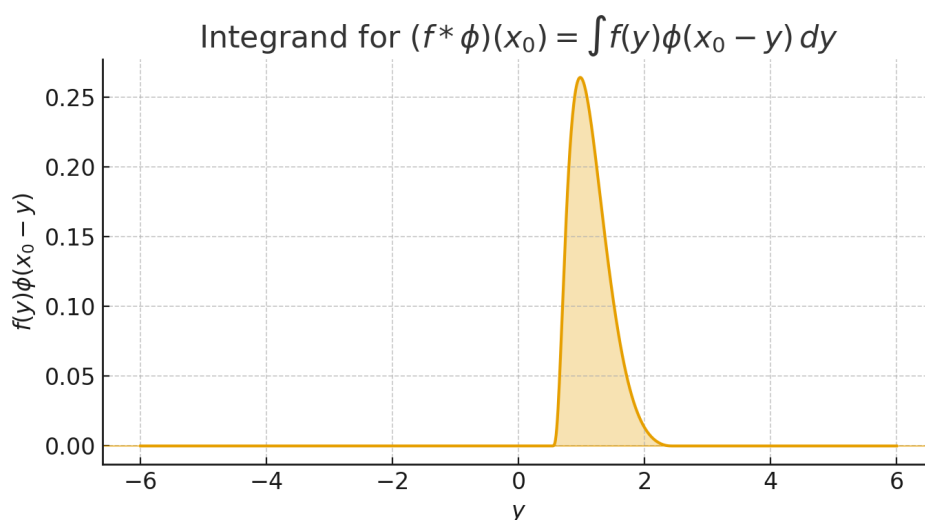


Figure 53: Integrand  $f(y)\phi(x_0 - y)$  for  $f(y) = e^{-y^2}$ . The area under this curve equals  $(f * \phi)(x_0)$ . Convolution at  $x_0$  is the weighted overlap of  $f$  with the shifted bump.

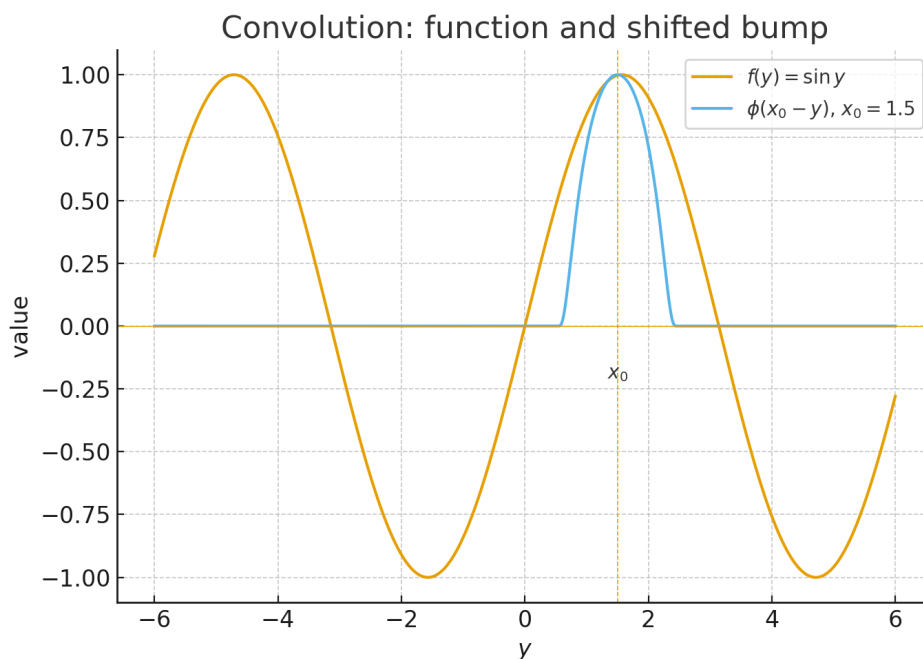


Figure 54: Convolution with a bump: the function  $f(y) = \sin y$  (blue) and the shifted test function  $y \mapsto \phi(x_0 - y)$  (orange) for a fixed  $x_0$ . As  $x_0$  moves, the bump samples local averages of  $\sin y$ .

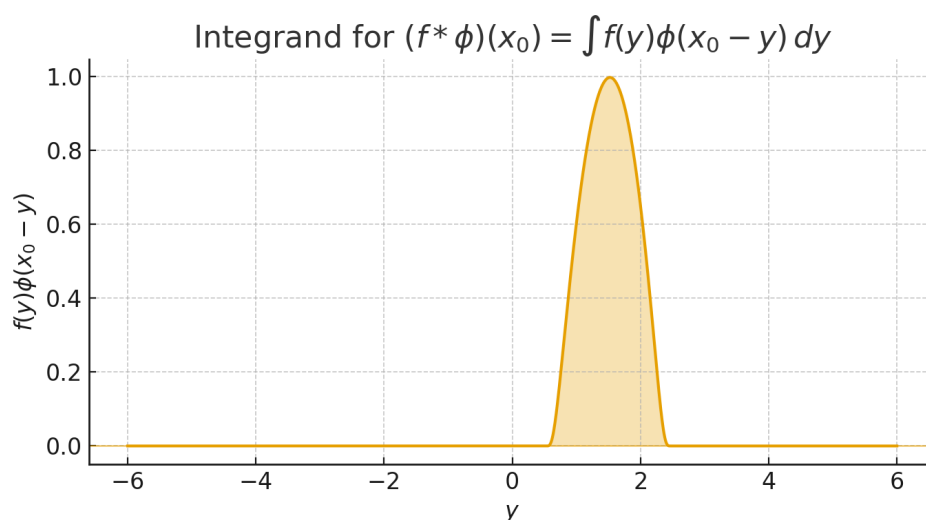


Figure 55: Integrand  $f(y)\phi(x_0 - y)$  for  $f(y) = \sin y$ . The signed area under this curve equals  $(f * \phi)(x_0)$ . Convolution smooths oscillations by averaging  $\sin y$  against the bump.

**Exercise 2.10. Identity property of the Dirac distribution**

(a) Show that if  $\phi \in \mathcal{D}(\mathbb{R}^n)$  then

$$\delta_0 * \phi = \phi.$$

(b) Show that if  $u \in \mathcal{D}'(\mathbb{R}^n)$  has compact support, then

$$\delta_0 * u = u.$$

**Solution**

(a) Let  $\phi \in \mathcal{D}(\mathbb{R}^n)$ . By definition of convolution between a distribution and a test function,

$$\langle \delta_0 * \phi, \psi \rangle = \langle \delta_0, x \mapsto \langle \phi, \psi(x + \cdot) \rangle \rangle, \quad \psi \in \mathcal{D}(\mathbb{R}^n).$$

Since  $\delta_0$  evaluates at 0,

$$\langle \delta_0 * \phi, \psi \rangle = \langle \phi, \psi(\cdot + 0) \rangle = \langle \phi, \psi \rangle.$$

Thus

$$\delta_0 * \phi = \phi.$$

(b) Let  $u \in \mathcal{D}'(\mathbb{R}^n)$  have compact support. The convolution  $\delta_0 * u$  is defined by

$$\langle \delta_0 * u, \psi \rangle = \langle \delta_0, x \mapsto \langle u, \psi(x + \cdot) \rangle \rangle, \quad \psi \in \mathcal{D}(\mathbb{R}^n).$$

Again using the defining property of  $\delta_0$ ,

$$\langle \delta_0 * u, \psi \rangle = \langle u, \psi(\cdot + 0) \rangle = \langle u, \psi \rangle.$$

Therefore,

$$\delta_0 * u = u.$$

□